

Simon's Multidimensional L^2 Conjecture and the Failure of Fixed-Potential Selected-Channel Entropy

Plücker–Schur reductions and a reciprocal Riesz–Talbot construction

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Abstract

Let $H = -\Delta + V$ on $L^2(\mathbb{R}^d)$, $d \geq 2$, with V bounded, real, and square integrable against the radial weight $|x|^{1-d}$. We examine a finite-angular-channel entropy strategy for Simon's multidimensional L^2 conjecture. For a finite angular model we first derive an exact Plücker–Jacobi identity which expresses the selected spectral density through the reciprocal volume of an analytic inverse Jost frame. We also record the corresponding full-column Hilbert–Schmidt cost, strong-resolvent completion, and double-buffered high-channel estimates.

The main result is negative for the proposed proof mechanism. In dimension three we construct a single bounded, real, mean-zero zonal potential $V \in X_3$ and a compact energy interval $K \subseteq (0, \infty)$ such that, for every cofinal positively smoothed, radially truncated, double-buffered Galerkin diagonal,

$$\int_K \log^- \|J_n(k)^{-1} e_0\|^2 dk \longrightarrow +\infty.$$

The construction combines a translated-barrier Hardy identity, zonal quarter-Talbot Riesz products, fractional-moment synchronization of a reciprocal Möbius phase, and uniform lacunary placement of radial cells at summable charged cost. It disproves the universal fixed-potential selected-channel entropy estimate and the stronger endpoint estimates that would imply it. Several natural alternatives are also decided: unrestricted matrix low-column estimates, coefficientwise quadratic expansion, raw reflection bounds, and one-point Poisson–Jensen replacements fail, whereas the exact Plücker, Riccati, interpolation, and buffered-completion identities remain valid.

This result does not determine the absolutely continuous spectral multiplicity of the limiting Schrödinger operator. In particular, it neither proves nor disproves Simon's conjecture: spectral density may persist in angular directions which move with the approximation. Any continuation of this entropy programme must therefore be basis-free, moving-probe, or full-operator, or must use a different spectral mechanism.

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Scope of the claims. The unconditional main conclusion is the failure of the fixed selected-channel entropy route. Statements used only conditionally are labelled as such at the point of use, and failed intermediate estimates are retained as explicit counterexamples. Simon's multidimensional L^2 conjecture remains open.

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1 The conjecture and the exact standard of proof

Let $V \in L^\infty(\mathbb{R}^d; \mathbb{R})$ and

$$H = -\Delta + V, \quad \text{dom}(H) = H^2(\mathbb{R}^d). \quad (1)$$

Set

$$\|V\|_{X_d}^2 := \int_{\mathbb{R}^d} |x|^{1-d} |V(x)|^2 dx. \quad (2)$$

In polar coordinates $x = r\omega$,

$$\|V\|_{X_d}^2 = \int_0^\infty \|V(r, \cdot)\|_{L^2(\mathbb{S}^{d-1})}^2 dr. \quad (3)$$

The bounded-real formulation of Simon’s Conjecture 20.2 is

$$m_{\text{ac}}(E; H) = \infty \quad \text{for Lebesgue-a.e. } E > 0. \quad (4)$$

It does not assert that the spectrum on $(0, \infty)$ is purely absolutely continuous. Embedded point or singular-continuous spectrum may coexist with the asserted absolutely continuous component.

The unitary polar flattening

$$(\mathcal{U}f)(r, \omega) = r^{(d-1)/2} f(r\omega) \quad (5)$$

gives, on the exterior half-line,

$$\tilde{H} = -\partial_r^2 + \frac{A_d}{r^2} + M_{V(r)}, \quad A_d = -\Delta_{\mathbb{S}^{d-1}} + \frac{(d-1)(d-3)}{4}, \quad (6)$$

acting on $L^2((1, \infty); \mathcal{K})$ with $\mathcal{K} = L^2(\mathbb{S}^{d-1})$. Formula (6) exposes the central tension: $M_{V(r)}$ is controlled in angular L^2 , whereas its operator norm is $\|V(r, \cdot)\|_\infty$.

The primary literature still describes the conjecture as unresolved; see Simon [1], Denisov’s survey [5], and his later operator-valued discussion [13]. The radial case follows from the one-dimensional L^2 theorem of Deift and Killip [2]. Known multidimensional theorems require additional decay, sign, angular regularity, operator-norm control, or a generic coupling parameter [8, 10, 14, 15].

2 Audit baseline and non-negotiable obstructions

The earlier finite-channel audit establishes the following ledger.

Statement	Status	Consequence
Strong-resolvent convergence of controlled radial/angular truncations	PROVED HERE	Herglotz matrices and matrix spectral measures converge vaguely.
Finite-model Jost/Gram formula	PROVED HERE	The canonical boundary density is positive definite; compact-probe density is positive definite a.e. and has finite local entropy.
Lower semicontinuity of $\int \log^- \det W$	PROVED HERE	A uniform entropy estimate survives the truncation limit.
Rank/countability finish	CONDITIONAL	Uniform N -channel entropy for every N implies (4).
Leakage-only abstract FCEN	FALSE	Schrödinger locality and propagation are indispensable.
Truncated leakage $(P_L - P_N)M_V P_N$	DIAGNOSTIC	Angular frequencies above L can alias into retained high channels.
Uniform full-leakage PSR_N endpoint at $\alpha = 1$	FALSE	Already for $N = 1$, a cutoff-edge scalar rotator and pure-degree barrier have bounded full-column cost but divergent quotient entropy.
Coefficientwise quotient estimate QBC_1	FALSE	A separated radial two-slab family has bounded L^2 cost but a divergent positive quadratic coefficient.
Exact scalar physical-coupling entropy	KNOWN	The one-dimensional transmission trace inequality controls the radial $m = N = 1$ Jost entropy uniformly in support radius.
Dual inverse frame and differential quotient flux	PROVED HERE	$\Gamma_P = \ \wedge^N (J^{-1}P)\ ^{-1}$, and the high-high cocycle generator cancels exactly.
Unrestricted cocycle low-column bound	FALSE	A high block preloads amplification e^H , while a later quarter-turn costs only $\pi^2/(4L)$.
Physical Riccati descent	PROVED HERE	The flux-normalized outgoing frame gives an exact Riccati formula for Γ_P and its differential quotient flux.
Cutoff-compatible scalar Riccati/Feshbach replacement	FALSE	The fixed Riesz–Talbot potential defeats every cofinal positive double-buffered diagonal, so a fixed selected probe is still insufficient.
Positive fixed-potential buffered diagonal and completion	PROVED HERE	Positive angular smoothing preserves L^∞ and X_3 , the buffered models converge strongly resolvent, and one liminf entropy estimate suffices.
Fixed-potential buffered selected $N = 1$ entropy	FALSE	One bounded real mean-zero X_3 potential has divergent selected entropy along every cofinal positive smoothing, radial truncation, and double-buffered Galerkin diagonal.

Statement	Status	Consequence
Outgoing high-channel edge removal	PROVED HERE	If $J/R \rightarrow \infty$, the outgoing high-sector Schur complement is $O(R^2/J^2)$; the double buffer places the Galerkin edge beyond the active block.
Broadband wrapped-mask turning-scale test	PROVED HERE	Increasing pulse area gives uniform high-band mass, but vanishing charged cost and clean transport force the mask far outside the generated modes' turning radius.
Outer unitary converter ordering	PROVED HERE	A reflectionless outer right factor leaves the scalar Plücker quotient unchanged; suppression requires conversion within or inside the turning dynamics.
Zero-background dimension-normalized matrix sum rule	PROVED HERE	The determinant trace identity is linear in matrix dimension after normalization and matches the scalar X_3 cost.
Naive insertion of the Bessel background	FALSE	The background-subtracted quadratic invariant contains an explicit angular-mean term with coefficient of order L^2 .
Radial-mean Bessel cancellation and signed relative sum rule	PROVED HERE	Taking $D_L/r^2 + q(r)I$ as reference removes every cross term and gives a dimension-uniform signed logarithmic identity.
Star-Feshbach charged compressed resolvent	PROVED HERE	For $N = 1$, retaining the full high-high block gives a positive trace-class negative-energy correction bounded only by $\int \ Q_0 M_{V \circ e_0}\ ^2$.
Star-relative bound-state divisor	PROVED HERE	Off-diagonal sign-flip convexity makes the relative negative 3/2-moment nonnegative, so the signed second trace is bounded by twice the charged column cost.
Scalar Schur high-minor divisor	OPEN	Complex zeros of the chosen high minor have indefinite cubic trace contribution; the pole-free radiation column must retain their reciprocal refocusing.
Radial-mean two-port and reciprocal refocusing	PROVED HERE	Exact unitary gauges remove the forward blocks; the physical radiation column changes only through $\Theta'R$, where $ \Theta'_n = 2k/ f_{q,n} ^2$.
Bessel-balanced scalar generator	PROVED HERE	Sandwiching the return coupling by the reciprocal phase velocity gives a unitary conjugate of W_L° , exactly preserving the charged low column and angular bandwidth.

Statement	Status	Consequence
Signed pole-free work reduction	PROVED HERE	Rationally weighted spectral mass controls density amplification, so a signed bound on the exact refocusing work already implies the one-sided $N = 1$ entropy estimate.
Poisson/interior-row shortcut	FALSE	A fixed-band mean-zero slab can have vanishing charged cost and vanishing compact real-band forward loss while its forward Jost row at $-\kappa^2$ grows exponentially; the singular mass at infinity is not L^2 -controlled.
Transport-free scalar $N = 1$ entropy	PROVED HERE	After deleting A_3/r^2 , angular fibers are scalar one-dimensional problems; Jensen and the scalar Deift–Killip transmission inequality give the selected-density entropy with exactly the charged cost.
Universal signed centrifugal endpoint	FALSE	The interpolation identity and endpoint cancellation are exact, but the universal estimate SCI_1 is false because it would imply the disproved fixed selected-channel entropy bound.
Negative-energy centrifugal action	PROVED HERE	The compressed resolvent decreases in λ and its total D_L/r^2 action is bounded by the charged star column; a monotone Stieltjes-density example proves that this does not by itself control boundary entropy.
Universal radially scaled angular heat endpoint	FALSE	The heat-flow identity and budget $\frac{1}{2}\ V^\circ\ _{X_3}^2$ are exact, but the universal product-Fisher estimate HFI_1 is false by the fixed Riesz–Talbot potential.
Raw directional reflection/bilinear density action	DIAGNOSTIC	It is sufficient but over-strong: an ideal scalar shell has arbitrarily large evanescent reflection while leaving the observation-plane density unchanged.
Cutoff-compatible refocusing-work entropy	FALSE	The signed reduction remains exact, but a universal bound is contradicted by the reciprocal Riesz–Talbot cascade.
Arbitrary matrix channel rotator/barrier	FALSE	The low-column cost tends to zero while the positive Grassmannian entropy diverges; the cutoff-edge scalar analogue is proved below.
Exact cutoff-edge scalar rotator/barrier	PROVED HERE	Every real 2×2 block has a reducing scalar lift; a pure degree- $2L$ harmonic is an exact high-only barrier of cost $\asymp L^{-1/2}$.
Scalar PSR_1	FALSE	The exact cutoff-edge lift gives bounded charged cost and fixed-band quotient entropy tending to infinity.

Statement	Status	Consequence
Full-space scalar converter	PROVED HERE	A zonal quarter-Talbot Riesz product concentrates the selected state exponentially, a localized reciprocal reflector charges exponentially little, and the reverse train refocuses it.
Physically scaled scalar nonlinear Plancherel SSNP $_\ell$	OPEN	A uniform fixed-band bound would exclude the frozen converter before restoring radial second-order errors.
Pure-detuning nonlinear Plancherel budget	PROVED HERE	Fixed-band transition probability has an exact nonlinear L^2 cost.
Abstract scaled-ensemble extension	FALSE	A first-order matrix rotator gives uniform band conversion at vanishing charged cost.
Scalar ground-state adiabatic swap	PROVED HERE	Positivity fixes the ground branch, and a small endpoint potential at $R = O(\ell)$ has vanishing overlap with h_ℓ .
Monotone chirped scalar passage	PROVED HERE	Uniform adiabaticity across a fixed energy band forces a divergent charged cost.
Rotating-frame ground ladder	PROVED HERE	At a free endpoint its ground angular momentum depends on s ; no fixed rotation rate selects one common ℓ across a nondegenerate band.
Gap-cancelled finite-depth composite passage	PROVED HERE	Vanishing cost forces angular frequency $\gg \ell^{3/2}$; fixed-depth and $o(\sqrt{\ell})$ -depth bounded-gradient designs cannot convert.
Exact scalar star-frame reduction	PROVED HERE	The charged column becomes an off-diagonal vector of exactly the same norm; all uncharged danger is isolated in its energy-dependent high-sector rotation.
Scalar two-plane mixer–barrier duality	PROVED HERE	The mixer leakage is Sb^2G_h , the cheapest high-only barrier costs $Q^2/(SG_h)$, and their product is exactly b^2Q^2 .
Charged pulse-area and scalar-roughness toll	PROVED HERE	Gap cancellation forces $\mathcal{E}_\ell \gtrsim \ell^{-1/4}$ and average angular gradient $\gtrsim \ell/\mathcal{E}_\ell^2$; target-scale Lipschitz controls have a positive cost floor.
Born high-band recoil	PROVED HERE	A direct fixed-band launch to degrees $\geq L$ costs at least order L^2/R^2 at first order; only a nonlinear roughness bootstrap remains.
Nonlinear charged spectral gate	PROVED HERE	Even with arbitrary high–high amplification retained, first escape from degrees $\leq N$ has angle at most $s(N+1) \int \ M_v e_0\ $; a small charged component acts strongly only on a preconcentrated state.

Statement	Status	Consequence
Density-concentration action	PROVED HERE	Conversion to h_ℓ forces $\int \psi ^2 \nabla \psi ^2 \gtrsim R^2 \sqrt{\ell}$; scalar phase masks leave density unchanged and cannot supply this action directly.
Wrapped-phase gate additivity	FALSE	A bounded azimuthal sawtooth injects uniform fixed-band mass into degrees $\geq L$ at vanishing charged quadratic cost when centrifugal transport is deleted.
Radial energy/BV obstruction	PROVED HERE	With centrifugal transport restored, angular current is at most $\sqrt{\Gamma_v} \log(R_2/R_1)$; stationary and radially low-variation weak slabs cannot convert.
Radial superoscillation toll	PROVED HERE	Gap cancellation forces radial L^∞ variation, and hence bounded-amplitude layer count, $\gtrsim \ell/\mathcal{E}_\ell^2$.
Raw radial variation as a cost proxy	FALSE	Tiny rapid radial oscillations make $\Gamma_v \rightarrow \infty$ while their charged cost, angular gradient norm, and full propagator effect vanish.
Effective radial-work bridge	PROVED HERE	Conversion forces state-correlated work of order one and $\Re_v \mathfrak{D}_s \gtrsim R^2$; for angular bandwidth L , $\mathcal{E} \mathfrak{D}_s \gtrsim R^2/L^2$.
Moving-support effective-work bridge	OPEN	The remaining profile is doubly super-rough; no charged-cost estimate yet controls its state-correlated work with broadband reciprocal refocusing.
Frozen resonant-drive test	DIAGNOSTIC	Single-energy transfer is cheap, but the Born and pure-detuning nonlinear models impose fixed-band costs.
Exterior/full-space bridge after inserting a free collar	KNOWN	For $q_d = \lceil d/2 \rceil$ the resolvent-power difference is trace class, so a.c. multiplicity is preserved.

The table is only an index. The hypotheses, equations, and logical content of every row are recorded next so that none of the status labels has to be accepted on trust.

2.1 Controlled truncations: strong resolvent and measure convergence

On the exterior space set

$$H_0 = -\partial_r^2 + r^{-2} A_d, \quad \hat{H}_{R,L} = H_0 + P_L V_R P_L, \quad V_R = \mathbf{1}_{[1,R]} M_V. \quad (7)$$

Here $P_L \rightarrow \mathbf{I}$ strongly, $V_R \rightarrow M_V$ strongly, and all three multiplication operators have norm at most $\|V\|_\infty$. Hence, for every f ,

$$(P_L V_R P_L - M_V) f = P_L V_R (P_L - \mathbf{I}) f + P_L (V_R - M_V) f + (P_L - \mathbf{I}) M_V f \rightarrow 0. \quad (8)$$

If $H_{\text{ext}} = H_0 + M_V$ and $z \in \mathbb{C} \setminus \mathbb{R}$, the second resolvent identity gives

$$\begin{aligned} & (\hat{H}_{R,L} - z)^{-1} - (H_{\text{ext}} - z)^{-1} \\ &= -(\hat{H}_{R,L} - z)^{-1} (P_L V_R P_L - M_V) (H_{\text{ext}} - z)^{-1}, \end{aligned} \quad (9)$$

and the two resolvents have norm at most $|\Im z|^{-1}$. Equations (8)–(9) prove strong resolvent convergence. For a fixed finite probe Φ_N this yields

$$M_N^{R,L}(z) := \Phi_N^*(\hat{H}_{R,L} - z)^{-1}\Phi_N \longrightarrow \Phi_N^*(H_{\text{ext}} - z)^{-1}\Phi_N =: M_N(z) \quad (10)$$

locally uniformly on $\mathbb{C}_+$. Equivalently, the representing matrix measures converge vaguely:

$$\int f(E) d\mu_N^{R,L}(E) \longrightarrow \int f(E) d\mu_N(E), \quad f \in C_c(\mathbb{R}). \quad (11)$$

This step alone says nothing about preservation of absolutely continuous density; that requires the entropy bound below.

2.2 Finite-model Jost/Gram formula and local entropy

For fixed R, L , let $m = m_L$ and let S and F solve

$$\begin{aligned} -S'' + Q_{R,L}S &= k^2S, & S(1, k) &= 0, & S'(1, k) &= I_m, \\ -F'' + Q_{R,L}F &= k^2F, & F(r, k) &\sim F_{0,L}(r, k) \quad (r \rightarrow \infty), \end{aligned} \quad (12)$$

where $Q_{R,L} = D_L/r^2 + P_L V_R P_L$, and put $J(k) = F(1, k)$. With the Wronskian of $F_{0,L}$ normalized to $2ikI_m$, the Dirichlet Weyl matrix satisfies

$$\Im m_{R,L}(k^2 + i0) = kJ(k)^{-*}J(k)^{-1}, \quad h_{R,L}(k^2) = \frac{k}{\pi}J(k)^{-*}J(k)^{-1}. \quad (13)$$

For a compact probe Φ_N define, with a transpose rather than an adjoint in the analytic variable,

$$T_N(k) = \int_1^\infty S(r, k)^T \Phi_N(r) dr, \quad \mathcal{A}_N(k) = J(k)^{-1}T_N(k). \quad (14)$$

Then the Gram formula and Cauchy–Binet identity give

$$W_N^{R,L}(k^2) = \frac{k}{\pi} \mathcal{A}_N(k)^* \mathcal{A}_N(k), \quad (15)$$

$$\det W_N^{R,L}(k^2) = \left(\frac{k}{\pi}\right)^N \left\| \bigwedge^N \mathcal{A}_N(k) \right\|^2 = \left(\frac{k}{\pi}\right)^N \sum_{|S|=N} |\det \mathcal{A}_{N,S}(k)|^2. \quad (16)$$

The large-imaginary- k Volterra asymptotic makes $\bigwedge^N \mathcal{A}_N$ nonzero as an analytic vector. Its real zeros on a compact positive interval are therefore isolated and of finite order. Thus

$$\det W_N^{R,L}(E) > 0 \quad \text{for a.e. } E > 0, \quad \int_I \log^- \det W_N^{R,L}(E) dE < \infty \quad (17)$$

for each fixed model. No constant uniform in R or L follows from this argument.

2.3 Entropy lower semicontinuity

Write $\mu_n = W_n dE + \mu_n^s$, $\mu = W dE + \mu^s$, and suppose $\mu_n \rightarrow \mu$ vaguely near $I = [a, b]$. On Hermitian matrices define

$$\varphi(A) = \begin{cases} \max\{0, -\log \det A\}, & A > 0, \\ +\infty, & A \not> 0. \end{cases} \quad (18)$$

The map φ is convex, lower semicontinuous, and decreasing in the Loewner order. If ρ_ε is a positive approximate identity and $I_\delta = [a + \delta, b - \delta]$, positivity of $\rho_\varepsilon * \mu_n^s$ and Jensen's inequality imply

$$\int_{I_\delta} \varphi((\rho_\varepsilon * \mu_n)(x)) dx \leq \int_I \varphi(W_n(E)) dE. \quad (19)$$

Vague convergence passes the convolved matrices to the limit. Fatou's lemma, differentiation of finite matrix measures as $\varepsilon \downarrow 0$, and then $\delta \downarrow 0$ give

$$\int_I \log^- \det W(E) dE \leq \liminf_{n \rightarrow \infty} \int_I \log^- \det W_n(E) dE. \quad (20)$$

The inner interval is indispensable: vague convergence does not prevent mass from approaching an endpoint.

2.4 The rank and countability finish

If the limiting entropy is finite, then a positive-measure set on which $\det W_N(E) = 0$ is impossible. Hence

$$\int_I \log^- \det W_N < \infty \implies \text{rank } W_N(E) = N \quad \text{for a.e. } E \in I. \quad (21)$$

In the direct-integral representation of the absolutely continuous part, a density generated by N probes has rank no larger than the fiber multiplicity. Therefore

$$\text{rank } W_N(E) = N \implies m_{\text{ac}}(E; H) \geq N \quad \text{for a.e. } E \in I. \quad (22)$$

Intersecting the full-measure sets over $N \in \mathbb{N}$ and compact positive intervals with rational endpoints yields

$$m_{\text{ac}}(E; H) = \infty \quad \text{for a.e. } E > 0. \quad (23)$$

This finish is rigorous but conditional on a uniform finite-model entropy estimate and a valid exterior/full-space bridge.

2.5 Why leakage and Herglotz positivity alone are false

The first exact counterexample is the fiberwise projection

$$\mathcal{H}(x) = \begin{pmatrix} x & \sqrt{x(1-x)} \\ \sqrt{x(1-x)} & 1-x \end{pmatrix}, \quad \mathcal{H}(x)^2 = \mathcal{H}(x), \quad (24)$$

on $L^2(0,1) \oplus L^2(0,1)$. Its off-diagonal leakage has finite cost

$$\int_0^1 x(1-x) dx = \frac{1}{6}, \quad (25)$$

but the first coordinate has spectral measure

$$d\mu_{e_1} = \frac{1}{2}\delta_0 + \frac{1}{2}\delta_1. \quad (26)$$

Thus its density vanishes a.e. on $(0,1)$. Nevertheless the block self-energy obeys

$$\Im \left[B^* (H_{QQ} - z)^{-1} B \right] \succeq 0, \quad z \in \mathbb{C}_+. \quad (27)$$

Consequently neither finite Hilbert–Schmidt leakage nor the Herglotz sign implies finite-channel entropy in abstract block theory. The family obtained by multiplying the off-diagonal entries by α has varying eigenbranches for generic α but collapses at $\alpha = 1$. Thus almost-every-coupling spectral averaging cannot simply be evaluated at the physical coupling.

2.6 Why truncated leakage is unsafe

On $\mathbb{S}^1$, with

$$v = a + c \cos(\ell\omega) + d \cos(2\ell\omega), \quad (28)$$

the retained block on $\text{span}\{Y_0, C_\ell\}$ is

$$\begin{pmatrix} a & c/\sqrt{2} \\ c/\sqrt{2} & a + d/2 \end{pmatrix}. \quad (29)$$

Writing $b = c/\sqrt{2}$, the truncated column sees $a^2 + b^2$, whereas the full column sees

$$\|QM_v Y_0\|^2 = b^2 + \frac{d^2}{2}. \quad (30)$$

The choice $a = c = 0$, $d = 2Q$ therefore creates a $+Q$ retained-channel barrier at zero truncated-column cost. Every proposed estimate below uses the full complement or the full scalar cost.

2.7 The corrected full-leakage endpoint

With $Q_N = I - P_N$, orthogonality gives the exact identity

$$\|P_N M_V P_N\|_{\mathfrak{S}_2}^2 + \|Q_N M_V P_N\|_{\mathfrak{S}_2}^2 = \|M_V P_N\|_{\mathfrak{S}_2}^2 = \sum_{j=1}^N \|V Y_j\|_2^2. \quad (31)$$

If $Y_1 = |\mathbb{S}^{d-1}|^{-1/2}$, then

$$|\mathbb{S}^{d-1}|^{-1} \|V(r, \cdot)\|_2^2 \leq \|M_{V(r)} P_N\|_{\mathfrak{S}_2}^2 \leq \left(\sum_{j=1}^N \|Y_j\|_\infty^2 \right) \|V(r, \cdot)\|_2^2. \quad (32)$$

The safe, still open endpoint is therefore

$$\sup_{R,L} \int_I \log^- \det W_N^{R,L}(E) dE \leq C_{I,N,\Phi} \left(1 + \int_1^\infty \|M_{V(r)} P_N\|_{\mathfrak{S}_2}^2 dr \right). \quad (33)$$

Equation (32) proves that the corrected cost is exactly of the strength of Simon's hypothesis. It does not prove (33).

2.8 The arbitrary matrix rotator/barrier obstruction is exact

In the two-channel matrix model, let

$$V_{\text{rot}}(r) = \delta_\Lambda u_2(r) u_2(r)^*, \quad u_2 = (-\sin \theta, \cos \theta), \quad \delta_\Lambda = \Lambda^{-2/3}, \quad (34)$$

on an interval of length Λ , with $\theta(r) = \pi r/(2\Lambda)$. Its low-column cost is exactly

$$\int_{\text{rot}} \|V_{\text{rot}}(r) e_1\|^2 dr = \frac{1}{2} \Lambda^{-1/3}. \quad (35)$$

The rotating-frame equation has constant coefficients and gives a full modal quarter-turn with error $O_K(\Lambda^{-1/3})$. Follow it by $\text{diag}(0, Q_0)$, which has zero low column, and choose a shorter logarithmic barrier so that its amplification grows like Λ^β with $0 < \beta < 1/3$ small enough. Lemma B.2 shows that forward/backward cancellation fails on a fixed positive fraction of every compact energy interval. Theorem B.3 therefore proves

$$\int_K \log^+ \Gamma_P(F_\Lambda(0, k)) dk \geq c_K \log \Lambda - C_K, \quad \int \|\mathcal{V}_\Lambda(r) P\|_{\mathfrak{S}_2}^2 dr \rightarrow 0. \quad (36)$$

This rejects the arbitrary matrix low-column endpoint, not yet the scalar angular endpoint (33). However, scalar compression does not rescue the estimate at the algebraic level. Proposition C.1 gives an exact scalar lift of every real 2×2 compressed block. For concentrated highest-weight harmonics on $\mathbb{S}^2$, the extra full constant-column cost tends to zero. More decisively, Proposition C.4 constructs a uniformly bounded scalar high-only barrier with

$$\|M_{b_{\ell,Q}}e_0\|_2^2 = O_Q(\ell^{-\alpha}), \quad \|(I - P_{\{e_0, h_\ell\}})M_{b_{\ell,Q}}h_\ell\|_2^2 \leq C_Q e^{-c\ell^{1-2\alpha}}. \quad (37)$$

Thus the barrier stage can be scalarized at vanishing full low-column cost. The unresolved obstruction has narrowed to the converter: can a bounded real scalar evolution rotate the constant harmonic into h_ℓ with sufficiently small error and quadratic cost when the entire angular space and the centrifugal flow are retained?

2.9 The exterior/full-space bridge as a separate theorem

Let $\Sigma = \{|x| = a\}$ lie inside a collar on which the tail potential vanishes, and let

$$H_{\text{dec}} = H_{\text{int}}^D \oplus H_{\text{ext}}^D. \quad (38)$$

The exact sufficient bridge is the existence of an integer q and $c > \|V\|_\infty$ such that

$$(H_{\text{tail}} + c)^{-q} - (H_{\text{dec}} + c)^{-q} \in \mathfrak{S}_1. \quad (39)$$

Birman–Kuroda then gives

$$(H_{\text{tail}})_{\text{ac}} \simeq (H_{\text{dec}})_{\text{ac}} \simeq (H_{\text{ext}}^D)_{\text{ac}}, \quad (40)$$

because H_{int}^D has compact resolvent. The needed theorem is available with exactly local, rather than global, regularity: Lotoreichik [11, Theorem 4.17 and Corollary 4.19] proves the weak Schatten estimate for the coupled/Dirichlet-decoupled pair when the bounded potential has finitely many bounded derivatives only near the cutting hypersurface. A free collar supplies derivatives of every order there, regardless of roughness farther out. The precise power and implication are recorded in Proposition 3.3 below.

3 Three exact reductions

3.1 A compact core can be discarded

Proposition 3.1 (Small-tail reduction). *Fix a compact energy interval $I \Subset (0, \infty)$ and $N \in \mathbb{N}$. Suppose there is $\varepsilon_{I,N} > 0$ with the following property: every bounded real exterior potential W satisfying*

$$\int_R^\infty \|W(r, \cdot)\|_2^2 dr < \varepsilon_{I,N}^2 \quad (41)$$

has absolutely continuous multiplicity at least N for a.e. $E \in I$. Then the same conclusion holds for every bounded real V satisfying (3).

Proof. Choose R so large that $V_{\text{tail}} = \mathbf{1}_{\{r > R\}}V$ satisfies (41). The difference $V - V_{\text{tail}}$ is bounded and compactly supported. Standard short-range scattering, equivalently the Birman–Kuroda theorem applied to a sufficiently high resolvent power, gives unitary equivalence of the absolutely continuous parts of $-\Delta + V$ and $-\Delta + V_{\text{tail}}$. Absolutely continuous multiplicity is therefore preserved. Applying the assumed small-tail statement to V_{tail} proves the claim. $\square$

Remark 3.2. Proposition 3.1 removes a major distraction. The endpoint $\alpha = 1$ need not be reached by continuation from small coupling; after removing a compact core, the physical potential itself has arbitrarily small X_d tail norm. A proof may therefore focus on a perturbative theorem in the correct norm while allowing $\|V\|_\infty$ to remain large on sparse sets.

Proposition 3.3 (Free-collar Dirichlet decoupling). *Let $V_{\text{tail}} \in L^\infty(\mathbb{R}^d; \mathbb{R})$ vanish on a neighbourhood of the sphere $\Sigma = \{|x| = a\}$. Let*

$$H_{\text{tail}} = -\Delta + V_{\text{tail}}, \quad H_{\text{dec}} = H_{\text{int}}^D \oplus H_{\text{ext}}^D \quad (42)$$

be the coupled and Dirichlet-decoupled realizations of the same differential expression. Set

$$q_d = \left\lceil \frac{d}{2} \right\rceil, \quad (43)$$

then, for every $c > \|V_{\text{tail}}\|_\infty$,

$$(H_{\text{tail}} + c)^{-q_d} - (H_{\text{dec}} + c)^{-q_d} \in \mathfrak{S}_{\frac{d-1}{2q_d}, \infty} \subset \mathfrak{S}_1. \quad (44)$$

Consequently the absolutely continuous parts of H_{tail} and H_{ext}^D are unitarily equivalent, including multiplicity.

Proof. The collar hypothesis implies

$$V_{\text{tail}} \in W_\Sigma^{m, \infty}(\mathbb{R}^d) \quad \text{for every } m \in \mathbb{N}_0, \quad (45)$$

where $W_\Sigma^{m, \infty}$ means bounded Sobolev regularity in some neighbourhood of Σ only. Apply Lotoreichik [11, Theorem 4.17] with $\ell = m = q_d$ and $z = -c$. The choice $c > \|V_{\text{tail}}\|_\infty$ puts $-c$ in the resolvent sets of the coupled, Dirichlet-decoupled, and auxiliary Neumann-decoupled operators. The theorem gives

$$(H_{\text{tail}} + c)^{-q_d} - (H_{\text{dec}} + c)^{-q_d} \in \mathfrak{S}_{\frac{d-1}{2q_d}, \infty}. \quad (46)$$

Because $(d-1)/(2q_d) < 1$, its singular values are

$$O\left(j^{-2q_d/(d-1)}\right), \quad (47)$$

with summable exponent. Hence (46) is trace class and proves (44).

Kato–Rosenblum applied to the bounded self-adjoint resolvent powers makes their absolutely continuous parts unitarily equivalent. The function $t \mapsto (t+c)^{-q_d}$ is one-to-one on the spectra of these semibounded operators, so the invariance principle transfers the equivalence back to H_{tail} and H_{dec} . The interior Dirichlet summand has compact resolvent, leaving precisely $(H_{\text{ext}}^D)_{\text{ac}}$. $\square$

The free collar also permits the probes of Section 4.2 to be placed where their regular transform is an explicit low-channel diagonal factor.

3.2 The full column is the scalar cost

Let P_N be a finite-rank projection commuting with A_d , with real orthonormal eigenharmonics $Y_1, \dots, Y_N$, and assume

$$Y_1 = |\mathbb{S}^{d-1}|^{-1/2}. \quad (48)$$

Put $Q_N = I - P_N$.

Proposition 3.4 (Exact full-column identity). *For a.e. r ,*

$$\begin{aligned} \|P_N M_V P_N\|_{\mathfrak{S}_2}^2 + \|Q_N M_V P_N\|_{\mathfrak{S}_2}^2 &= \|M_V P_N\|_{\mathfrak{S}_2}^2 \\ &= \sum_{j=1}^N \|V(r, \cdot) Y_j\|_{L^2(\mathbb{S}^{d-1})}^2. \end{aligned} \quad (49)$$

Consequently,

$$\frac{1}{|\mathbb{S}^{d-1}|} \|V(r, \cdot)\|_2^2 \leq \|M_V P_N\|_{\mathfrak{S}_2}^2 \leq C_N \|V(r, \cdot)\|_2^2, \quad C_N = \sum_{j=1}^N \|Y_j\|_\infty^2. \quad (50)$$

Proof. The first equality is Pythagoras in the orthogonal decomposition $\mathcal{K} = P_N \mathcal{K} \oplus Q_N \mathcal{K}$. Expanding the Hilbert–Schmidt norm in the basis Y_j gives the second equality. The upper bound follows from $|Y_j(\omega)| \leq \|Y_j\|_\infty$. The lower bound is the $j = 1$ term and (48). $\square$

Define the finite-channel energy

$$\mathcal{E}_N(V) := \int_1^\infty \|M_{V(r)} P_N\|_{\mathfrak{S}_2}^2 dr. \quad (51)$$

For fixed N , Proposition 3.4 says

$$|\mathbb{S}^{d-1}|^{-1} \|V\|_{X_{d,\text{ext}}}^2 \leq \mathcal{E}_N(V) \leq C_N \|V\|_{X_{d,\text{ext}}}^2. \quad (52)$$

Thus the corrected full leakage does not introduce a stronger hypothesis. It is simply a finite-channel representation of the original scalar angular cost.

3.3 Finite models and normalization

Let P_L be the complete spectral projection of A_d through angular degree L , let $m_L = \text{rank } P_L$, and assume $P_N \leq P_L$. With compact radial truncation $V_R = \mathbf{1}_{[1,R]} V$, define

$$H_{R,L} = -\partial_r^2 + \frac{D_L}{r^2} + P_L M_{V_R} P_L \quad (53)$$

on $L^2((1, \infty); \mathbb{C}^{m_L})$, with Dirichlet boundary condition at $r = 1$. The Bessel diagonal D_L/r^2 is treated as background, not as perturbation.

Let $F(r, k)$ be the outgoing Jost matrix with the unambiguous normalization

$$F(r, k) = F_{0,L}(r, k)(\mathbf{I} + o(1)), \quad r \rightarrow \infty, \quad (54)$$

where $F_{0,L}$ is the outgoing diagonal Bessel matrix and is normalized so that its real-axis Wronskian is $2ik\mathbf{I}$. Write

$$J_{R,L}(k) = F(1, k). \quad (55)$$

After the harmless channelwise Bessel normalization, the canonical Dirichlet density is

$$h_{R,L}(k^2) = \frac{k}{\pi} J_{R,L}(k)^{-*} J_{R,L}(k)^{-1}, \quad k > 0. \quad (56)$$

All later formulas use this normalization.

4 The Plücker–Schur reduction

This section gives the main new algebraic reduction.

Theorem 4.1 (Exact low-channel Plücker formula). *Let $J \in \mathbb{C}^{m \times m}$ be invertible, let P be an orthogonal projection of rank N , and let $Q = \mathbf{I} - P$. Put*

$$h = \frac{k}{\pi} J^{-*} J^{-1}, \quad h_P = P h P|_{P\mathbb{C}^m}. \quad (57)$$

Then

$$\det h_P = \left(\frac{k}{\pi}\right)^N \frac{\det(Q J J^* Q|_{Q\mathbb{C}^m})}{|\det J|^2} \quad (58)$$

$$= \left(\frac{k}{\pi}\right)^N \frac{\|\Lambda^{m-N}(QJ)\|^2}{|\det J|^2}. \quad (59)$$

Here the exterior norm is the Euclidean norm of the vector of all $(m - N) \times (m - N)$ minors of QJ .

Proof. Set $G = JJ^* > 0$. Since $J^{-*}J^{-1} = G^{-1}$, Jacobi's complementary-minor identity gives

$$\det(PG^{-1}P|_{P\mathbb{C}^m}) = \frac{\det(QGQ|_{Q\mathbb{C}^m})}{\det G}. \quad (60)$$

Now $\det G = |\det J|^2$. Moreover, $QGQ = (QJ)(QJ)^*$ on $Q\mathbb{C}^m$. The Cauchy–Binet identity gives

$$\det((QJ)(QJ)^*) = \sum_{|S|=m-N} |\det(QJ)_S|^2 = \left\| \bigwedge^{m-N} (QJ) \right\|^2. \quad (61)$$

Multiplication by $(k/\pi)^N$ proves both formulas. $\square$

Define the Grassmannian amplification factor

$$\Gamma_{P;R,L}(k) := \frac{|\det J_{R,L}(k)|}{\left\| \bigwedge^{m_L-N} (QJ_{R,L}(k)) \right\|}. \quad (62)$$

Proposition 4.2 (Dual inverse-frame Plücker identity). *For every $J \in \text{GL}(m, \mathbb{C})$ and every coordinate projection P of rank N ,*

$$\frac{\left\| \bigwedge^{m-N} (QJ) \right\|}{|\det J|} = \left\| \bigwedge^N (J^{-1}P) \right\|. \quad (63)$$

Consequently

$$\Gamma_P(J) = \left\| \bigwedge^N (J^{-1}P) \right\|^{-1} = \det(PJ^{-*}J^{-1}P|_{P\mathbb{C}^m})^{-1/2}. \quad (64)$$

Proof. For every N -element row set T , Jacobi's complementary-minor identity gives

$$\det(J^{-1})_{T,P} = \pm \frac{\det J_{Q,T^c}}{\det J}. \quad (65)$$

Square the moduli and sum over T . Cauchy–Binet identifies the left collection with the coordinates of $\bigwedge^N (J^{-1}P)$ and the right collection with those of $\bigwedge^{m-N} (QJ)/\det J$. The Gram determinant formula gives (64). $\square$

Thus the natural analytic exterior vector is

$$\mathfrak{g}_P(k) = \bigwedge^N (J(k)^{-1}P), \quad \log \Gamma_P(k) = -\log \|\mathfrak{g}_P(k)\|. \quad (66)$$

It has no distinguished-minor denominator and works for every fixed N .

Proposition 4.3 (Exact differential Grassmannian flux). *Let $\mathcal{J}(r) \in \text{GL}(m, \mathbb{C})$ solve*

$$\mathcal{J}' = \mathcal{A}\mathcal{J}, \quad \mathcal{A} = \begin{pmatrix} \mathcal{A}_{PP} & \mathcal{A}_{PQ} \\ \mathcal{A}_{QP} & \mathcal{A}_{QQ} \end{pmatrix}. \quad (67)$$

Assume $N < m$; when $N = m$, omit the Q rows and take $C = 0$. Put

$$X = P\mathcal{J}, \quad Y = Q\mathcal{J}, \quad S = YY^*, \quad (68)$$

$$C = XY^*S^{-1}, \quad E = X - CY, \quad \Sigma = EE^*. \quad (69)$$

Then

$$EY^* = 0, \quad \Gamma_P(\mathcal{J})^2 = \det \Sigma, \quad (70)$$

$$\Sigma' = \mathcal{B}\Sigma + \Sigma\mathcal{B}^*, \quad \mathcal{B} = \mathcal{A}_{PP} - C\mathcal{A}_{QP}, \quad (71)$$

$$\frac{d}{dr} \log \Gamma_P(\mathcal{J}(r)) = \Re \text{tr}(\mathcal{A}_{PP} - C\mathcal{A}_{QP}). \quad (72)$$

In particular, neither $\mathcal{A}_{QQ}$ nor $\mathcal{A}_{PQ}$ occurs directly in the logarithmic increment. The displayed generator contains only the charged column $\mathcal{A}P$; the high–high evolution is absorbed into the dressed Grassmannian state C .

Proof. The definition of C gives $EY^* = 0$. The Schur complement of $S = YY^*$ in $\mathcal{J}\mathcal{J}^*$ is $EE^* = \Sigma$, so

$$|\det \mathcal{J}|^2 = \det S \det \Sigma. \quad (73)$$

This proves (70). From (67),

$$X' = \mathcal{A}_{PP}X + \mathcal{A}_{PQ}Y, \quad Y' = \mathcal{A}_{QP}X + \mathcal{A}_{QQ}Y. \quad (74)$$

Substitution of $X = CY + E$ into $E' = X' - C'Y - CY'$ gives

$$E' = (\mathcal{A}_{PP} - C\mathcal{A}_{QP})E + \mathcal{K}Y \quad (75)$$

for a matrix $\mathcal{K}$. Since $EY^* = 0$, the $\mathcal{K}Y$ terms vanish in $E'E^* + EE'^*$, proving (71). Taking one half of the logarithmic determinant derivative proves (72). $\square$

Proposition 4.4 (No low-column estimate for unrestricted cocycles). *Pointwise cancellation of the high-high generator in (72) does not imply a bound for the Grassmannian increment in terms of $\int \|\mathcal{A}P\|_{\mathfrak{S}_2}^2$ alone. More precisely, for $m = 2$, $N = 1$, and $P = e_1 e_1^*$ there are piecewise constant left cocycles with $\|\mathcal{A}(r)\| \leq 1$ such that*

$$\int_0^{H+L} \|\mathcal{A}(r)P\|_{\mathfrak{S}_2}^2 dr = \frac{\pi^2}{4L}, \quad \log \Gamma_P(\mathcal{J}(H+L)) = H. \quad (76)$$

Consequently there is no uniform estimate of the terminal increment by a constant plus this low-column cost.

Proof. Let

$$\mathbf{K} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \omega_L = \frac{\pi}{2L}, \quad (77)$$

and, with $L \geq \pi/2$, set

$$\mathcal{A}(r) = \begin{cases} \text{diag}(0, 1), & 0 < r < H, \\ \omega_L \mathbf{K}, & H < r < H+L. \end{cases} \quad (78)$$

For $\mathcal{J}(0) = \mathbf{I}$, the first interval gives $\mathcal{J}(H) = D_H := \text{diag}(1, e^H)$. The second gives the exact quarter-turn

$$\mathcal{J}(H+L) = e^{(\pi/2)\mathbf{K}} D_H = \begin{pmatrix} 0 & -e^H \\ 1 & 0 \end{pmatrix}. \quad (79)$$

The first generator has zero P -column, while the second has P -column norm ω_L . This proves the cost identity in (76). Moreover $\Gamma_P(\mathbf{I}) = \Gamma_P(D_H) = 1$, whereas the final determinant has modulus e^H and its Q row has norm one. Hence the entire logarithmic growth H occurs during a rotation whose charged cost tends to zero as $L \rightarrow \infty$. $\square$

Remark 4.5 (What the counterexample does and does not reject). The dressed state C in (72) remembers the uncharged amplification. A later charged rotation can therefore make the product $C\mathcal{A}_{QP}$ large even when $\mathcal{A}_{QP}$ is small. Thus any successful flux correction must retain quantitative control of the dressed state. Proposition 4.4 concerns arbitrary matrix cocycles; it does not yet reject the Schrödinger endpoint, where $V_m = P_L M_V P_L$ is a compression of one scalar multiplication operator and the cost includes its leakage into every omitted angular channel.

Theorem 4.1 is equivalently

$$-\log \det h_P(k^2) = -N \log \frac{k}{\pi} + 2 \log \Gamma_{P;R,L}(k). \quad (80)$$

Thus high-channel volume is not to be bounded separately: it is divided out exactly.

Corollary 4.6 (Sharp sufficient sum rule). *Let $K = [k_0, k_1] \Subset (0, \infty)$. If*

$$\sup_{R,L: P_N \leq P_L} \int_K \log^+ \Gamma_{P_N;R,L}(k) dk \leq C_{K,N}(1 + \mathcal{E}_N(V)) \quad (81)$$

then the boundary N -channel entropy is uniformly bounded on K^2 .

Proof. On K , the term $-N \log(k/\pi)$ is bounded. Equation (80) and $x^+ \geq x$ yield the desired estimate. $\square$

Remark 4.7 (Why this is sharper than a distinguished minor). Replacing the denominator in (62) by one chosen minor produces artificial poles whenever that minor vanishes. The Plücker norm sums all minors and cannot vanish on the real axis when J is invertible. The appropriate analytic object is therefore an exterior vector or its scalar outer factor, not the quotient by a preselected high-channel determinant.

Corollary 4.8 (Schur–Jost sufficient target). *Write, at points where the Q block is invertible,*

$$J = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad S_P = A - BD^{-1}C. \quad (82)$$

Then

$$\det h_P(k^2) = \left(\frac{k}{\pi}\right)^N \frac{\det(I + C^* D^{-*} D^{-1} C)}{|\det S_P|^2} \geq \left(\frac{k}{\pi}\right)^N |\det S_P|^{-2}. \quad (83)$$

Consequently it is sufficient to prove the meromorphic relative estimate

$$\sup_{R,L} \int_K \log^+ |\det S_{P;R,L}(k)| dk \leq C_{K,N}(1 + \mathcal{E}_N(V)), \quad (84)$$

provided the zeros of D are included in a complete pole/Blaschke audit.

Proof. Block inversion gives

$$J^{-1}P = \begin{pmatrix} I \\ -D^{-1}C \end{pmatrix} S_P^{-1}. \quad (85)$$

Taking the Gram determinant of these N columns and using (56) proves (83). The last assertion follows on compact K from the bounded free factor. $\square$

4.1 Stage 1 scalar test: a single normalized radiation column

Take

$$P_0 = |Y_0\rangle\langle Y_0|, \quad Y_0 = |\mathbb{S}^{d-1}|^{-1/2}, \quad e_0 = (1, 0, \dots, 0)^T. \quad (86)$$

The Plücker quotient has a particularly simple form which removes every irrelevant high-channel volume:

Proposition 4.9 (Exact scalar Plücker identities). *For every finite model and every $k > 0$,*

$$h_0(k^2) = \frac{k}{\pi} \|J(k)^{-1}e_0\|^2, \quad (87)$$

$$\Gamma_0(k) = \|J(k)^{-1}e_0\|^{-1}. \quad (88)$$

If $J = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and D is invertible, then

$$\Gamma_0(k) = \frac{|a - bD^{-1}c|}{\sqrt{1 + \|D^{-1}c\|^2}} \leq |a - bD^{-1}c|. \quad (89)$$

Moreover, the charged column is exactly

$$\int_1^\infty \|M_{V(r)}Y_0\|_2^2 dr = \frac{1}{|\mathbb{S}^{d-1}|} \int_1^\infty \|V(r, \cdot)\|_2^2 dr. \quad (90)$$

Proof. Equation (87) is the $(0, 0)$ entry of (56). Comparing it with (59) for $N = 1$ proves (88). Block inversion gives

$$J^{-1}e_0 = (a - bD^{-1}c)^{-1} \begin{pmatrix} 1 \\ -D^{-1}c \end{pmatrix}, \quad (91)$$

which proves (89). Finally, $M_V Y_0 = |\mathbb{S}^{d-1}|^{-1/2} V$ proves (90). $\square$

Thus the compactly supported small-tail problem for $N = 1$ is precisely

$$\sup_{R,L} \int_K \log^+ \frac{1}{\|J_{R,L}(k)^{-1}e_0\|} dk \leq C_K \left(1 + \frac{1}{|\mathbb{S}^{d-1}|} \|V\|_{X_{d,\text{ext}}}^2 \right). \quad (92)$$

Theorem D.3 below disproves this target under its stated uniform finite-cutoff quantifiers. The denominator in (89) is beneficial and must not be discarded in any cutoff-compatible replacement. The following scalar extraction lemma records a rigorous sufficient route, although its hypotheses cannot hold uniformly for the counterexample family.

Proposition 4.10 (Scalar Hardy extraction). *Let $K \Subset (0, \infty)$ and choose a bounded piecewise $C^{1,\alpha}$ Jordan domain Ω_K in the first quadrant whose boundary contains K as a straight segment away from the corners. Let $k_* \in \Omega_K$ and let ω_* be harmonic measure at k_* . Its density with respect to arclength is then bounded below on K . Let $j_0(k)$ be the free constant-channel Jost factor and set*

$$f_{R,L}(k) = j_0(k) \langle e_0, J_{R,L}(k)^{-1}e_0 \rangle. \quad (93)$$

Suppose uniformly in R, L that

$$|f_{R,L}(k_*)| \geq c_* > 0, \quad \int_{\partial\Omega_K} |f_{R,L}(k)|^2 d\omega_*(k) \leq C_*^2. \quad (94)$$

Then

$$\sup_{R,L} \int_K \log^+ \Gamma_{0;R,L}(k) dk \leq C(K, j_0, c_*, C_*). \quad (95)$$

Proof. For $k \in \Omega_K$ one has $\Im(k^2) = 2\Re k \Im k > 0$, so k^2 lies in the resolvent set of the self-adjoint finite model and $J(k)$ is invertible. Thus (93) is analytic. The $H^2(\Omega_K)$ norm below is understood through nontangential boundary values in $L^2(\partial\Omega_K, \omega_*)$. Since

$$\|J^{-1}e_0\| \geq |\langle e_0, J^{-1}e_0 \rangle| = |j_0|^{-1} |f_{R,L}|, \quad (96)$$

the free factor, which is bounded above and below on K , gives

$$\log^+ \Gamma_0 \leq C_K + \log^- |f_{R,L}|. \quad (97)$$

Subharmonicity of $\log |f|$ and (94) imply

$$\begin{aligned} \int_{\partial\Omega_K} \log^- |f| d\omega_* &\leq \int_{\partial\Omega_K} \log^+ |f| d\omega_* - \log |f(k_*)| \\ &\leq \frac{1}{2} C_*^2 + \log(c_*^{-1}). \end{aligned} \quad (98)$$

The lower bound for the harmonic-measure density on K converts (98) to Lebesgue measure there. Combining with (97) proves the claim. $\square$

Proposition 4.10 gives a sufficient scalar-coordinate criterion. For a compactly supported tail with

$$\varepsilon^2 = |\mathbb{S}^{d-1}|^{-1} \int_R^\infty \|V(r, \cdot)\|_2^2 dr, \quad (99)$$

the first attempted Stage 1 obligation was to prove, uniformly in the angular cutoff,

$$\|f_{R,L}\|_{H^2(\Omega_K)} \leq C_K, \quad |f_{R,L}(k_*) - 1| \leq C_K \varepsilon, \quad C_K \varepsilon < 1. \quad (100)$$

The second inequality would supply $c_* = 1 - C_K \varepsilon$ and the first would supply C_* . Thus RCH_1 would imply the scalar Plücker estimate at the physical coupling without spectral averaging. The following exact slab test shows, however, that this unrenormalized condition is false.

Proposition 4.11 (L^2 -small tails can destroy the interior normalization). *Fix $k_* \in \mathbb{C}$ with $\Re k_*, \Im k_* > 0$. In dimension $d = 3$, already in the radial constant channel there are bounded compactly supported potentials q_ℓ such that*

$$\|q_\ell\|_2 \rightarrow 0, \quad |f_\ell(k_*)| \rightarrow 0. \quad (101)$$

Consequently neither the interior lower bound in (94) nor the closeness assertion in (100) can hold under the L^2 tail hypothesis alone.

Proof. Let

$$A_\ell = \ell^2, \quad a_\ell = \ell^{-3/4}, \quad q_\ell(r) = a_\ell \mathbf{1}_{[A_\ell, A_\ell + \ell]}(r). \quad (102)$$

Then

$$\|q_\ell\|_2^2 = a_\ell^2 \ell = \ell^{-1/2} \rightarrow 0, \quad M_\ell := \int q_\ell = a_\ell \ell = \ell^{1/4} \rightarrow \infty. \quad (103)$$

For $d = 3$ the flattened constant channel has no centrifugal term. Put

$$p_\ell(k) = \sqrt{k^2 - a_\ell}, \quad p_\ell(k) \rightarrow k \quad (a_\ell \downarrow 0), \quad (104)$$

with the analytic branch near k_* . Matching the solution and its derivative at the two edges of the constant slab gives, after division by the free Jost factor,

$$\begin{aligned} \frac{J_\ell(k)}{J_0(k)} &= \frac{(k + p_\ell)^2}{4kp_\ell} e^{i(k-p_\ell)\ell} - \frac{(k - p_\ell)^2}{4kp_\ell} e^{i(k+p_\ell)\ell} + O_{k_*}(e^{-2\Im k_* A_\ell}) \\ &= \exp\left(\frac{iM_\ell}{2k} + O_{k_*}(a_\ell^2 \ell + a_\ell)\right) (1 + o(1)). \end{aligned} \quad (105)$$

Here the second transmitted term and the reflected wave are exponentially small at k_* , while $k - p_\ell = a_\ell/(2k) + O_{k_*}(a_\ell^2)$. Since $f_\ell = J_0/J_\ell$ in the scalar model,

$$f_\ell(k_*) = \exp\left(\frac{M_\ell}{2ik_*} + o(1)\right), \quad \Re \frac{1}{2ik_*} < 0. \quad (106)$$

Equation (101) follows. $\square$

The failed factor is a WKB phase: it has unit modulus on the physical axis but exponential size inside the quadrant. With angular mixing it becomes an operator evolution rather than a scalar exponential, exactly the phenomenon identified in Denisov's WKB analysis [6]. For a tail supported in $[R_0, T]$ and a finite angular cutoff, define

$$\begin{aligned} 2ik \partial_r U_{V,L}(k; r) &= \left(\frac{D_L}{r^2} + P_L M_{V(r)} P_L\right) U_{V,L}(k; r), & U_{V,L}(k; R_0) &= \text{I}, \\ 2ik \partial_r U_{0,L}(k; r) &= \frac{D_L}{r^2} U_{0,L}(k; r), & U_{0,L}(k; R_0) &= \text{I}, \end{aligned} \quad (107)$$

$$\mathcal{U}_{T,L}(k) = U_{0,L}(k; T)^{-1} U_{V,L}(k; T). \quad (108)$$

For real $k \neq 0$ both evolutions are unitary, so $\mathcal{U}_{T,L}(k)$ is unitary. For k in the first quadrant they are analytic and invertible. In the radial $d = 3$ model,

$$\mathcal{U}_T(k) = \exp\left(\frac{1}{2ik} \int_{R_0}^T q(r) dr\right), \quad (109)$$

which is precisely the factor exposed by (106).

Let $\mathcal{J}_{T,L} = J_{0,L}^{-1} J_{T,L}$, put

$$w_{T,L}(k) = \mathcal{J}_{T,L}(k)^{-1} e_0, \quad g_{T,L}(k) = \mathcal{U}_{T,L}(k)^{-1} w_{T,L}(k). \quad (110)$$

The following vector criterion is invariant under the harmless real-axis rotation and avoids zeros of a single coordinate.

Proposition 4.12 (Gauge-invariant vector Hardy extraction). *Suppose, uniformly in T, L , that*

$$\|g_{T,L}(k_*)\| \geq c_* > 0, \quad \int_{\partial\Omega_K} \|g_{T,L}(k)\|^2 d\omega_*(k) \leq C_*^2. \quad (111)$$

Then

$$\sup_{T,L} \int_K \log^+ \tilde{\Gamma}_{0;T,L}(k) dk \leq C(K, c_*, C_*). \quad (112)$$

Proof. On K , unitarity of (108) gives

$$\|g_{T,L}(k)\| = \|w_{T,L}(k)\|, \quad \log^+ \tilde{\Gamma}_{0;T,L}(k) = \log^- \|g_{T,L}(k)\|. \quad (113)$$

The function $u(k) = \log \|g_{T,L}(k)\|$ is subharmonic. Hence

$$\begin{aligned} \int_{\partial\Omega_K} u^-(k) d\omega_*(k) &\leq \int_{\partial\Omega_K} u^+(k) d\omega_*(k) - u(k_*) \\ &\leq \frac{1}{2} C_*^2 + \log(c_*^{-1}). \end{aligned} \quad (114)$$

The positive lower bound for the harmonic-measure density on K and (113) prove the result. $\square$

The replacement for RCH_1 is therefore the gauged vector condition (111), denoted GVH_1 . Its linear term is genuinely controlled. In the radial $d = 3$ model, writing $g_T = 1 + g_T^{(1)} + O(V^2)$ gives the exact cancellation

$$g_T^{(1)}(k) = -\frac{1}{2ik} \int_{R_0}^T e^{2ik(r-R_0)} q(r) dr. \quad (115)$$

Proposition 4.13 (First-order gauged Hardy bound). *For a rectangular Ω_K with its lower side $K \Subset (0, \infty)$, rounded at the corners if desired,*

$$\sup_T \|g_T^{(1)}\|_{H^2(\Omega_K)}^2 + \int_K \sup_{T > R_0} |g_T^{(1)}(k)|^2 dk \leq C_K \int_{R_0}^\infty |q(r)|^2 dr. \quad (116)$$

The same estimate holds for any fixed Bessel order; the $O_K(r^{-1})$ amplitude correction is absorbed by Cauchy–Schwarz.

Proof. On the lower side of the rectangle, (115) is the maximal Fourier integral controlled by Carleson–Hunt. On the upper side, Plancherel applies after the damping factor $e^{-2r\Im k}$. On a vertical side $k = k_0 + iy$,

$$\begin{aligned} &\int_0^\eta \left| \int_{R_0}^\infty e^{2ik_0 r} e^{-2yr} q(r) dr \right|^2 dy \\ &= \int_{R_0}^\infty \int_{R_0}^\infty e^{2ik_0(r-s)} q(r) \overline{q(s)} \frac{1 - e^{-2\eta(r+s)}}{2(r+s)} dr ds \leq C \|q\|_2^2, \end{aligned} \quad (117)$$

by Hilbert’s integral inequality. Since $|k|$ is bounded below on Ω_K , the factor $(2ik)^{-1}$ is harmless. The fixed-order Bessel remainder is bounded by $\|q\|_2 \|r^{-1}\|_{L^2(R_0, \infty)}$. $\square$

The classical Christ–Kiselev multilinear bounds [4] require an input exponent strictly below 2 and therefore do not settle the endpoint needed here. Equations (107)–(116) remove the zero-phase evolution before that endpoint is addressed. The first nontrivial perturbative diagnostic is the quadratic coefficient of the gauged vector norm, whose real-axis value is still the quotient combination below because the gauge is unitary there. Theorem 8.4 will show that this diagnostic is not uniformly bounded coefficientwise, even though the exact physical evolution remains bounded.

There is an exact perturbative diagnostic for where GVH_1 first becomes nontrivial. Let $J_0(k)$ be the diagonal free Bessel Jost matrix and use the *left* normalization

$$\begin{aligned} \mathcal{J}_\alpha(k) &= J_0(k)^{-1} J_\alpha(k) = \mathbf{I} + \alpha A(k) + \alpha^2 B(k) + O(\alpha^3), \\ w_\alpha &= \mathcal{J}_\alpha^{-1} e_0, \quad \tilde{\Gamma}_\alpha := \frac{\Gamma_\alpha}{|j_0|} = \|w_\alpha\|^{-1}. \end{aligned} \quad (118)$$

If $\beta = \langle e_0, A e_0 \rangle$, direct inversion gives

$$\|w_\alpha\|^2 = 1 - 2\alpha \Re \beta + \alpha^2 \left(\|A e_0\|^2 + 2\Re \langle e_0, (A^2 - B) e_0 \rangle \right) + O(\alpha^3), \quad (119)$$

$$\log \tilde{\Gamma}_\alpha = \alpha \Re \beta + \alpha^2 \left((\Re \beta)^2 - \frac{1}{2} \|A e_0\|^2 - \Re \langle e_0, (A^2 - B) e_0 \rangle \right) + O(\alpha^3). \quad (120)$$

At first order only the low-low matrix element occurs. At second order the full outgoing column appears through

$$\|A e_0\|^2 = |\beta|^2 + \|Q_0 A e_0\|^2. \quad (121)$$

The summands in the quadratic coefficient must not be estimated separately. Long weak radial potentials can make $\|A e_0\|^2$ and $\Re \langle e_0, (A^2 - B) e_0 \rangle$ individually large while they cancel in $\log \tilde{\Gamma}$. The correct bilinear object is the full quotient-normalized combination

$$\mathcal{Q}_2(k) := (\Re \beta)^2 - \frac{1}{2} \|A e_0\|^2 - \Re \langle e_0, (A^2 - B) e_0 \rangle. \quad (122)$$

Indeed, if

$$\mathcal{J}_\alpha = \begin{pmatrix} 1 + \alpha a_1 + \alpha^2 a_2 & \alpha b_1 + O(\alpha^2) \\ \alpha c_1 + O(\alpha^2) & \mathbf{I} + \alpha D_1 + O(\alpha^2) \end{pmatrix}, \quad (123)$$

then the exact Schur quotient (89) gives

$$\mathcal{Q}_2 = \Re \left(a_2 - b_1 c_1 - \frac{1}{2} a_1^2 \right) - \frac{1}{2} \|c_1\|^2. \quad (124)$$

The high-high coefficient D_1 cancels completely at this order. This made a gauged $H^2(\Omega_K)$ bound for the *combined positive part* of (122) the natural next candidate. The radial two-slab test in Theorem 8.4 disproves that coefficientwise candidate. Thus (122) remains a useful cancellation diagnostic, but it cannot be promoted to the endpoint lemma.

4.2 Ordinary compact probes

The boundary density may be converted into an L^2 probe density without changing the open estimate. Remove a compact core so that $V = 0$ on a collar $(1, c)$. Let $g \in C_c^\infty((1, c))$ be real and nonzero, and in flattened coordinates take

$$\Phi_N e_j = g(r) Y_j(\omega). \quad (125)$$

In physical coordinates the same probe is

$$\phi_j(x) = |x|^{-(d-1)/2} g(|x|) Y_j(x/|x|). \quad (126)$$

Because the collar dynamics is diagonal in angular momentum, its regular transform is an $N \times N$ diagonal matrix $D_g(k)$ independent of R and L . Hence

$$W_N^{R,L}(k^2) = D_g(k)^* h_P(k^2) D_g(k), \quad \det W_N^{R,L} = |\det D_g(k)|^2 \det h_P(k^2). \quad (127)$$

The zeros of $\det D_g$ are isolated and contribute a fixed, locally integrable logarithm. Consequently (81) implies the finite-channel entropy theorem for genuine L^2 probes.

5 A second route: normalized angular entropy

Let P_L contain every spherical harmonic of degree at most L . The addition theorem gives

$$\sum_{j=1}^{m_L} |Y_j(\omega)|^2 = \frac{m_L}{|\mathbb{S}^{d-1}|}. \quad (128)$$

Proposition 5.1 (Exact normalized trace identity). *For every $v \in L^2(\mathbb{S}^{d-1})$,*

$$\frac{1}{m_L} \|M_v P_L\|_{\mathfrak{S}_2}^2 = \frac{1}{|\mathbb{S}^{d-1}|} \|v\|_2^2, \quad (129)$$

$$\frac{1}{m_L} \|P_L M_v P_L\|_{\mathfrak{S}_2}^2 \leq \frac{1}{|\mathbb{S}^{d-1}|} \|v\|_2^2, \quad (130)$$

$$\frac{1}{m_L} \|Q_L M_v P_L\|_{\mathfrak{S}_2}^2 \leq \frac{1}{|\mathbb{S}^{d-1}|} \|v\|_2^2. \quad (131)$$

Proof. Expand the Hilbert–Schmidt norm and apply (128):

$$\|M_v P_L\|_{\mathfrak{S}_2}^2 = \sum_{j=1}^{m_L} \int_{\mathbb{S}^{d-1}} |v(\omega)|^2 |Y_j(\omega)|^2 d\omega = \frac{m_L}{|\mathbb{S}^{d-1}|} \|v\|_2^2. \quad (132)$$

The last two claims follow by orthogonal compression. $\square$

This identity suggests auditing the finite-width matrix L^2 sum rule of Molchanov–Vainberg [7] with constants explicit in m_L . With $h_{0,L}$ denoting the diagonal Bessel reference density, the desired dimension-linear statement is

$$\frac{1}{m_L} \int_I \log^- \det \left(h_{0,L}(E)^{-1/2} h_{R,L}(E) h_{0,L}(E)^{-1/2} \right) dE \leq C_I \left(1 + \|V\|_{X_d}^2 \right) \quad (133)$$

The perturbative cost in (133) is compatible with Proposition 5.1; the Bessel normalization prevents high angular momentum from being charged as a potential.

5.1 Why normalized entropy is not yet the theorem

Even a proof of (133) would not, by itself, justify the passage $L \rightarrow \infty$. A positive fraction of good spectral directions can drift to higher angular modes while every fixed direction degenerates. The exact matrix example

$$W_m = e^{-\sqrt{m}} P_{\lfloor \sqrt{m} \rfloor} + (P_m - P_{\lfloor \sqrt{m} \rfloor}) \quad (134)$$

has

$$\frac{1}{m} \log \det W_m \rightarrow -1, \quad (135)$$

yet the density in every fixed coordinate tends to zero. This is the *dimension-escape obstruction*.

The normalized route therefore requires the following separate statement.

Proof obligation 5.2 (Angular tightness/no escape). Derive from the scalar multiplication structure, the centrifugal term, and a buffered version of (133) that for each fixed N there are a finite-dimensional subspace $\mathcal{E}_N \subset \mathcal{K}$ and L_0 such that

$$\sup_{R, L \geq L_0} \int_I \log^- \det W_{\mathcal{E}_N}^{R, L}(E) dE < \infty \quad (136)$$

The subspace $\mathcal{E}_N$ must be fixed before the limit $L \rightarrow \infty$.

If both (133) and (136) are proved, the established entropy lower-semicontinuity theorem finishes the limit. The normalized route is therefore viable, but the no-escape lemma is a genuine theorem rather than a compactness slogan.

6 The full Feshbach self-energy and its exact energy average

For the $P_N \oplus Q_N$ decomposition, write

$$H = \begin{pmatrix} H_{PP} & B^* \\ B & H_{QQ} \end{pmatrix}, \quad B = Q_N M_V P_N. \quad (137)$$

For $z \in \mathbb{C}_+$,

$$P_N(H - z)^{-1}P_N = (H_{PP} - z - \Sigma_N(z))^{-1}, \quad \Sigma_N(z) = B^*(H_{QQ} - z)^{-1}B. \quad (138)$$

The familiar sign is exact:

$$\Im \Sigma_N(E + i\eta) = \eta B^*((H_{QQ} - E)^2 + \eta^2)^{-1}B \succeq 0. \quad (139)$$

Proposition 6.1 (Poisson average of the full self-energy). *Let A be self-adjoint and B bounded. Then, in the strong operator topology,*

$$\int_{\mathbb{R}} \Im [B^*(A - E - i\eta)^{-1}B] dE = \pi B^*B, \quad \eta > 0. \quad (140)$$

Proof. The spectral theorem reduces the claim to

$$\int_{\mathbb{R}} \frac{\eta}{(\lambda - E)^2 + \eta^2} dE = \pi. \quad (141)$$

Sandwiching by B^* and B proves (140). $\square$

6.1 The exact $2m$ -dimensional phase-space quotient

The differential quotient identity has a direct phase-space lift. In a finite angular model write

$$W(r) = \frac{D}{r^2} + V_m(r), \quad -F'' + WF = k^2 F. \quad (142)$$

Order the phase coordinates as (u_P, u'_P, u_Q, u'_Q) . The first-order system is

$$\Psi' = \mathbb{A}\Psi, \quad \mathbb{A} = \begin{pmatrix} \mathbb{A}_{PP} & \mathbb{A}_{PQ} \\ \mathbb{A}_{QP} & \mathbb{A}_{QQ} \end{pmatrix}, \quad (143)$$

where

$$\mathbb{A}_{PP} = \begin{pmatrix} 0 & \mathbb{I}_P \\ W_{PP} - k^2 \mathbb{I}_P & 0 \end{pmatrix}, \quad \mathbb{A}_{PQ} = \begin{pmatrix} 0 & 0 \\ W_{PQ} & 0 \end{pmatrix}, \quad (144)$$

$$\mathbb{A}_{QP} = \begin{pmatrix} 0 & 0 \\ W_{QP} & 0 \end{pmatrix}, \quad \mathbb{A}_{QQ} = \begin{pmatrix} 0 & \mathbb{I}_Q \\ W_{QQ} - k^2 \mathbb{I}_Q & 0 \end{pmatrix}. \quad (145)$$

Let $\mathbb{P}$ be the rank- $2N$ low-channel phase projection and let $\mathbb{T}' = \mathbb{A}\mathbb{T}$ be an invertible fundamental transfer.

Corollary 6.2 (Phase-space high–high cancellation). *Define the phase-space regression state $\mathbb{C}$ from $\mathbb{T}$ exactly as C was defined from $\mathcal{J}$ in (68)–(69). Then*

$$\frac{d}{dr} \log \Gamma_{\mathbb{P}}(\mathbb{T}(r, k)) = \Re \operatorname{tr}(\mathbb{A}_{PP} - \mathbb{C}\mathbb{A}_{QP}). \quad (146)$$

The full block W_{QQ} cancels. After conjugating the known free and centrifugal low evolution, the variable part of the displayed charged column contains only V_{PP} and V_{QP} , hence only $M_V P_N$.

Proof. Apply Proposition 4.3 to the decomposition $\mathbb{P} \oplus (\mathbb{I} - \mathbb{P})$ and use (144)–(145). The only potential entries in $\mathbb{A}_{PP}$ and $\mathbb{A}_{QP}$ are V_{PP} and V_{QP} . $\square$

For $m = 2$ and $N = 1$, Corollary 6.2 is the promised 4×4 low/high test. It passes algebraically even when the high-channel potential is arbitrary: V_{22} changes the dressed state $\mathbb{C}$ but does not appear as a separately charged term.

The outgoing solution is not the full transfer matrix. It is the $2m \times m$ flux-normalized Lagrangian frame

$$\mathbb{L}(r, k) = \begin{pmatrix} F(r, k) \\ F'(r, k) \end{pmatrix}, \quad \mathbb{L}(r, k)^* \begin{pmatrix} 0 & \mathbb{I} \\ -\mathbb{I} & 0 \end{pmatrix} \mathbb{L}(r, k) = 2ik\mathbb{I} \quad (147)$$

on the positive axis, while the physical Jost matrix is the configuration block $J(k) = F(1, k)$. The following identity removes this apparent mismatch.

Proposition 6.3 (Physical Riccati descent). *For every real $k > 0$, the outgoing configuration matrix $F(r, k)$ is invertible. Assume $N < m$; the case $N = m$ is obtained by omitting the Q rows and setting $C_F = 0$. Define its logarithmic derivative and its low-row regression by*

$$\mathcal{R} = F'F^{-1}, \quad C_F = (PF)(QF)^*((QF)(QF)^*)^{-1}. \quad (148)$$

Then

$$\mathcal{R}' = W - k^2\mathbb{I} - \mathcal{R}^2, \quad (149)$$

$$\Im \mathcal{R} = kF^{-*}F^{-1} > 0, \quad (150)$$

$$\det(P\Im \mathcal{R}P|_{P\mathbb{C}^m}) = k^N \Gamma_P(F)^{-2}, \quad (151)$$

$$\frac{d}{dr} \log \Gamma_P(F(r, k)) = \Re \operatorname{tr}(\mathcal{R}_{PP} - C_F \mathcal{R}_{QP}). \quad (152)$$

Moreover, the charged Riccati column obeys

$$(\mathcal{R}P)' = \left(\frac{D}{r^2} - k^2\mathbb{I} \right) P + V_m P - \mathcal{R}^2 P. \quad (153)$$

Thus $V_m P$ is the only variable inhomogeneous source. The high–high block V_{QQ} changes the dressed state $\mathcal{R}$ but is not charged separately.

Proof. If $F(r, k)c = 0$, the flux identity (147), tested on c , gives $0 = 2ik\|c\|^2$; hence F is invertible. Differentiating $F'F^{-1}$ and using $F'' = (W - k^2\mathbb{I})F$ proves (149). Multiplication of the flux identity on the left by F^{-*} and on the right by F^{-1} gives (150). Proposition 4.2 then gives (151). Finally apply Proposition 4.3 to the exact left cocycle $F' = \mathcal{R}F$; this proves (152). Right multiplication of (149) by P gives (153). $\square$

This proves the algebraic descent, including the 4×4 one-low/one-high test, without Airy asymptotics. It also identifies the genuinely analytic remainder.

Proof obligation 6.4 (Riccati/Feshbach energy square function). After conjugating the known free centrifugal flow and the real-axis WKB phase, prove for each fixed N that the energy-positive shell increment generated by (152) satisfies

$$\int_K [\log \Gamma_P(F(R_1, k)) - \log \Gamma_P(F(R_2, k))]_+ dk \leq C_{K,N} \int_{R_1}^{R_2} \|M_{V(r)} P_N\|_{\mathfrak{S}_2}^2 dr + \mathcal{F}(R_1) - \mathcal{F}(R_2), \quad (154)$$

with $\mathcal{F} \geq 0$, uniformly in the angular cutoff and the support radius. The proof must identify elimination of buffered elliptic channels with the Feshbach complement (138) and keep the energy-dependent dressed state inside the estimate. The unrestricted one-low/one-high matrix estimate fails by Theorem B.3, and the stronger uniform scalar statement in (154) fails by Theorem D.3. The cutoff-edge scalar lift shows that a degree- $2L$ multiplier can create the exact rotator and barrier inside the degree- L model at bounded full constant-column cost. Any viable replacement must therefore impose a cutoff-compatible bandwidth buffer or a justified diagonal order of limits. The full-space converter in Problem 6.5 remains a stronger diagnostic, not the unresolved test for the displayed uniform estimate.

The orientation in (154) is essential. The outgoing solution is normalized outside the support, so attaching the shell $[R_1, R_2]$ changes the boundary value by propagating from R_2 inward to R_1 . In the scalar case $\Gamma(F(R_2, k)) = 1$ outside the support, while the inner Jost modulus can be larger than one. The opposite orientation would therefore miss precisely the entropy loss that must be controlled.

The finite laboratory. For $m = 2$ and $N = 1$, write

$$\mathcal{R} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad V_m = \begin{pmatrix} v_{11} & \overline{v_{21}} \\ v_{21} & v_{22} \end{pmatrix}. \quad (155)$$

After the known diagonal centrifugal term is separated, the charged column of (149) is exactly

$$a' = v_{11} + \text{known free term} - a^2 - bc, \quad (156)$$

$$c' = v_{21} - ca - dc, \quad (157)$$

$$\frac{d}{dr} \log \Gamma_P(F) = \Re(a - C_F c). \quad (158)$$

The truncated variable source cost is $|v_{11}|^2 + |v_{21}|^2$. The coefficient v_{22} acts only through the dressed variables d, c, C_F , but Proposition 4.4 shows that this algebraic fact alone gives no estimate: a high block may preload the state and a later small rotation may expose it. Allowing an arbitrary bounded v_{22} is thus a deliberately stronger matrix stress test, not the actual scalar endpoint. For the physical compression $V_m = P_L M_V P_L$, the charged norm is $\|M_V P_N\|_{\mathfrak{S}_2}$ in the full angular space. It is tempting to conclude that creating v_{22} must therefore be expensive. Propositions C.1 and C.4 show that this is false: high-harmonic concentration produces an exact $v_{22} = Q$ block with vanishing full constant-column cost, and the high state can be made almost invariant during the barrier. What remains genuinely non-finite is the converter, because the scalar lift of its 2×2 compression also couples the selected high harmonic to the rest of the high sector.

Proof obligation 6.5 (Stronger full-space scalar converter dichotomy). Let $e_0 = (4\pi)^{-1/2}$ and let h_ℓ be the real highest-weight harmonic (268). Decide between the following two alternatives for the full angular Schrödinger evolution, including A_3/r^2 :

- (i) prove a quantitative lower bound excluding any uniformly bounded real scalar converter which maps both directional modes $e_0^\pm$ to $h_\ell^\pm$ with error $o(1)$ while $\int \|M_{v(r)} e_0\|_2^2 dr = o(1)$; or

- (ii) construct such converters with modal error $o(e^{-\kappa T_\ell})$ and concatenate them with the thin-belt barrier of Proposition C.4.

A proof of (i) would supply structural input for a cutoff-stable relative sum rule. A construction in (ii) would give a full-space version of the rotator/barrier obstruction. Theorem D.3 already disproves PSR_1 by the weaker finite-cutoff mechanism, so this dichotomy is retained as a stronger question rather than as the logical gate to PSR_1 . Propositions F.3 and F.4 prove the first part of (i) in the centrifugal-perturbative regime: any successful converter must generate order-one angular current and cannot be close to the decomposable evolution obtained by deleting A_3/r^2 . What remains is to control or construct that nonperturbative angular transport using the charged scalar column. Propositions F.1 and F.2 also exclude a near-invariant two-state scalar realization of the matrix rotator. The sole constructive escape is therefore a genuinely multimode excursion followed by uniform refocusing into h_ℓ . Proposition F.5 shows that high angular frequency can itself be generated at vanishing cost; Proposition F.6 then shows that a clean cheap mask cannot be placed at the same radial scale as degree- ℓ centrifugal transport. The remaining core is therefore the coupled weak-slab version of the achromatic bidirectional refocusing problem F.25, not the ideal phase-mask product by itself. The resonant-drive diagnostic (673) further shows that the obvious weak resonant pulse is cheap only pointwise in energy: at first order, uniform transfer over a fixed band already has order-one charged cost. Proposition F.7 makes that budget exact and nonlinear for a pure-detuning ensemble. Proposition F.8 shows why it cannot simply be imported into the physically scaled system. Propositions F.9, F.10, and F.11 exclude respectively the direct positive-ground rotator, the standard monotone highest-weight chirp, and the adjacent-rung rotating ground ladder. Corollary F.14 forces any gap-cancelled cheap design to use scalar angular bandwidth $\gtrsim \ell/\mathcal{E}_\ell^2$, while Proposition F.16 excludes direct broadband injection into that rough band at first order. What remains is a genuinely nonlinear bootstrap: lower-mode concentration, rough complementary-sector rotation, and reciprocal refocusing in the exact scaled radial evolution.

The formula sees the full leakage and is immune to cutoff aliasing. It does not yet prove entropy, because the low-channel boundary solution depends on the same energy being integrated. If

$$\mathcal{U}_{P,R,L}(z) : \mathbb{C}^N \longrightarrow L^2((1, \infty); P_N \mathcal{K}) \quad (159)$$

denotes the low solution generated by the boundary data, the missing nonlinear embedding has the schematic form

$$\sup_{0 < \eta < 1} \int_I \text{tr}[\mathcal{U}_P(E + i\eta)^* \Im \Sigma_N(E + i\eta) \mathcal{U}_P(E + i\eta)] \, dE \leq C_{I,N} \mathcal{E}_N(V) \quad (160)$$

The fixed-vector Poisson identity (140) is the linear seed of (160); the energy dependence of $\mathcal{U}_P$ is the entire difficulty.

7 A rigorous free spacetime estimate in the conjectured norm

The weight in Simon's conjecture is exactly the weight selected by free finite-channel propagation.

Proposition 7.1 (Restricted free smoothness). *Let $I = [E_0, E_1] \Subset (0, \infty)$, let P_N be as above, and let $H_0 = -\partial_r^2 + A_d/r^2$ on the exterior half-line. There is $C_{I,N} < \infty$ such that*

$$\int_{\mathbb{R}} \left\| M_V e^{-itH_0} \mathbf{1}_I(H_0) P_N f \right\|_{L^2((1, \infty) \times \mathbb{S}^{d-1})}^2 \, dt \leq C_{I,N} \mathcal{E}_N(V) \|f\|^2. \quad (161)$$

Proof. Let $k \in K = [\sqrt{E_0}, \sqrt{E_1}]$. In each of the finitely many channels, let $\varphi_j(r, k)$ be the Bessel generalized eigenfunction in flattened coordinates, normalized for the spectral transform. Since $r \geq 1$ and $K \Subset (0, \infty)$,

$$\sup_{1 \leq j \leq N} \sup_{r \geq 1, k \in K} |\varphi_j(r, k)| \leq C_{I,N}. \quad (162)$$

Writing the spectral coefficients as $\widehat{f}_j(k)$ gives

$$u(t, r, \omega) = \sum_{j=1}^N Y_j(\omega) \int_K e^{-itk^2} \varphi_j(r, k) \widehat{f}_j(k) dk. \quad (163)$$

Plancherel in t , with $E = k^2$ and $dE = 2k dk$, followed by Cauchy–Schwarz in the finite channel index, yields

$$\int_{\mathbb{R}} |u(t, r, \omega)|^2 dt \leq C_{I,N} \sum_{j=1}^N |Y_j(\omega)|^2 \int_K |\widehat{f}_j(k)|^2 dk. \quad (164)$$

Multiply by $|V(r, \omega)|^2$, integrate in r and ω , and use (49). The spectral transform is unitary, so the resulting right-hand side is bounded by $C_{I,N} \mathcal{E}_N(V) \|f\|^2$. $\square$

Remark 7.2. Estimate (161) is an L_t^2 statement, whereas Cook’s method asks for L_t^1 . Applying Cauchy–Schwarz on an infinite time interval loses a divergent factor. The correct next step is therefore not to quote Cook’s theorem, but to prove a maximal or dressed square-function estimate. This is precisely the role of (160) and the nonlinear Carleson module below.

8 Auxiliary approaches and their limitations

8.1 Module A: a Plücker outer-factor sum rule

For the finite model, define the analytic exterior vector

$$p_Q(z) = \bigwedge^{m_L - N} (Q_N J_{R,L}(z)). \quad (165)$$

On the real axis its norm is the denominator of (62). A distinguished coordinate of p_Q is unsafe because it may have zeros unrelated to the spectral problem. The natural scalar replacement is an outer function $\mathcal{O}_Q$ whose boundary modulus is $\|p_Q\|$:

$$\log |\mathcal{O}_Q(z)| = \frac{1}{\pi} \int_{\mathbb{R}} \frac{\Im z}{|t - z|^2} \log \|p_Q(t)\| dt, \quad (166)$$

with the standard integrable regularization at infinity.

After removing bound-state zeros of $\det J$ by a Blaschke product $\mathcal{B}$, set formally

$$\mathcal{D}_P(z) = \frac{\det J(z)}{\mathcal{B}(z) \mathcal{O}_Q(z)}. \quad (167)$$

Since $|\mathcal{B}| = 1$ on $\mathbb{R}$,

$$\log |\mathcal{D}_P(k)| = \log \Gamma_{P;R,L}(k). \quad (168)$$

Proof obligation 8.1 (Grassmannian sum rule). Prove that $\mathcal{D}_P$ is of uniformly bounded characteristic and that its trace identity has quadratic coefficient

$$\int_1^\infty \operatorname{tr} \left(P_N M_{V(r)}^2 P_N \right) dr = \mathcal{E}_N(V), \quad (169)$$

with all zero and pole terms of favorable sign or bounded by the same cost. Deduce (81).

The cancellation in (169) is the central advantage of the Plücker quotient. A full determinant charges $\operatorname{tr}(P_L M_V P_L)^2$, while the relative P -volume should charge only the P corner of M_V^2 :

$$\operatorname{tr}(P_N M_V^2 P_N) = \|P_N M_V P_N\|_{\mathfrak{S}_2}^2 + \|Q_N M_V P_N\|_{\mathfrak{S}_2}^2. \quad (170)$$

This is exactly the safe cost.

8.2 Module B: dressed Carleson and wave-packet control

Proposition 7.1 gives the linear estimate. The desired nonlinear upgrade is a maximal bound for a flux-normalized interaction variable $Z_{R,L}(r, k)$:

$$\int_K \sup_{\rho \geq 1} \left\| \int_1^\rho e^{2iks} M_{V(s)} Z_{R,L}(s, k) P_N ds \right\|_{\mathfrak{S}_2}^2 dk \leq C_{K,N} \mathcal{E}_N(V) \quad (171)$$

The factor Z is constrained by flux conservation and the common second-order radial kinetic term; it is not an arbitrary bounded matrix. An estimate of the form (171) would control the logarithmic growth of the exterior volume in (62) shell by shell and imply (160).

Recent wave-packet work of Denisov [15] supplies a model for exploiting phase separation, but its conclusions are generic in a parameter and require additional decay/frequency hypotheses. The proposed use here is narrower: establish an energy square function for the exact full leakage, not uniform pointwise free asymptotics.

The first proposed harmonic-analysis milestone was already linear. Let

$$v_{\nu j}(r) = \langle Y_\nu, M_{V(r)} Y_j \rangle_{L^2(\mathbb{S}^{d-1})}, \quad \sum_\nu |v_{\nu j}(r)|^2 = \|V(r, \cdot) Y_j\|_2^2, \quad (172)$$

and let $\phi_\nu^\pm(k, r)$ be flux-normalized incoming and outgoing Bessel waves. The tempting raw estimate was

$$\int_K \sup_{T > R} \sum_\nu \left| \int_R^T \phi_\nu^-(k, r) v_{\nu j}(r) \phi_j^+(k, r) dr \right|^2 dk \leq C_{K,j} \int_R^\infty \|V(r, \cdot) Y_j\|_2^2 dr. \quad (173)$$

The full sum in (173) avoids angular aliasing, but that fact does not remove a radial forward mode.

Proposition 8.2 (The raw on-shell estimate is false). *For every nontrivial compact $K \Subset (0, \infty)$, estimate (173) fails already for a radial scalar potential in the constant angular channel.*

Proof. Uniformly for $k \in K$, the flux-normalized constant-channel waves have the large- r form

$$\phi_0^\pm(k, r) = (2k)^{-1/2} e^{\pm i(kr - \vartheta_0)} (1 + O_K(r^{-1})), \quad \phi_0^-(k, r) \phi_0^+(k, r) = \frac{1}{2k} + O_K(r^{-1}). \quad (174)$$

Let $A_\ell = \ell^2$ and choose the bounded compactly supported radial potential

$$q_\ell(r) = \varepsilon \ell^{-1/2} \mathbf{1}_{[A_\ell, A_\ell + \ell]}(r), \quad V_\ell(r, \omega) = q_\ell(r). \quad (175)$$

Its charged cost is $\int |q_\ell|^2 dr = \varepsilon^2$. Taking $j = \nu = 0$ and $T = A_\ell + \ell$, however, gives

$$\int_{A_\ell}^T \phi_0^- q_\ell \phi_0^+ dr = \frac{\varepsilon \sqrt{\ell}}{2k} + O_K\left(\frac{\varepsilon \sqrt{\ell}}{A_\ell}\right), \quad (176)$$

$$\int_K \sup_{T > A_\ell} \left| \int_{A_\ell}^T \phi_0^- q_\ell \phi_0^+ dr \right|^2 dk \geq c_K \varepsilon^2 \ell \quad (177)$$

for all sufficiently large ℓ . This contradicts (173) with a constant independent of the support location and length. $\square$

Remark 8.3. Proposition 8.2 does not contradict the free spacetime estimate in Proposition 7.1. Plancherel controls an L_t^2 evolution before restricting both waves to the same energy shell; the failure above is precisely the nonoscillatory diagonal created by that restriction.

The scalar Jost relation identifies the exact subtraction. Let f_0^+ be the free outgoing constant-channel solution, $f_0^- = \overline{f_0^+}$ on the positive axis, j_0 its Jost factor, and s_0 the regular solution. Then

$$s_0(r, k) = \frac{\overline{j_0(k)} f_0^+(r, k) - j_0(k) f_0^-(r, k)}{2ik}. \quad (178)$$

For the truncated first Born coefficient

$$\beta_T(k) = \frac{1}{j_0(k)} \int_R^T s_0(r, k) v_{00}(r) f_0^+(r, k) dr, \quad (179)$$

substitution of (178) gives

$$\beta_T(k) = \frac{\overline{j_0(k)}}{2ik j_0(k)} \int_R^T v_{00}(r) (f_0^+(r, k))^2 dr - \frac{1}{2ik} \int_R^T v_{00}(r) |f_0^+(r, k)|^2 dr, \quad (180)$$

$$\Re \beta_T(k) = \Re \left[\frac{\overline{j_0(k)}}{2ik j_0(k)} \int_R^T v_{00}(r) (f_0^+(r, k))^2 dr \right]. \quad (181)$$

The omitted term in (181) is purely imaginary. Since $(f_0^+)^2$ has phase e^{2ikr} and a fixed-order Bessel amplitude, the Carleson–Hunt maximal theorem [3], together with the $O_K(r^{-1})$ asymptotic remainder, proves the corrected scalar estimate

$$\int_K \sup_{T>R} |\Re \beta_T(k)|^2 dk \leq C_K \int_R^\infty |v_{00}(r)|^2 dr \leq \frac{C_K}{|\mathbb{S}^{d-1}|} \int_R^\infty \|V(r, \cdot)\|_2^2 dr. \quad (182)$$

Thus the scalar linear part is controlled after the exact forward phase has been removed; no Airy turning analysis is needed at this order because only the constant channel occurs.

The first proposed nonlinear obligation was quadratic and retained the Plücker quotient cancellation. Let $A_{R,L,T}$ and $B_{R,L,T}$ be the first two coefficients of the left-normalized Jost–Volterra expansion cut off at T , put $\beta_{R,L,T} = \langle e_0, A_{R,L,T} e_0 \rangle$, and define

$$\mathcal{Q}_{R,L,T}(k) := (\Re \beta_{R,L,T})^2 - \frac{1}{2} \|A_{R,L,T} e_0\|^2 - \Re \langle e_0, (A_{R,L,T}^2 - B_{R,L,T}) e_0 \rangle. \quad (183)$$

The proposed quotient Bessel–Carleson estimate QBC_1 asserted

$$\sup_L \int_K \sup_{T>R} [\mathcal{Q}_{R,L,T}(k)]_+ dk \leq \frac{C_K}{|\mathbb{S}^{d-1}|} \int_R^\infty \|V(r, \cdot)\|_2^2 dr. \quad (184)$$

It was deliberately imposed on the combined positive part. Nevertheless, it is false before any Airy turning channel is present.

In the scalar radial $d = 3$ model use

$$f_0(r, k) = e^{ik(r-1)}, \quad s_0(r, k) = \frac{\sin(k(r-1))}{k}. \quad (185)$$

The exact combined quadratic coefficient reduces to

$$\mathcal{Q}_{R,1,T}(k) = -\Re \int_{R < r < t < T} s_0(r, k)^2 e^{2ik(t-1)} q(r) q(t) dt dr. \quad (186)$$

Theorem 8.4 (Radial two-slab obstruction to QBC_1). *Fix $K = [k_0, k_1] \Subset (0, \infty)$, $A > R > 1$, $b > 0$, and $0 < \delta < \pi/k_1$. Put*

$$a_\Lambda = \Lambda^{-3/4}, \quad B_\Lambda = A + \Lambda + 1, \quad (187)$$

$$q_\Lambda(r) = a_\Lambda \mathbf{1}_{[A, A+\Lambda]}(r) + b \mathbf{1}_{[B_\Lambda, B_\Lambda+\delta]}(r). \quad (188)$$

Then

$$\int_R^\infty |q_\Lambda(r)|^2 dr = \Lambda^{-1/2} + b^2\delta, \quad (189)$$

but, with $T_\Lambda = B_\Lambda + \delta$,

$$\int_K [\mathcal{Q}_{R,1,T_\Lambda}(k)]_+ dk \longrightarrow \infty. \quad (190)$$

Thus (184) is false even for one radial channel and arbitrarily small fixed L^2 tails.

Proof. Decompose (186) into the two self-interactions and the inner–outer cross term. The cross term is

$$\mathcal{C}_\Lambda(k) = -a_\Lambda b I_\Lambda(k) \Re J_\Lambda(k), \quad (191)$$

where, uniformly on K ,

$$I_\Lambda(k) = \int_A^{A+\Lambda} \frac{\sin^2(k(r-1))}{k^2} dr = \frac{\Lambda}{2k^2} + O_K(1), \quad (192)$$

$$J_\Lambda(k) = \int_{B_\Lambda}^{B_\Lambda+\delta} e^{2ik(t-1)} dt = e^{i(2k(B_\Lambda-1)+k\delta)} \frac{\sin(k\delta)}{k}. \quad (193)$$

Hence

$$\mathcal{C}_\Lambda(k) = -\Lambda^{1/4} \frac{b \sin(k\delta)}{2k^3} \cos(\lambda_\Lambda k) + O_K(\Lambda^{-3/4}), \quad \lambda_\Lambda = 2(B_\Lambda - 1) + \delta. \quad (194)$$

The continuous weight is positive on K . Since the periodic function $[-\cos \theta]_+$ has mean $1/\pi$,

$$\int_K [\mathcal{C}_\Lambda(k)]_+ dk = \frac{b\Lambda^{1/4}}{2\pi} \int_K \frac{\sin(k\delta)}{k^3} dk + o(\Lambda^{1/4}). \quad (195)$$

The inner self-interaction has $L^1(K)$ norm $O_K(\Lambda^{-1/2})$ and the fixed outer self-interaction has norm $O_{K,b,\delta}(1)$, by integrating the outer oscillation first in (186). Since $[x+y]_+ \geq [x]_+ - |y|$, the cross growth proves (190). Multiplying every q_Λ by a fixed $\varepsilon > 0$ multiplies both sides of the proposed quadratic comparison by ε^2 , so small tail norm does not repair the estimate. $\square$

The obstruction is nonuniform perturbation theory, not physical growth. The long inner slab accumulates a forward phase of size $\Lambda^{1/4}$; expanding that phase in the coupling before composing it with the outer reflector creates the divergent coefficient.

Theorem 8.5 (Exact radial scalar entropy at physical coupling). *Let $q \in L^2(1, \infty)$ be real and compactly supported, let $J_q(k)$ be its half-line Jost function, and let $K = [k_0, k_1] \Subset (0, \infty)$. Then*

$$\int_K \log^+ |J_q(k)| dk \leq |K| \log 2 + \frac{\pi}{8k_0^2} \int_1^\infty |q(r)|^2 dr. \quad (196)$$

The constant is independent of the support radius.

Proof. Extend q by zero to the left of $r = 1$. To the left of its support, the outgoing solution has the form $a_q(k)e^{ik(r-1)} + b_q(k)e^{-ik(r-1)}$. Flux conservation gives $|a_q|^2 - |b_q|^2 = 1$, so

$$|J_q(k)| \leq |a_q(k)| + |b_q(k)| \leq 2|a_q(k)|. \quad (197)$$

The Deift–Killip transmission inequality [2] states

$$\int_{\mathbb{R}} k^2 \log |a_q(k)| dk \leq \frac{\pi}{8} \int_{\mathbb{R}} |q(r)|^2 dr. \quad (198)$$

Since $\log |a_q| \geq 0$ and $k \geq k_0$ on K , the claimed estimate follows. $\square$

In particular, the exact entropy of the two-slab family in Theorem 8.4 is uniformly bounded although its quadratic Taylor coefficient diverges. Therefore no coefficientwise coupling series whose second-order bound implies (184) can be the endpoint proof.

The direct replacement is the physical-coupling exterior frame. With the relative Jost matrix $\mathcal{J}_{T,L}$ and WKB evolution $\mathcal{U}_{T,L}$ from (108), define

$$\tilde{\mathbf{g}}_{P;T,L}(k) := \bigwedge^N \left(\mathcal{U}_{T,L}(k)^{-1} \mathcal{J}_{T,L}(k)^{-1} P_N \right). \quad (199)$$

On the real axis the gauge is unitary, so

$$\log \tilde{\Gamma}_{P;T,L}(k) := -\log \left\| \bigwedge^N (\mathcal{J}_{T,L}(k)^{-1} P_N) \right\| = -\log \|\tilde{\mathbf{g}}_{P;T,L}(k)\|. \quad (200)$$

The free diagonal normalization changes this from $\log \Gamma_P$ only by a fixed bounded term on K .

Proof obligation 8.6 (Physical-coupling exterior Hardy theorem). For every fixed N and compact $K \Subset (0, \infty)$, prove under the small-tail hypothesis, uniformly in R, L, T , an interior lower bound

$$\|\tilde{\mathbf{g}}_{P;T,L}(k_*)\| \geq c_{K,N} > 0 \quad (201)$$

and the boundary estimate

$$\int_{\partial\Omega_K} \|\tilde{\mathbf{g}}_{P;T,L}(k)\|^2 d\omega_*(k) \leq C_{K,N} (1 + \mathcal{E}_N(V)). \quad (202)$$

Subharmonicity and Proposition 4.2 would then give PSR_N . The first route to (202) is Proposition 6.3 followed by Problem 6.4 and the dressed estimate (160); every estimate must be made directly at $\alpha = 1$.

8.3 Module C: buffered impact-parameter channels

Let $K = [k_0, k_1]$ and divide the exterior into dyadic shells

$$\mathcal{S}_j = [R_j, 2R_j], \quad R_j = 2^j R_0. \quad (203)$$

Choose a buffered spectral projector

$$P_j = \mathbf{1}_{\{A_d \leq \Lambda^2 R_j^2\}}(A_d), \quad \Lambda^2 > 4(k_1^2 + \|V\|_\infty + 1). \quad (204)$$

For $r \in \mathcal{S}_j$ and $Q_j = \mathbf{I} - P_j$,

$$Q_j \left(\frac{A_d}{r^2} + M_V - k^2 \right) Q_j \succeq Q_j, \quad k \in K, \quad (205)$$

as a quadratic form. Thus modes above the buffer are elliptic on the shell and can be integrated out without a real-axis resonance. Projectors are held constant on each shell, so no derivative $P'_j(r)$ is created.

Proposition 6.3 identifies the exact shell increment, but it does not estimate its positive energy part. Let $\Delta_j(k)$ denote the inward shell increment

$$\Delta_j(k) := \log \Gamma_P(F(R_j, k)) - \log \Gamma_P(F(2R_j, k)). \quad (206)$$

The dimension-free target is

$$\int_K \Delta_j(k)^+ dk \leq C_{K,N} \int_{R_j}^{2R_j} \|V(r, \cdot)\|_2^2 dr + \mathcal{F}_j - \mathcal{F}_{j+1} \quad (207)$$

where $\mathcal{F}_j \geq 0$ is a flux correction that telescopes. This is the dyadic form of (154). Here $\mathcal{F}_j$ must be built from the dressed Riccati/Feshbach state; the exact cancellation says that V_{QQ} may influence this state but may not appear as an additional cost. Summing (207) gives (81).

Remark 8.7. The buffer must be applied to the dynamics, but the cost must remain the full scalar norm. Replacing Q_N by $P_j - P_N$ on the right of (207) would reintroduce the exact aliasing failure (28)–(30).

8.4 Module D: normalized entropy plus angular tightness

This module begins with the exact identity (129). Its two steps are deliberately separated:

$$\text{dimension-linear matrix sum rule} \implies \text{NMSR}, \quad (208)$$

$$\text{NMSR} + \text{buffered angular tightness} \implies \text{TIGHT}_N \implies \text{FCEN}_N. \quad (209)$$

The likely tightness mechanism is an impact-parameter buffer combined with a determinant chain rule. If $W_L > 0$ and $P_a < P_b$, then

$$\log \det W_{P_b} = \log \det W_{P_a} + \log \det(W_{P_b}/W_{P_a}), \quad (210)$$

where the quotient is the positive Schur complement. One must show that entropy cannot be carried indefinitely by successive high-degree Schur complements without spending the scalar energy in (129). Equation (134) shows why a general matrix argument cannot do this; the proof must use the centrifugal ordering and scalar multiplication structure.

8.5 Module E: the solved raywise operator and one unbounded short-range theorem

Remove the angular kinetic term but retain the potential exactly:

$$H_{\text{ray}} = -\partial_r^2 + M_{V(r,\omega)} = \int_{\mathbb{S}^{d-1}}^{\oplus} (-\partial_r^2 + V(r,\omega)) d\omega, \quad (211)$$

with a Dirichlet boundary condition at $r = R$. This comparison problem is already solved.

Proposition 8.8 (Infinite multiplicity for the raywise comparison). *If V is real, bounded, and satisfies (3), then*

$$m_{\text{ac}}(E; H_{\text{ray}}) = \infty \quad \text{for a.e. } E > 0. \quad (212)$$

Proof. Fubini gives $V(\cdot, \omega) \in L^2(R, \infty)$ for a.e. ω . The scalar half-line L^2 theorem gives a positive absolutely continuous density for a.e. $E > 0$ in almost every such fiber. A second application of Fubini makes the exceptional set independent of ω : for a.e. $E > 0$, the density is positive for a.e. ω . The multiplicity fiber of (211) then contains the infinite-dimensional space $L^2(\mathbb{S}^{d-1}, w_\omega(E) d\omega)$. $\square$

The original exterior operator is the form sum

$$H_{\text{ext}} = H_{\text{ray}} + \frac{A_d}{r^2}. \quad (213)$$

The coefficient r^{-2} is integrable at infinity, but A_d is unbounded and M_V has no angular regularity. The entire conjecture would therefore follow from one comparison theorem.

Proof obligation 8.9 (Unbounded angular short-range comparison). Prove existence and completeness of

$$W_{\pm}(H_{\text{ray}} + r^{-2}A_d, H_{\text{ray}}). \quad (214)$$

For bounded angular cutoffs the perturbation is short range. The new content is a cutoff-uniform estimate through the domains of A_d and H_{ray} ; an evanescent-channel bound of the type (205) is the natural compactness mechanism.

8.6 Module F: stochastic angular paths and quadratic variation

Let $(\Omega_t)_{t \geq 0}$ be spherical Brownian motion started in its invariant uniform distribution, and set

$$\tau(r) = \int_R^r s^{-2} ds = R^{-1} - r^{-1}. \quad (215)$$

Stationarity and Tonelli give the exact path-cost identity

$$\mathbb{E} \int_R^\infty |V(r, \Omega_{\tau(r)})|^2 dr = \frac{1}{|\mathbb{S}^{d-1}|} \int_R^\infty \|V(r, \cdot)\|_2^2 dr. \quad (216)$$

Thus Simon's norm has exactly the form of an expected quadratic-variation budget along a stationary angular path. This is the continuum analogue of path averaging in tree sum rules and operates directly at coupling one.

Proof obligation 8.10 (Stochastic exterior-power entropy). Construct the antisymmetric N -path evolution associated with (215) and prove an inequality of the form

$$\int_I \log \det W_N(E) dE \geq -C_{I,N} - C_I \mathbb{E} \sum_{j=1}^N \int_R^\infty |V(r, \Omega_{\tau(r)}^{(j)})|^2 dr. \quad (217)$$

The obstruction is conceptual and explicit: Feynman–Kac controls imaginary-time propagation, whereas (217) concerns real-energy boundary values. A successful martingale or Girsanov argument must retain the exterior-power phase rather than replacing the unitary dynamics by a heat estimate.

9 Endpoint device: fixed coupling without spectral averaging

The fixed-coupling counterexample (24) prevents the inference “almost every α implies $\alpha = 1$.” A different use of the coupling parameter may nevertheless be viable. For a Hilbert–Schmidt Birman–Schwinger operator $K(z)$,

$$D_2(\zeta, z) = \det {}_2(\mathbf{I} + \zeta K(z)) \quad (218)$$

is entire in ζ and satisfies

$$\log |D_2(\zeta, z)| \leq \frac{|\zeta|^2}{2} \|K(z)\|_{\mathfrak{S}_2}^2. \quad (219)$$

Subharmonic mean-value at the point $\zeta = 1$ can therefore turn a complex-neighbourhood estimate into a fixed-coupling estimate; no specialization of an almost-every-real-coupling statement is involved.

For the present problem, a formal relative perturbation is

$$K_{N,L}(r) = \begin{pmatrix} V_{PP}(r) & V_{PQ}(r) \\ V_{QP}(r) & 0 \end{pmatrix}, \quad \|K_{N,L}(r)\|_{\mathfrak{S}_2}^2 = \|V_{PP}\|_{\mathfrak{S}_2}^2 + 2\|V_{QP}\|_{\mathfrak{S}_2}^2. \quad (220)$$

The high-high block is placed in the reference operator. This bookkeeping is attractive, but a general Hilbert–Schmidt perturbation determinant can fail to have nontangential boundary values [12]. Moreover Theorem B.3 proves that an arbitrary high block recreates the channel-rotator obstruction. The determinant route is therefore conditional on a Schrödinger-specific boundary theorem that uses the scalar angular compression and its full leakage.

There is nevertheless an exact reason to test this route. With

$$\mathcal{V} = \begin{pmatrix} V_{PP} & V_{PQ} \\ V_{QP} & V_{QQ} \end{pmatrix}, \quad \mathcal{V}_0 = \begin{pmatrix} 0 & 0 \\ 0 & V_{QQ} \end{pmatrix}, \quad (221)$$

the quadratic trace difference is

$$\mathrm{tr}(\mathcal{V}^2) - \mathrm{tr}(\mathcal{V}_0^2) = \|V_{PP}\|_{\mathfrak{S}_2}^2 + 2\|V_{QP}\|_{\mathfrak{S}_2}^2 = \|K_{N,L}\|_{\mathfrak{S}_2}^2. \quad (222)$$

Thus the high-energy coefficient of a formal relative matrix sum rule deletes V_{QQ} exactly and leaves a quantity equivalent, with dimension-independent constants, to the charged column $\|\mathcal{V}P\|_{\mathfrak{S}_2}^2 = \|V_{PP}\|_{\mathfrak{S}_2}^2 + \|V_{QP}\|_{\mathfrak{S}_2}^2$. Existing matrix L^2 sum rules charge the full matrix potential [7]; they do not supply this relative boundary statement with an arbitrary high-channel reference.

Proposition C.1 now shows that even imposing scalar compression does not validate this bookkeeping by itself. A concentrated high harmonic permits an arbitrary V_{QQ} entry with an $o(1)$ addition to the full constant-column cost, and Proposition C.4 makes the high state almost invariant. Hence the relative trace coefficient contains no purely algebraic obstruction to the rotator/barrier. Any positive relative theorem must use the dynamics of the converter, including the centrifugal flow and the scalar dilation into the whole high sector.

Proof obligation 9.1 (Relative matrix Schrödinger sum rule). Assume specifically that $\mathcal{V} = P_L M_v P_L$ is the finite angular compression of one real scalar multiplication potential, and let $\mathcal{V}_0$ retain only its QQ block. Construct the Schrödinger-specific relative Jost or perturbation determinant for $(\mathcal{V}, \mathcal{V}_0)$ and prove that its boundary entropy controls $\log^+ \Gamma_P$ while its trace side is bounded by the full scalar column $\|M_v P_N\|_{\mathfrak{S}_2}^2$; the finite part is (222). The constants must be uniform in support radius and angular model dimension. The proof must show that scalar compression and the full angular dynamics prevent the converter in Problem 6.5; compressed trace algebra and the barrier cost alone are insufficient. It must also retain the full Grassmannian quotient.

Proof obligation 9.2 (Fixed-coupling subharmonic transfer). Construct a scalar analytic quantity $D_P(\zeta, z)$ such that

- (i) $\log \Gamma_{P;R,L}(k) \leq \log^+ |D_P(1, k + i0)| + C$;
- (ii) D_P has a complex-coupling growth bound controlled by $\mathcal{E}_N(V)$;
- (iii) its boundary values and zero terms are uniform in R, L .

Then apply the subharmonic mean-value inequality in a disk centred at $\zeta = 1$ to obtain the physical endpoint directly.

10 Consequences for the selected-channel approach

The shortest fully rigorous completion route is now

$$\|V\|_{X_d}^2 < \infty \xrightarrow{\text{Prop. 3.1}} \text{small physical tail}, \quad (223)$$

$$\text{small tail} + \text{PSR}_N \xrightarrow{\text{Thm. 4.1}} \sup_{R,L} \int_I \log^- \det W_N^{R,L} < \infty, \quad (224)$$

$$\text{strong resolvent convergence} + \text{entropy l.s.c.} \implies \int_I \log^- \det W_N < \infty, \quad (225)$$

$$\det W_N(E) > 0 \text{ a.e.} \implies m_{\text{ac}}(E; H) \geq N \text{ a.e.}, \quad (226)$$

$$N \in \mathbb{N} \text{ and rational } I \implies m_{\text{ac}}(E; H) = \infty \text{ a.e. on } (0, \infty). \quad (227)$$

The alternative normalized route replaces PSR_N by $\text{NMSR} + \text{TIGHT}_N$. The raywise route bypasses entropy entirely if Problem 8.9 is solved, while Problem 8.10 would supply the entropy estimate directly from the exact path cost (216).

Theorem G.4, proved in the final appendix, now shows that even the buffered fixed-potential selected $N = 1$ replacement for the first route is false. Thus (223)–(227) remains a correct conditional chain, but its selected-probe input cannot hold universally. The normalized, moving-probe, full-operator, raywise, and stochastic routes are not decided by that counterexample.

10.1 Remaining open directions

The following order maximizes falsifiability and minimizes dependence on unproved machinery.

- Stage 1. Free collar and exact $N = 1$ column complete.** Proposition 3.3 proves the multiplicity-preserving bridge with $q_d = \lceil d/2 \rceil$, while Proposition 4.9 identifies the exact scalar column and cost $|\mathbb{S}^{d-1}|^{-1} \|V\|_{X_d}^2$.
- Stage 2. Zero-phase audit and WKB gauge complete.** Propositions 8.2 and 4.11 rule out the raw same-shell estimate and the unrenormalized interior normalization. Equations (107)–(110) give the operator correction, Proposition 4.12 gives the invariant entropy extraction, and Proposition 4.13 proves its first-order H^2 bound.
- Stage 3. Coefficientwise quadratic route decided.** Theorem 8.4 disproves QBC_1 even in the scalar radial model, while Theorem 8.5 proves that the corresponding exact physical entropy remains uniformly bounded. Coupling-Taylor control is therefore removed from the endpoint route.
- Stage 4. Dual geometry, physical descent, and the generic no-go theorem complete.** Proposition 4.2 removes the distinguished-minor denominator, and Propositions 4.3 and 6.3 identify the exact high–high cancellation in the physical outgoing frame. Proposition 4.4 proves that this cancellation alone cannot yield a low-column estimate.
- Stage 5. Finite matrix rotator/barrier decided.** Lemma B.1, the rapid-phase good-set lemma, and Theorem B.3 prove that the unrestricted two-channel matrix low-column endpoint is false.
- Stage 6. Scalar PSR_1 decided negatively.** Proposition D.1, Lemma D.2, and Theorem D.3 give an exact scalar rotator/barrier in the finite cutoff with bounded full-column cost and divergent quotient entropy. The proposed uniform PSR_N route is therefore closed already at $N = 1$.
- Stage 7. Fixed-potential buffering decided negatively.** Theorem G.4 produces one bounded mean-zero X_3 potential whose selected entropy diverges along every cofinal positive smoothing, radial truncation, and double-buffered cutoff. Buffering removes the algebraic cutoff edge but does not prevent a genuine full-space reciprocal cascade.
- Stage 8. Full-space scalar converter decided positively.** Lemma G.1, Propositions G.6 and G.8, and Proposition G.2 give concentration, localized reflection, reciprocal synchronization, radial realization, and refocusing at summable charged cost.
- Stage 9. Do not replace the boundary theorem by one imaginary-energy row.** Proposition E.26 gives both vanishing-cost single slabs and a fixed finite-cost slab train for which the negative-energy forward row diverges. The rational second-trace boundary localization is essential; Poisson subharmonicity carries an uncontrolled singular mass at infinity.
- Stage 10. The transport-free scalar endpoint is complete but cannot be continued uniformly.** Proposition E.27 proves the selected entropy after deleting A_3/r^2 . The Riesz–Talbot construction shows that centrifugal angular transport is not a perturbative remainder: it enables the counterexample.
- Stage 11. Signed centrifugal interpolation is an identity, not an endpoint bound.** Proposition E.29 and its pure-gauge cancellation remain exact, but the universal estimate SCI_1 is false by Corollary G.5. The absolute-variation replacement was already algebraically false.

- Stage 12. Keep negative-energy action as a budget, not an entropy theorem.** Proposition E.31 proves a dimension-free charged bound for the full positive centrifugal resolvent action. Remark E.32 gives a monotone Stieltjes family with divergent entropy, so the missing step must retain real-axis local propagation.
- Stage 13. The angular heat endpoint is also decided negatively.** Proposition E.33 and its exact X_3 heat budget remain valid, but the universal product-Fisher estimate HFI_1 would imply the false selected entropy theorem.
- Stage 14. Audit the alternative routes under the new falsification test.** The normalized entropy plus tightness, raywise comparison, stochastic path, and direct full-operator determinant routes must be formulated without uniform cutoff-edge compression.
- Stage 15. Move to a basis-free endpoint.** Entropy lower semicontinuity, Proposition 3.1, and the free-collar Birman–Kuroda bridge remain valid finishing tools, but no theorem controlling one fixed selected probe can supply their hypothesis. The next viable targets are moving-probe or full-operator entropy, or a raywise/non-entropic mechanism which cannot lose all mass by rotation of the good angular directions.

11 Obstructions to stronger intermediate estimates

Every module must pass the following tests before it can be promoted to a lemma.

Candidate step	Required test	Failure signal
Plücker outer factor	Treat the norm of all minors; audit zeros, poles, and high- k coefficients.	Division by one minor or an uncontrolled destructive zero term.
Hardy interior normalization	Remove the operator WKB evolution and use the full vector norm as in (111).	The long weak slab (102) makes an unrenormalized interior coordinate exponentially small at vanishing L^2 cost.
Coefficientwise Born expansion	Test the combined quotient coefficient on two separated radial slabs before adding angular or Airy structure.	Theorem 8.4 gives divergent positive quadratic mass at bounded, arbitrarily small L^2 cost; this route is rejected.
Physical Riccati descent	Recover $\Gamma_P(F)$ from $P\mathfrak{S}RP$ and audit the differential generator in the 4×4 model.	A direct charge involving V_{QQ} , or failure to match the physical Jost quotient.
Unrestricted cocycle square function	Test a high-channel amplifier followed by a slow quarter-turn.	Proposition 4.4 produces growth H at charged cost $\pi^2/(4L)$; this route is rejected.
Uniform scalar-compression square function	Test a degree- $2L$ multiplier against the degree- L model.	Proposition D.1 and Lemma D.2 give the exact scalar rotator/barrier counterexample.
Self-energy averaging	Keep the energy-dependent dressed solution inside the estimate.	Applying (140) as if the test vector were independent of E .

Moving cutoff	Charge the full scalar potential, use a bandwidth buffer, and justify the order of limits for one fixed potential.	Full-column charging alone does not stop a pure degree- $2L$ multiplier from producing an exact high-only barrier in $P_{\leq L}$.
Normalized entropy	Prove TIGHT_N for a fixed subspace.	Good directions drift with L as in (134).
Raywise comparison	Control the unbounded A_d uniformly in angular cutoff.	Treating $r^{-2}A_d$ as a bounded short-range potential.
Stochastic path sum rule	Preserve real-energy exterior-power phases.	Substituting an imaginary-time Feynman–Kac estimate for the required boundary entropy.
Complex coupling	Use a mean-value estimate centred at 1.	Specializing an almost-every-real- α theorem to $\alpha = 1$.
Relative determinant	Establish Schrödinger boundary values.	Appealing to general $\det_2$ theory, contradicted by known Hilbert–Schmidt examples.
Full-space scalar channel rotator	Test the full converter into a concentrated highest-weight packet; verify the nodal leakage, angular-current, radial-action, nonlinear Fourier, pulse-area, roughness, and high-band recoil bounds.	These tests constrain a cutoff-stable converter, but they cannot rescue uniform PSR_1 from the exact cutoff-edge lift.
Limit $R, L \rightarrow \infty$	Use entropy lower semicontinuity on inner intervals.	Claiming that strong resolvent convergence preserves a.c. spectrum.

12 A conditional implication for Simon’s conjecture

Theorem 12.1 (Completion from the Grassmannian sum rule). *Assume the following for every N and every compact $K \Subset (0, \infty)$:*

- (a) *the finite models (53) satisfy PSR_N ;*
- (b) *their matrix spectral measures converge vaguely to the exterior measure;*
- (c) *$\int \log^- \det$ is lower semicontinuous under that convergence;*
- (d) *the exterior and full operators have unitarily equivalent absolutely continuous parts.*

Then Simon’s conclusion (4) holds.

Proof. Corollary 4.6 and (127) give a uniform finite-model entropy bound. Assumptions (b)–(c) pass it to the exterior spectral density W_N :

$$\int_I \log^- \det W_N(E) \, dE < \infty. \quad (228)$$

Therefore $\det W_N(E) > 0$ for a.e. $E \in I$, so $\text{rank } W_N(E) = N$. In a direct-integral representation, the rank of a density generated by N vectors is bounded by the a.c. fiber multiplicity; hence $m_{\text{ac}}(E) \geq N$ a.e. on I . Assumption (d) transfers the conclusion to the full operator. Intersect the full-measure sets for all N and all compact intervals with rational endpoints. $\square$

13 Conclusion

The audit has moved the project beyond the unsafe truncated-column formulation. The central open object is no longer an unspecified determinant of a large matrix. It is the low-channel Grassmannian amplification

$$\Gamma_P(k) = \frac{|\det J(k)|}{\|\wedge^{m-N}(QJ(k))\|}, \quad (229)$$

whose logarithm is exactly the entropy loss of the low spectral density, up to the free factor. Its expected quadratic trace coefficient is

$$\mathrm{tr}(PM_V^2P) = \|M_V P\|_{\mathfrak{S}_2}^2, \quad (230)$$

which is precisely Simon's scalar angular L^2 cost when P contains the constant harmonic.

The latest audit changes the principal route. The raw same-shell estimate, RCH_1 , and now the combined coefficientwise estimate QBC_1 are all false. The two-slab obstruction is especially decisive: its quadratic Taylor coefficient diverges although the exact scalar physical-coupling entropy is uniformly bounded by Theorem 8.5. Therefore the conjecture has not been disproved; rather, any endpoint proof must resum the forward phase and work directly at physical coupling.

The exact replacement is now more concrete. The dual inverse frame (66) removes zeros and poles of a selected minor, and the physical Riccati identities (151)–(153) show that the high–high potential is absorbed into a dressed state while the only variable source is the charged column $V_m P_N$. Proposition 4.4 now supplies a crucial negative result: for an unrestricted left cocycle, a high block can preload amplification and a later arbitrarily cheap rotation can transfer it into Γ_P . Therefore the differential cancellation is an identity, not by itself an estimate.

Theorem B.3 completes the next negative step: the exact converter and a shorter barrier make the arbitrary two-channel matrix Grassmannian estimate false, with vanishing low-column cost. Proposition C.1 then removes the hoped-for purely algebraic rescue: every real 2×2 block has an exact scalar lift, and highest-weight concentration makes the extra full constant-column cost vanish. Proposition C.4 goes further by scalarizing the high-only barrier with exponentially small high-state leakage.

The cutoff quantifier supplies the decisive scalar result. Proposition D.1 proves that degrees $L+1$ through $2L$ can cancel every unwanted retained component and make $\mathrm{span}\{e_0, h_L\}$ exactly reducing. Lemma D.2 identifies the closed-form barrier $p_{2L} = \sin^{2L} \theta \cos(2L\phi)$: it has zero compressed constant column, an order-one diagonal entry on h_L , and full charged density $\asymp L^{-1/2}$. Theorem D.3 combines these facts with the exact rotator and rapid-phase lemma to disprove scalar PSR_1 with uniformly bounded scalar amplitude and bounded full constant-column cost.

This does not touch Simon's conjecture. It rejects the proposed uniform sufficient strengthening and identifies the precise failure: multiplier bandwidth was allowed to track and overtake the Galerkin cutoff. Any repaired entropy route must fix one scalar potential first and justify a bandwidth-buffered or diagonal limit, or work directly in the full operator. The conditional completion theorem remains correct, but its assumption (a) is false as presently formulated. The full-space converter analysis below is a stronger diagnostic rather than the gate to PSR_1 .

The first part of the converter fork is now decided. Proposition F.3 shows that scalar multiplication without angular transport has conversion error asymptotic to $\sqrt{2}$, and Proposition F.4 forces every successful converter to carry an order-one angular flux. Thus the previously convenient far-field placement, where A_3/r^2 was intended to be a small error, cannot validate the compressed rotator. The remaining alternatives are singular but concrete: prove that the charged scalar column cannot sustain the required current, including any frequency cascade, or build a converter at the centrifugal scale where that current is genuinely nonperturbative.

The compressed lift itself is now decided more sharply. Proposition F.1 shows that its high-state leakage exceeds the intended adiabatic gap by a factor of order $\ell^{1/4}$, while Proposition F.2

proves that every uniformly bounded alternative scalar lift has leakage comparable to its off-diagonal coupling. Thus no perturbation of the two-channel adiabatic argument can close the construction while keeping the chosen plane approximately invariant. A counterexample must instead exploit the entire complementary sector and then refocus it into h_ℓ for both directional modes, uniformly on an energy set of positive measure. Conversely, a positive theorem need only exclude this cascade-and-refocusing mechanism, not arbitrary compressed rotations.

Proposition F.5 prevents one final premature positive conclusion: the low-column cost does not control angular frequency. A slab of height ϵ and length ϵ^{-1} has vanishing cost but imprints a fixed bounded angular phase, even when that phase oscillates at an arbitrarily high degree after centrifugal transport is deleted. Proposition F.6 now supplies the missing radial check: at $R = O(L)$, making that mask cheap forces unbounded degree- L centrifugal action during the slab, while making it clean forces its scalar cost to diverge. Thus a succession of ideal masks and free propagators is only an auxiliary controllability test, not a radial construction. The decisive object is the coupled scalar–centrifugal evolution inside the weak slabs, uniform in k and reciprocal for the two directional modes. This is now the narrowest unresolved converter mechanism. The resonant-drive calculation (671) confirms that this mechanism can be cheap at a single energy while remaining at $R \asymp \ell$, so no pointwise-in-energy converter bound can be sufficient. But its linearized fixed-band amplitude is a Fourier transform, and (673) forces an order-one charged cost. The nonlinear Fourier calculation in Proposition F.7 upgrades this to an exact theorem when the detuning varies but the coupling potential is fixed. The physical radial parameter scales both terms, and Proposition F.8 shows that no such theorem survives for arbitrary matrix controls: the adiabatic rotator is already a uniform-band counterexample. Positivity is now the scalar dividing line. Proposition F.9 proves that the scalar angular ground branch cannot rotate from e_0 to h_ℓ , and even a vanishing endpoint matching potential has $o(1)$ overlap with h_ℓ at $R = O(\ell)$. The next rotating-frame candidate, a monotone chirped highest-weight pulse, is also excluded in its standard uniformly adiabatic regime by Proposition F.10: sweeping a continuum of resonances forces divergent charged cost. Proposition F.11 also excludes the tempting cheap adjacent-rung escape: the terminal rotating-frame ground momentum depends on s and cannot equal one common ℓ on a fixed band. Proposition F.12 rules out finite-depth composites, and Corollary F.14 adds the missing star-frame pulse area. A gap-cancelled converter must satisfy $\mathcal{E}_\ell \gtrsim \ell^{-1/4}$ and, if its cost still vanishes, its scalar bandwidth must be at least $\ell/\mathcal{E}_\ell^2 \gg \ell$; target-scale Lipschitz controls have a positive cost floor. Proposition F.16 then proves that the forced rough scale cannot be populated in one broadband Born transition. Proposition F.13 isolates what is left: an energy-dependent nonlinear bootstrap which first concentrates mass in lower modes, applies the rough scalar component only after that concentration, and reverses the entire excursion for both directions. Proposition F.17 now proves that the first crossing of every fixed spectral cut is charged even after retaining the full nonlinear high-sector dynamics, and Proposition F.19 forces centrifugal density action $\gtrsim R^2\sqrt{\ell}$, which scalar phase masks themselves cannot produce. Proposition F.18 makes the first failure exact: one bounded wrapped phase crosses all cuts below L with fixed-band high-mode mass and vanishing quadratic cost once centrifugal transport is deleted. Multiscale charged-gate additivity is therefore false. Proposition F.20 then restores the radial centrifugal term and excludes all stationary or radially low-variation weak slabs by monotone angular energy. Corollary F.21 sharpens this to radial variation $\gtrsim \ell/\mathcal{E}_\ell^2$ under gap cancellation, the same scale already forced in angular bandwidth. Proposition F.22 then shows that raw BV cannot itself be charged: dynamically inert oscillations inflate it without changing the propagator. Proposition F.23 isolates the state-correlated radial work and proves the first exact product bridge $\mathfrak{R}_v\mathfrak{D}_s \gtrsim R^2$, or $\mathcal{E}\mathfrak{D}_s \gtrsim R^2/L^2$ at angular bandwidth L . These statements remain a rigorous analysis of the stronger full-space problem, but the unresolved doubly super-rough refocusing fork is ancillary after the finite-cutoff disproof.

The buffered completion theorem is correct as a conditional statement: positive spherical Fejér smoothing and a Galerkin cutoff placed beyond both bandwidth and support give the

correct strong-resolvent limit, and Corollary E.6 removes the cutoff edge. The final Riesz–Talbot construction nevertheless shows that its selected $N = 1$ entropy hypothesis is false for every such diagonal. It combines an exact translated-barrier Hardy isometry, exponentially concentrating zonal Talbot products, fractional-moment reciprocal synchronization, summable-cost radial realization, and uniform lacunary phase placement. The resulting single bounded real mean-zero $V \in X_3$ satisfies (766). Thus SCI_1 , HFI_1 , and every other universal estimate which would bound that fixed selected channel are closed routes, although their interpolation identities remain valid.

This is not a counterexample to Simon’s conjecture. A fixed probe can lose all of its density while absolutely continuous spectral mass survives in directions which move with the angular scale. The theorem therefore changes the target rather than the conjecture’s status: the next endpoint must control a basis-free portion of the spectral density, permit the good subspace to move, or avoid selected-channel entropy entirely. The normalized full-operator, moving-subspace, raywise, and stochastic routes remain possible; the fixed selected-channel route does not.

A Entropy lower semicontinuity used in the limit

Let $\mu_n \rightarrow \mu$ vaguely be positive semidefinite $N \times N$ matrix measures on a neighbourhood of $I = [a, b]$, and write

$$\mu_n = W_n dE + \mu_n^s, \quad \mu = W dE + \mu^s. \quad (231)$$

Define on Hermitian matrices

$$\varphi(A) = \begin{cases} \max\{0, -\log \det A\}, & A > 0, \\ +\infty, & \text{otherwise.} \end{cases} \quad (232)$$

The function φ is convex, lower semicontinuous, and decreasing in the Loewner order. For a smooth approximate identity ρ_ε and an inner interval $I_\delta = [a + \delta, b - \delta]$, Jensen’s inequality and positivity of the convolved singular part give

$$\int_{I_\delta} \varphi((\rho_\varepsilon * \mu_n)(x)) dx \leq \int_I \varphi(W_n(E)) dE. \quad (233)$$

Vague convergence gives pointwise convergence of the mollified matrices. Fatou’s lemma, followed by differentiation of finite matrix measures and then $\delta \downarrow 0$, yields

$$\int_I \log^- \det W(E) dE \leq \liminf_{n \rightarrow \infty} \int_I \log^- \det W_n(E) dE. \quad (234)$$

The inner interval is essential because vague convergence alone does not control mass arriving at an endpoint.

B Exact matrix rotator counterexample and the scalar fork

The following calculation closes the channel-rotator mechanism for arbitrary two-channel matrix potentials. In a two-channel matrix model, put on $[0, \Lambda]$

$$V_{\text{rot}}(r) = \delta_\Lambda R(\theta(r)) \text{diag}(0, 1) R(\theta(r))^*, \quad \theta(r) = \omega_\Lambda r, \quad (235)$$

where

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \quad \omega_\Lambda = \frac{\pi}{2\Lambda}, \quad \delta_\Lambda = \Lambda^{-2/3}. \quad (236)$$

For $P = e_1 e_1^*$ its exact truncated-column cost is

$$\int_0^\Lambda \|V_{\text{rot}}(r)P\|_{\mathfrak{S}_2}^2 dr = \delta_\Lambda^2 \int_0^\Lambda \sin^2\left(\frac{\pi r}{2\Lambda}\right) dr = \frac{1}{2} \Lambda^{-1/3}. \quad (237)$$

Lemma B.1 (Uniform two-channel conversion). *Let $K \Subset (0, \infty)$. For all sufficiently large Λ , the scattering matrix of (235) maps a unit incoming forward wave in channel one to a transmitted forward wave in channel two, up to a scalar phase, with total reflected and wrong-channel amplitude*

$$O_K \left(\delta_\Lambda + \frac{\omega_\Lambda}{\delta_\Lambda} + \frac{\Lambda \omega_\Lambda^2}{\delta_\Lambda} \right) = O_K(\Lambda^{-1/3}), \quad (238)$$

uniformly for $k \in K$. More strongly, in the normalized free modal basis

$$e_j^\pm = \begin{pmatrix} e_j \\ \pm i k e_j \end{pmatrix}, \quad (239)$$

the full 4×4 transfer map differs by the same bound from a direction-preserving channel swap $e_j^\pm \mapsto e^{i\phi_{j,\pm}(k)} e_{3-j}^\pm$.

Proof. Let $K = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and write $u(r) = R(\theta(r))y(r)$. The equation $-u'' + V_{\text{rot}}u = k^2u$ becomes exactly

$$-y'' - 2\omega_\Lambda K y' + \omega_\Lambda^2 y + \delta_\Lambda \text{diag}(0, 1)y = k^2 y. \quad (240)$$

Thus the rotating frame has constant coefficients. Its plane-wave pencil is

$$p^2 I - 2i\omega_\Lambda p K + \omega_\Lambda^2 I + \delta_\Lambda \text{diag}(0, 1) - k^2 I. \quad (241)$$

At $\omega_\Lambda = 0$, the two forward wave numbers are k and $\sqrt{k^2 - \delta_\Lambda}$; their separation is comparable to δ_Λ , uniformly on K . Riesz projection perturbation of the constant 4×4 first-order generator therefore changes the forward modal projections by $O_K(\omega_\Lambda/\delta_\Lambda)$. The first-order eigenvalue shift vanishes because K is off diagonal, so propagation over length Λ creates phase error $O_K(\Lambda \omega_\Lambda^2/\delta_\Lambda)$. Matching the constant coefficient modes to the free leads contributes $O_K(\delta_\Lambda + \omega_\Lambda)$. Finally, the zero-eigenvalue potential direction is $R(\theta)e_1$; it starts at e_1 and ends at e_2 . The same projection calculation applies to all four signs in (239), and the forward/backward spectral clusters remain separated by a constant depending only on K . Combining these bounds gives (238) and its full modal version. $\square$

Lemma B.2 (Rapid-phase good set). *Let $K = [k_0, k_1]$. Suppose $A_\Lambda, B_\Lambda \in C^1(K)$, $M_\Lambda > 0$, and $\psi_\Lambda \in C^2(K)$ obey*

$$cM_\Lambda \leq \max\{|A_\Lambda|, |B_\Lambda|\} \leq CM_\Lambda, \quad |(\log M_\Lambda)'| \leq C \log \Lambda, \quad (242)$$

$$|A'_\Lambda| + |B'_\Lambda| \leq C(\log \Lambda)M_\Lambda, \quad c\Lambda \leq |\psi'_\Lambda| \leq C\Lambda, \quad |\psi''_\Lambda| \leq C\Lambda. \quad (243)$$

Then there are $c_0, c_1 > 0$, independent of Λ , such that for all large Λ ,

$$\left| \left\{ k \in K : |A_\Lambda(k)e^{i\psi_\Lambda(k)} + B_\Lambda(k)| \geq c_0 M_\Lambda(k) \right\} \right| \geq c_1 |K|. \quad (244)$$

Proof. Partition K , apart from two endpoint pieces, into intervals on which ψ_Λ traverses 2π . Their lengths are comparable to Λ^{-1} . On each such interval the three coefficients vary by $O((\log \Lambda)/\Lambda)$ relative to M_Λ . If $\max\{|A|, |B|\} \geq cM$, elementary circle geometry shows that $|Ae^{i\vartheta} + B| \geq c_0 M$ on a fixed positive proportion of $\vartheta \in [0, 2\pi]$, uniformly in the ratio A/B . The change of variables $\vartheta = \psi_\Lambda(k)$ and (243) preserve a fixed proportion. Summing the complete intervals proves (244); the two endpoint pieces have total length $O(\Lambda^{-1})$. $\square$

Theorem B.3 (Failure of the arbitrary matrix low-column bound). *Let $K = [k_0, k_1] \Subset (0, \infty)$ and $P = e_1 e_1^*$. There is a uniformly bounded sequence of compactly supported real symmetric 2×2 matrix potentials $\mathcal{V}_\Lambda$ such that*

$$\int_0^\infty \|\mathcal{V}_\Lambda(r)P\|_{\mathfrak{S}_2}^2 dr = \frac{1}{2} \Lambda^{-1/3} \longrightarrow 0, \quad (245)$$

but for their outgoing Jost matrices,

$$\int_K \log^+ \Gamma_P(F_\Lambda(0, k)) dk \geq c_K \log \Lambda - C_K \longrightarrow \infty. \quad (246)$$

Thus no uniform Grassmannian entropy estimate for arbitrary matrix Schrödinger potentials can be charged only to the low matrix column.

Proof. Use (235) on $[0, \Lambda]$ and follow it by $\text{diag}(0, Q_0)$, where $Q_0 > k_1^2$. Write

$$\kappa_{\min} = \sqrt{Q_0 - k_1^2}, \quad \kappa_{\max} = \sqrt{Q_0 - k_0^2}, \quad (247)$$

choose

$$0 < \beta < \frac{\kappa_{\min}}{3\kappa_{\max}}, \quad T_\Lambda = \frac{\beta}{\kappa_{\min}} \log \Lambda, \quad (248)$$

and put the barrier on $[\Lambda, \Lambda + T_\Lambda]$. The barrier has zero P -column, so (245) follows from (237).

For $\kappa(k) = \sqrt{Q_0 - k^2}$, backward propagation of the outgoing high channel through the barrier gives, in the free directional basis at its left edge, coefficients

$$A_\Lambda(k) = e^{ikT_\Lambda} \left[\cosh(\kappa T_\Lambda) + \frac{i}{2} \left(\frac{\kappa}{k} - \frac{k}{\kappa} \right) \sinh(\kappa T_\Lambda) \right], \quad (249)$$

$$B_\Lambda(k) = -\frac{i}{2} e^{ikT_\Lambda} \left(\frac{\kappa}{k} + \frac{k}{\kappa} \right) \sinh(\kappa T_\Lambda). \quad (250)$$

In particular,

$$c_K e^{\kappa(k)T_\Lambda} \leq \max\{|A_\Lambda(k)|, |B_\Lambda(k)|\} \leq C_K e^{\kappa(k)T_\Lambda}. \quad (251)$$

The full modal statement in Lemma B.1 transforms the high Jost column at $r = \Lambda$ into the low configuration entry

$$G_\Lambda(k) = A_\Lambda(k) e^{i\phi_{+, \Lambda}(k)} + B_\Lambda(k) e^{i\phi_{-, \Lambda}(k)} + O_K(\Lambda^{-1/3} e^{\kappa(k)T_\Lambda}). \quad (252)$$

The constant-coefficient pencil (241) gives

$$\left| \frac{d}{dk} (\phi_{+, \Lambda} - \phi_{-, \Lambda}) \right| \asymp_K \Lambda, \quad \left| \frac{d^2}{dk^2} (\phi_{+, \Lambda} - \phi_{-, \Lambda}) \right| = O_K(\Lambda). \quad (253)$$

Equations (249)–(250) have relative derivatives $O_K(T_\Lambda)$, so Lemma B.2 applies with $M_\Lambda = e^{\kappa T_\Lambda}$. Moreover the choice of β gives

$$\Lambda^{-1/3} e^{\kappa(k)T_\Lambda} \leq \Lambda^{-1/3 + \beta \kappa_{\max}/\kappa_{\min}} = o(1). \quad (254)$$

On the good set from (244), the configuration Jost matrix is therefore an anti-diagonal channel swap with one entry of size at least $c_K e^{\kappa T_\Lambda}$ and the other bounded above and below, plus $o(1)$ in the complementary entries. Hence

$$\Gamma_P(F_\Lambda(0, k)) \geq c_K e^{\kappa(k)T_\Lambda} \geq c_K \Lambda^\beta \quad (255)$$

on a subset of K of measure at least $c_K |K|$. Integrating its positive logarithm proves (246). $\square$

The theorem rejects an arbitrary matrix replacement of the scalar angular problem. It does not disprove Simon's conjecture. A scalar realization necessarily creates a structured high-sector dilation, but the next section shows that angular concentration can make both its full low-column cost and its barrier leakage vanish. For a full-space, cutoff-stable counterexample one would still have to lift the rotating converter in the complete scalar angular dynamics. Section D shows that the uniform finite-cutoff statement PSR_1 has weaker quantifiers and can already be disproved without solving that stronger problem.

C Scalar lifts of the two-channel block

Angular concentration can make a large high-channel matrix element very cheap in scalar L^2 . The exact minimum-cost calculation is as follows.

Proposition C.1 (Minimum-cost scalar lift of a real 2×2 block). *Let (X, μ) have finite measure S , set $e_0 = S^{-1/2}$, and let $h \in L^\infty(X; \mathbb{R})$ satisfy*

$$\int_X h \, d\mu = 0, \quad \|h\|_2 = 1, \quad \int_X h^3 \, d\mu = 0. \quad (256)$$

Put

$$g_h = h^2 - S^{-1}, \quad G_h = \|g_h\|_2^2 = \|h\|_4^4 - S^{-1}. \quad (257)$$

For real a, b, d and $G_h > 0$, define

$$v_{a,b,d} = a + \sqrt{S} b h + \frac{d-a}{G_h} g_h. \quad (258)$$

Then

$$P_{\{e_0, h\}} M_{v_{a,b,d}} P_{\{e_0, h\}} = \begin{pmatrix} a & b \\ b & d \end{pmatrix} \quad (259)$$

and

$$\|v_{a,b,d}\|_2^2 = S(a^2 + b^2) + \frac{(d-a)^2}{G_h}, \quad (260)$$

$$\|M_{v_{a,b,d}} e_0\|_2^2 = a^2 + b^2 + \frac{(d-a)^2}{SG_h}. \quad (261)$$

Moreover, (258) is the unique minimum- L^2 real scalar potential having the compressed block (259).

Proof. The functions $1, h, g_h$ are pairwise orthogonal; the third-moment condition is used for $h \perp g_h$. The three prescribed matrix entries fix respectively the coefficients of these three functions, giving (258). Equations (259)–(261) follow by direct substitution. Any other lift differs by a function orthogonal to $1, h, g_h$, so Pythagoras proves minimality and uniqueness. $\square$

Proposition C.2 (Exact scalar mixer–barrier duality). *Under the hypotheses of Proposition C.1, let*

$$m_{h,b} = \sqrt{S} b h, \quad d_{h,Q} = \frac{Q}{G_h} (h^2 - S^{-1}). \quad (262)$$

The first multiplier has the exact compressed mixer block

$$P_{\{e_0, h\}} M_{m_{h,b}} P_{\{e_0, h\}} = \begin{pmatrix} 0 & b \\ b & 0 \end{pmatrix}, \quad (263)$$

and its leakage from the selected high state is

$$\|(I - P_{\{e_0, h\}}) M_{m_{h,b}} h\|_2^2 = S b^2 G_h. \quad (264)$$

The second is the unique minimum- L^2 high-only barrier with compressed block $\text{diag}(0, Q)$, and its charged constant-column cost is

$$\|M_{d_{h,Q}} e_0\|_2^2 = \frac{Q^2}{SG_h}. \quad (265)$$

Consequently the exact product law is

$$\|(I - P_{\{e_0, h\}})M_{m_h, b}h\|_2^2 \|M_{d_h, Q}e_0\|_2^2 = b^2 Q^2. \quad (266)$$

Thus concentration cannot make a high-only scalar barrier cheap while also making the corresponding pure scalar two-state mixer invariant. This is a two-plane obstruction; it does not exclude a genuinely multimode cascade.

Proof. The functions $1, h, g_h$ are pairwise orthogonal and $M_{m_h, b}e_0 = bh$. Also

$$M_{m_h, b}h = \sqrt{S}bh^2 = be_0 + \sqrt{S}bg_h. \quad (267)$$

Since $g_h \perp \{e_0, h\}$, this proves (263)–(264). Proposition C.1 with $(a, b, d) = (0, 0, Q)$ proves (265); multiplication gives (266). $\square$

Proposition C.3 (Highest-weight concentration on $\mathbb{S}^2$). *Let $S = 4\pi$ and, for $\ell \geq 1$, let*

$$h_\ell(\theta, \phi) = C_\ell(\sin \theta)^\ell \cos(\ell\phi), \quad \|h_\ell\|_2 = 1. \quad (268)$$

Define

$$J_n = \sqrt{\pi} \frac{\Gamma(n+1)}{\Gamma(n+3/2)}. \quad (269)$$

Then $C_\ell^2 = (\pi J_\ell)^{-1}$ and

$$H_{4, \ell} := \int_{\mathbb{S}^2} h_\ell^4 d\omega = \frac{3}{4\pi} \frac{J_{2\ell}}{J_\ell^2} \sim \frac{3}{4\pi\sqrt{2\pi}} \sqrt{\ell}, \quad (270)$$

$$H_{6, \ell} := \int_{\mathbb{S}^2} h_\ell^6 d\omega = \frac{5}{8\pi^2} \frac{J_{3\ell}}{J_\ell^3} \sim \frac{5}{8\pi^3\sqrt{3}} \ell. \quad (271)$$

In particular $G_\ell = H_{4, \ell} - (4\pi)^{-1} \asymp \sqrt{\ell}$. The exact high-only lift

$$v_{\ell, Q} = \frac{Q}{G_\ell} \left(h_\ell^2 - \frac{1}{4\pi} \right) \quad (272)$$

satisfies

$$P_{\{e_0, h_\ell\}} M_{v_{\ell, Q}} P_{\{e_0, h_\ell\}} = \text{diag}(0, Q), \quad (273)$$

$$\|M_{v_{\ell, Q}}e_0\|_2^2 = \frac{Q^2}{4\pi G_\ell} \sim \frac{\sqrt{2\pi}}{3} \frac{Q^2}{\sqrt{\ell}}, \quad (274)$$

$$\limsup_{\ell \rightarrow \infty} \|v_{\ell, Q}\|_\infty \leq \frac{4\sqrt{2}}{3} |Q|. \quad (275)$$

The selected two-dimensional space is not invariant. Its high-state leakage has the nonzero limit

$$\lim_{\ell \rightarrow \infty} \frac{\|(I - P_{\{e_0, h_\ell\}})M_{v_{\ell, Q}}h_\ell\|_2^2}{Q^2} = \frac{20}{9\sqrt{3}} - 1. \quad (276)$$

All harmonics in this leakage have degree between ℓ and 3ℓ .

Proof. The ϕ integrals are $\int_0^{2\pi} \cos^2(\ell\phi) d\phi = \pi$, $\int_0^{2\pi} \cos^4(\ell\phi) d\phi = 3\pi/4$, and $\int_0^{2\pi} \cos^6(\ell\phi) d\phi = 5\pi/8$. The remaining integrals are $J_\ell, J_{2\ell}, J_{3\ell}$. The gamma-ratio asymptotic gives $J_n \sim \sqrt{\pi/n}$ and proves (270)–(271). Since $\int h_\ell^3 = 0$, Proposition C.1 gives (273)–(274). Also

$$\frac{\|h_\ell\|_\infty^2}{H_{4, \ell}} \longrightarrow \frac{4\sqrt{2}}{3}, \quad (277)$$

which proves (275). Finally,

$$\|(I - P_{\{e_0, h_\ell\}})M_{v_\ell, Q}h_\ell\|_2^2 = Q^2 \left[\frac{H_{6, \ell} - H_{4, \ell}/(2\pi) + 1/(16\pi^2)}{G_\ell^2} - 1 \right]. \quad (278)$$

The displayed asymptotics give (276). The identity $\cos^3(\ell\phi) = (3\cos(\ell\phi) + \cos(3\ell\phi))/4$ shows that $h_\ell^3 - (4\pi)^{-1}h_\ell$ has azimuthal frequencies ℓ and 3ℓ ; its spherical-harmonic degrees therefore lie between ℓ and 3ℓ . $\square$

The fixed leakage in (276) is not unavoidable. A slightly non-minimal lift makes the high state almost invariant.

Proposition C.4 (Thin-belt high-only scalar barrier). *Fix $0 < \alpha < 1/2$ and let*

$$E_\ell = \{|\theta - \pi/2| \leq \ell^{-\alpha}\}, \quad \mu_\ell = |E_\ell|, \quad \epsilon_\ell = \int_{E_\ell^c} h_\ell^2 d\omega. \quad (279)$$

There are real axisymmetric $b_{\ell, Q}$ such that

$$P_{\{e_0, h_\ell\}}M_{b_{\ell, Q}}P_{\{e_0, h_\ell\}} = \text{diag}(0, Q), \quad (280)$$

$$\|M_{b_{\ell, Q}}e_0\|_2^2 = O_Q(\ell^{-\alpha}), \quad \|b_{\ell, Q}\|_\infty = O_Q(1), \quad (281)$$

$$\|(I - P_{\{e_0, h_\ell\}})M_{b_{\ell, Q}}h_\ell\|_2^2 \leq C_Q e^{-c\ell^{1-2\alpha}}. \quad (282)$$

Proof. Put $\rho_\ell = \mu_\ell/(4\pi - \mu_\ell)$ and

$$b_{\ell, Q} = q_{\ell, Q}(\mathbf{1}_{E_\ell} - \rho_\ell \mathbf{1}_{E_\ell^c}), \quad q_{\ell, Q} = \frac{Q}{(1 - \epsilon_\ell) - \rho_\ell \epsilon_\ell}. \quad (283)$$

It has mean zero, and its inner product with h_ℓ vanishes because it is axisymmetric while h_ℓ has azimuthal frequency ℓ . Its expectation in h_ℓ is Q , proving (280). Since $\mu_\ell = 4\pi \sin(\ell^{-\alpha}) = O(\ell^{-\alpha})$ and the Gaussian-beam tail obeys

$$\epsilon_\ell \leq C e^{-c\ell^{1-2\alpha}}, \quad (284)$$

one has $q_{\ell, Q} = Q(1 + o(1))$ and

$$\|M_{b_{\ell, Q}}e_0\|_2^2 = q_{\ell, Q}^2 \frac{\mu_\ell}{4\pi - \mu_\ell}. \quad (285)$$

This proves (281). The leakage is exactly the variance of this two-valued function under the probability density h_ℓ^2 :

$$\|(I - P_{\{e_0, h_\ell\}})M_{b_{\ell, Q}}h_\ell\|_2^2 = q_{\ell, Q}^2 (1 + \rho_\ell)^2 \epsilon_\ell (1 - \epsilon_\ell), \quad (286)$$

which gives (282). $\square$

Corollary C.5 (Algebraic scalar lift of the rotator/barrier profile). *Let the 2×2 rotator block have entries*

$$a = \delta_\Lambda \sin^2 \theta, \quad b = -\delta_\Lambda \sin \theta \cos \theta, \quad d = \delta_\Lambda \cos^2 \theta \quad (287)$$

on an interval of length Λ , with $\delta_\Lambda = \Lambda^{-2/3}$ and one quarter-turn. If $\ell_\Lambda \rightarrow \infty$, Proposition C.1 gives an exact scalar compressed lift whose full constant-column cost is

$$\frac{1}{2} \delta_\Lambda^2 \Lambda \left(1 + \frac{1}{4\pi G_{\ell_\Lambda}} \right). \quad (288)$$

Follow it by the belt barrier of Proposition C.4 for time $T_\Lambda = O(\log \Lambda)$. Choosing, for example, $\ell_\Lambda = (\log \Lambda)^p$ with $p\alpha > 1$ makes the barrier cost and its cumulative high-state leakage tend to zero. Since $\|h_\ell\|_\infty = O(\ell^{1/4})$ for (268), every fixed polylogarithmic choice also keeps the rotator lift uniformly bounded: $\delta_\Lambda \|h_{\ell_\Lambda}\|_\infty \rightarrow 0$.

This is an exact scalar lift of the desired compressed matrices, not yet a scalar dynamical counterexample: multiplication during the converter does not preserve $\text{span}\{e_0, h_{\ell_\Lambda}\}$.

D Cutoff-edge scalar counterexample to PSR_1

The preceding full-space converter question is stronger than the quantifiers in PSR_1 . That estimate is required uniformly over the finite angular cutoff. A scalar multiplier may therefore contain degrees above the retained cutoff, provided its *full* constant-column cost is charged. The omitted degrees can cancel every unwanted retained component of the converter. This cutoff-edge freedom decides scalar PSR_1 negatively.

Write

$$\mathcal{H}_L = \bigoplus_{n=0}^L \mathcal{Y}_n, \quad P_{\leq L} : L^2(\mathbb{S}^2) \longrightarrow \mathcal{H}_L, \quad \mathcal{S}_L = \text{span}\{e_0, h_L\}, \quad (289)$$

where h_L is the normalized real highest-weight harmonic. All spaces and maps in the next proposition are taken over the reals.

Proposition D.1 (Exact cutoff-edge scalar lift). *For every $L \geq 1$ and every real symmetric matrix*

$$B = \begin{pmatrix} a & b \\ b & d \end{pmatrix}, \quad (290)$$

there is a real spherical polynomial $v_{L,B}$ of degree at most $2L$ such that $\mathcal{S}_L$ reduces $P_{\leq L} M_{v_{L,B}} P_{\leq L}$ and

$$P_{\leq L} M_{v_{L,B}} P_{\leq L} \Big|_{\mathcal{S}_L} = B \quad (291)$$

in the ordered basis (e_0, h_L) . The lift $B \mapsto v_{L,B}$ may be chosen linear. Consequently, for each fixed L there are finite constants $C_{2,L}, C_{\infty,L}$ such that

$$\|v_{L,B}\|_2 \leq C_{2,L} \|B\|, \quad \|v_{L,B}\|_\infty \leq C_{\infty,L} \|B\|. \quad (292)$$

No uniform estimate on these constants is needed below.

Proof. Put

$$\mathcal{W}_L = \bigoplus_{n=L+1}^{2L} \mathcal{Y}_n, \quad T_L : \mathcal{W}_L \longrightarrow \mathcal{H}_L, \quad T_L w = P_{\leq L}(h_L w). \quad (293)$$

We first prove

$$\text{Ran } T_L = e_0^\perp \cap \mathcal{H}_L. \quad (294)$$

Certainly $T_L^* e_0 = 0$, because $h_L e_0$ has pure degree L . Conversely, let $f \in \ker T_L^*$ and let $f_n \neq 0, n \geq 1$, be its highest nonconstant harmonic component. If H_L and F_n are the associated homogeneous harmonic polynomials, the degree- $(L+n)$ harmonic component of $H_L F_n$ is nonzero. Indeed, otherwise the Fischer decomposition would make $H_L F_n$ divisible by $|x|^2$; irreducibility of $|x|^2$ and uniqueness of the harmonic decomposition would force one of the nonzero harmonic factors to be divisible by $|x|^2$, which is impossible. Lower components of f cannot cancel this top degree. Thus $T_L^* f \neq 0$. Hence $\ker T_L^* = \mathbb{R}e_0$, proving (294).

Set

$$v_{L,B}^{\text{low}} = a\sqrt{4\pi} e_0 + b\sqrt{4\pi} h_L. \quad (295)$$

Then $P_{\leq L}(v_{L,B}^{\text{low}} e_0) = ae_0 + bh_L$. The remaining required column is

$$g_{L,B} = be_0 + dh_L - P_{\leq L}(v_{L,B}^{\text{low}} h_L). \quad (296)$$

Self-adjoint symmetry gives $\langle e_0, g_{L,B} \rangle = 0$. By (294), choose $w_{L,B} \in \mathcal{W}_L$ with $T_L w_{L,B} = g_{L,B}$, using once and for all a linear right inverse of T_L on $e_0^\perp$. Then $v_{L,B} = v_{L,B}^{\text{low}} + w_{L,B}$ has the two required columns. The compressed multiplier is self-adjoint, so invariance of $\mathcal{S}_L$ makes it reducing. Finite dimensionality gives (292). $\square$

The high-only barrier has a simpler closed form. Define

$$p_{2L}(\theta, \phi) = \sin^{2L} \theta \cos(2L\phi), \quad c_L = \langle h_L, p_{2L} h_L \rangle. \quad (297)$$

The function p_{2L} is a pure real harmonic of degree $2L$ and $\|p_{2L}\|_\infty = 1$.

Lemma D.2 (Exact high-only scalar barrier at the cutoff). *One has*

$$P_{\leq L} M_{p_{2L}} e_0 = 0, \quad P_{\leq L} M_{p_{2L}} h_L = c_L h_L, \quad (298)$$

$$c_L = \frac{I_{4L+1}}{2I_{2L+1}} \longrightarrow \frac{1}{2\sqrt{2}}, \quad \|p_{2L}\|_2^2 = \pi I_{4L+1} \asymp L^{-1/2}, \quad (299)$$

where $I_m = \int_0^\pi \sin^m \theta \, d\theta$. Hence, for every fixed $Q > 0$, the uniformly bounded scalar multiplier

$$b_{L,Q} = \frac{Q}{c_L} p_{2L} \quad (300)$$

compresses on $\mathcal{S}_L$ to $\text{diag}(0, Q)$, while

$$\|M_{b_{L,Q}} e_0\|_2^2 \asymp_Q L^{-1/2}. \quad (301)$$

Proof. The first identity in (298) is degree orthogonality. Moreover

$$p_{2L} h_L = \frac{C_L}{2} \sin^{3L} \theta (\cos(3L\phi) + \cos(L\phi)). \quad (302)$$

The first summand has azimuthal order $3L$; the second has order L , so its only component of degree at most L is a multiple of h_L . This proves the second block identity. Since $h_L = C_L \sin^L \theta \cos(L\phi)$ and $C_L^{-2} = \pi I_{2L+1}$, direct integration gives (299). The Wallis asymptotic for I_m and $e_0 = (4\pi)^{-1/2}$ finish the proof. $\square$

Theorem D.3 (Scalar PSR_1 is false). *Let $K = [k_0, k_1] \Subset (0, \infty)$. There are angular cutoffs $L \rightarrow \infty$ and uniformly bounded, compactly supported real scalar potentials V_L on $[1, \infty) \times \mathbb{S}^2$ such that*

$$\sup_L \int_1^\infty \|M_{V_L(r)} e_0\|_2^2 \, dr < \infty, \quad (303)$$

but for the finite angular models on $\mathcal{H}_L$,

$$\int_K \log^+ \Gamma_{0;V_L}(k) \, dk \longrightarrow \infty. \quad (304)$$

Thus (92), and hence scalar PSR_1 , fails. This does not disprove Simon's conjecture.

Proof. Fix $Q_0 > k_1^2$, put $\kappa_{\min} = \sqrt{Q_0 - k_1^2}$ and $\kappa_{\max} = \sqrt{Q_0 - k_0^2}$, and choose $0 < \gamma < 1/2$. Let

$$T_L = L^\gamma. \quad (305)$$

Use Proposition D.1 to lift the rotating block (235) with $\delta_\Lambda = \Lambda^{-2/3}$ and $\omega_\Lambda = \pi/(2\Lambda)$. For each L , choose $\Lambda = \Lambda_L$ so large that

$$C_{2,L}^2 \Lambda_L^{-1/3} \leq 1, \quad C_{\infty,L} \Lambda_L^{-2/3} \leq 1, \quad \left(\Lambda_L^{-1/3} + \frac{L^2}{\Lambda_L} + \frac{L^{7/2}}{\Lambda_L^3} \right) e^{2\kappa_{\max} T_L} \leq L^{-1}. \quad (306)$$

This is possible with no information about the growth of the lift constants.

Choose a fixed $A_K \geq 3$ larger than the uniform constant in the upper bound of (253), and place the rotator on a shell beginning at $R_L = A_K \Lambda_L$, replacing $\theta(r)$ by $\omega_{\Lambda_L}(r - R_L)$. On its

two-dimensional reducing space, cancel the centrifugal entry $d_L(r) = L(L+1)/r^2$ by adding $-d_L(r)c_L^{-1}p_{2L}$. On the following shell of length T_L , use

$$\frac{Q_0 - d_L(r)}{c_L} p_{2L}. \quad (307)$$

Lemma D.2 then makes the *total* compressed radial operator on $\mathcal{S}_L$ exactly the matrix rotator followed by $\text{diag}(0, Q_0)$. At radii comparable to Λ_L , the normalized free Bessel directional frame differs from the plane-wave frame, together with one k derivative, by $O_K(L^2/\Lambda_L)$. Moreover the charged cost of the centrifugal cancellation on the rotator is

$$O\left(\Lambda_L \frac{L^4}{\Lambda_L^4} \|p_{2L}e_0\|_2^2\right) = O\left(\frac{L^{7/2}}{\Lambda_L^3}\right). \quad (308)$$

Thus all matching, centrifugal, and converted-mode errors remain negligible after barrier amplification by (306).

The converter cost is bounded by

$$CC_{2,L}^2 \delta_{\Lambda_L}^2 \Lambda_L = CC_{2,L}^2 \Lambda_L^{-1/3} \leq C. \quad (309)$$

The barrier cost tends to zero:

$$\int_{\text{barrier}} \|M_{V_L(r)}e_0\|_2^2 dr \leq C_Q T_L L^{-1/2} = C_Q L^{\gamma-1/2} \rightarrow 0. \quad (310)$$

The choice of Λ_L , together with (299), also makes the full scalar amplitudes uniformly bounded. This proves (303).

Because $\mathcal{S}_L$ reduces every compressed scalar multiplier used above, the full finite Jost matrix is block diagonal relative to $\mathcal{S}_L \oplus \mathcal{S}_L^\perp$, and

$$\Gamma_0(k) = \|J(k)^{-1}e_0\|^{-1} \quad (311)$$

is exactly the two-channel quotient. Propagation from R_L through the inner free collar is diagonal in the angular channels. For real k , the incoming and outgoing degree- L Bessel factors at $r = 1$ are conjugates and hence have equal modulus; their common row size cancels from Γ_0 . In the degree zero row the collar merely appends the phases $e^{\pm ik(R_L-1)}$ to the two directional terms. Consequently the proof of Theorem B.3 applies with the present T_L : (306) makes the converted-mode and Bessel matching errors negligible after barrier amplification, while the total relative phase has first derivative comparable to Λ_L and second derivative $O_K(\Lambda_L)$. Lemma B.2 therefore supplies a subset of K of fixed positive measure on which

$$\Gamma_0(k) \geq c_K e^{\kappa_{\min} T_L}. \quad (312)$$

Consequently $\int_K \log^+ \Gamma_0 \geq c_K T_L - C_K \rightarrow \infty$, proving (304). Taking the radial truncation parameter beyond the support of V_L places this model among the terms in the supremum in (92), whereas its right-hand side stays bounded by (303). This completes the quantifier-level contradiction. $\square$

Remark D.4 (What has been disproved). The counterexample uses a degree- $2L$ scalar multiplier against a degree- L Galerkin model. Its full constant-column norm is included in (303); no truncated-column aliasing is hidden. What fails is uniformity of the proposed sufficient estimate under cutoff-edge scalar compression. A theorem which first fixes one scalar potential and passes to all sufficiently large angular cutoffs, or which imposes an independently justified relation between multiplier bandwidth and model cutoff, is a different statement. The full-space converter problem below remains mathematically open, but it is no longer needed to decide PSR₁.

E A cutoff-compatible fixed-potential diagonal

The failure of uniform PSR_1 does not force the same quantifiers on an approximation theorem. For one fixed scalar potential it is enough to choose a single cofinal sequence of angularly smoothed, radially truncated models, put the Galerkin boundary beyond the smoothing bandwidth, and retain a bounded entropy subsequence. This section makes that replacement exact and separates two issues: approximation is harmless, while the required entropy estimate remains substantive.

We first record an L^∞ -safe finite-band approximation. Let

$$\mathcal{D}_B(\omega, \eta) = \sum_{n=0}^B \sum_{m=-n}^n Y_{nm}(\omega) \overline{Y_{nm}(\eta)}, \quad m_B = (B+1)^2, \quad (313)$$

and define the positive zonal kernel

$$\mathcal{K}_B(\omega, \eta) = \frac{|\mathcal{D}_B(\omega, \eta)|^2}{m_B/(4\pi)}, \quad \mathcal{S}_B v(\omega) = \int_{\mathbb{S}^2} \mathcal{K}_B(\omega, \eta) v(\eta) d\eta. \quad (314)$$

Proposition E.1 (Positive finite-band approximation). *The operators $\mathcal{S}_B$ preserve real functions and constants, map into spherical polynomials of degree at most $2B$, and satisfy*

$$\|\mathcal{S}_B v\|_p \leq \|v\|_p, \quad 1 \leq p \leq \infty, \quad \mathcal{S}_B v \rightarrow v \quad \text{in } L^2(\mathbb{S}^2). \quad (315)$$

Consequently, if $V \in L^\infty \cap X_3$ and

$$V_j(r, \omega) = \mathbf{1}_{[1, R_j]}(r) \mathcal{S}_{B_j} V(r, \cdot)(\omega), \quad R_j, B_j \rightarrow \infty, \quad (316)$$

then V_j is real, compactly supported, and of angular degree at most $2B_j$, while

$$\|V_j\|_\infty \leq \|V\|_\infty, \quad \|V_j - V\|_{X_3} \rightarrow 0, \quad \|V_j\|_{X_3} \leq \|V\|_{X_3}. \quad (317)$$

Proof. The reproducing identity gives $\int |\mathcal{D}_B(\omega, \eta)|^2 d\eta = m_B/(4\pi)$. Hence $\mathcal{K}_B$ is a symmetric Markov kernel. Positivity, Jensen's inequality, and duality prove all L^p contractions. The product of two degree- B spherical polynomials has degree at most $2B$. Finally, the normalized squared reproducing kernels form an approximate identity: the Christoffel–Darboux formula gives concentration of their mass in every fixed neighbourhood of the diagonal. Thus $\mathcal{S}_B v \rightarrow v$ first for continuous functions and then in L^2 by density and contraction. Apply this fiberwise in r and use dominated convergence, followed by $R_j \rightarrow \infty$, to obtain (317). $\square$

Let $H_0 = -\partial_r^2 + A_3/r^2$ on the exterior half-line. Extend each finite model to the full angular space by leaving the complementary channels free:

$$\widehat{H}_j = H_0 + P_{L_j} M_{V_j} P_{L_j} = H_{R_j, L_j}(V_j) \oplus H_0|_{P_{L_j}^\perp}. \quad (318)$$

Proposition E.2 (Strong resolvent convergence along every buffered diagonal). *If $L_j \rightarrow \infty$, then*

$$\widehat{H}_j \rightarrow H_0 + M_V \quad \text{in the strong resolvent sense.} \quad (319)$$

In particular one may impose, without changing the limit, the double buffer

$$\frac{L_j}{B_j + R_j + 1} \rightarrow \infty. \quad (320)$$

The matrix spectral measures of every fixed finite collection of compact low-channel probes then converge vaguely.

Proof. The uniform L^∞ bound and (317) imply $M_{V_j} \rightarrow M_V$ strongly. Since $P_{L_j} \rightarrow I$ strongly and all three operator families are uniformly bounded, $P_{L_j} M_{V_j} P_{L_j} \rightarrow M_V$ strongly. The second resolvent identity for bounded self-adjoint perturbations of H_0 proves (319). Condition (320) can be enforced by choosing L_j after B_j and R_j . Vague convergence of fixed-probe spectral measures follows from the spectral theorem. $\square$

The buffer excludes the exact mechanism of Theorem D.3 for a purely algebraic reason.

Lemma E.3 (A nonconstant reducing cutoff-edge block consumes the buffer). *Let v be a real spherical polynomial of degree at most $B < L$, and let h_L have pure degree L . If $\text{span}\{e_0, h_L\}$ is invariant under $P_{\leq L} M_v P_{\leq L}$, then v is constant and its compressed block on that plane is a scalar multiple of the identity. In particular, neither the rotator nor the high-only barrier used in Theorem D.3 can occur under (320).*

Proof. Because $B < L$, $P_{\leq L}(v e_0) = v e_0$ has no degree- L component. Invariance forces it to be a multiple of e_0 , hence v is constant. The conclusion follows. $\square$

Remark E.4 (Why shell stacking does not restore the edge counterexample). The scalar PSR_1 examples may be placed on disjoint shells and their charged costs may be made small by lengthening the rotator. This observation can produce bad *unbuffered* diagonals for a shell-stacked potential, but it does not negate the quantifiers in Theorem E.9. At stage j the angular smoothing fixes the multiplier bandwidth and the radial truncation fixes the support radius before L_j is chosen. Condition (320) then puts both the multiplier edge and the whole interaction region far below the Galerkin edge. Any counterexample surviving this order of limits must be a genuine full-space scalar converter, not the compression identity in Theorem D.3.

The double buffer also gives a quantitative real-energy statement. It does not prove the entropy estimate, but it removes the Galerkin boundary from the remaining converter problem.

Proposition E.5 (Outgoing high-channel Schur complement is coercive). *Let $K = [k_0, k_1] \subseteq (0, \infty)$, let V be real, supported in $1 \leq r \leq R$, and put $M = \|V\|_\infty$. For $J < L$, set $Q_{J,L} = P_L - P_J$ and consider on $L^2((1, R); Q_{J,L} \mathcal{K})$*

$$\mathcal{A}_{J,L}(k) = -\partial_r^2 + \frac{D_L}{r^2} + Q_{J,L} M_V Q_{J,L} - k^2, \quad (321)$$

with Dirichlet condition at $r = 1$ and the free outgoing Bessel Dirichlet-to-Neumann condition at $r = R$. If

$$g_{J,R} := \frac{(J+1)(J+2)}{R^2} - M - k_1^2 > 0, \quad (322)$$

then $\mathcal{A}_{J,L}(k)$ is invertible for every $k \in K$ and

$$\sup_{k \in K} \|\mathcal{A}_{J,L}(k)^{-1}\| \leq g_{J,R}^{-1}, \quad (323)$$

uniformly in L . If $\mathcal{B}_{J,L} = Q_{J,L} M_V P_J$, the exact outgoing Feshbach correction obeys

$$\sup_{k \in K} \|\mathcal{B}_{J,L}^* \mathcal{A}_{J,L}(k)^{-1} \mathcal{B}_{J,L}\| \leq \frac{M^2}{g_{J,R}}. \quad (324)$$

Proof. For angular degree n , let $f_n(r, k) = k r h_n^{(1)}(kr)$ be the outgoing flattened free solution and set $\mathbf{m}_n(k, R) = f'_n(R, k)/f_n(R, k)$. The standard half-integer Hankel expansion gives

$$|x h_n^{(1)}(x)|^2 = \sum_{q=0}^n c_{nq} x^{-2q}, \quad c_{nq} > 0. \quad (325)$$

Hence $\Re \mathbf{m}_n(k, R) = \partial_R \log |f_n(R, k)| \leq 0$. If u satisfies the outgoing boundary condition, integration by parts gives

$$\begin{aligned} \Re \langle \mathcal{A}_{J,L}(k)u, u \rangle &= \int_1^R \left(\|u'\|^2 + \left\langle u, \left(\frac{D_L}{r^2} + Q_{J,L} M_V Q_{J,L} - k^2 \right) u \right\rangle \right) dr \\ &\quad - \Re \langle u(R), \mathbf{m}_{J,L}(k, R)u(R) \rangle \geq g_{J,R} \|u\|_2^2. \end{aligned} \quad (326)$$

The complex coercive form theorem proves invertibility and (323). Since $\|\mathcal{B}_{J,L}\| \leq M$, the Feshbach bound follows. $\square$

Corollary E.6 (The double buffer removes the cutoff edge). *For the approximants (316) satisfying (320), put*

$$J_j = \lfloor \sqrt{R_j L_j} \rfloor. \quad (327)$$

Then $J_j/R_j \rightarrow \infty$, $J_j/L_j \rightarrow 0$, and, for every fixed compact $K \Subset (0, \infty)$,

$$\sup_{k \in K} \|\mathcal{B}_{J_j, L_j}^* \mathcal{A}_{J_j, L_j}(k)^{-1} \mathcal{B}_{J_j, L_j}\| \rightarrow 0. \quad (328)$$

Moreover $J_j + 2B_j < L_j$ eventually, so every one-step coupling from the active block generated by V_j is retained. Thus any failure of (343) must occur inside a growing active block far from the Galerkin edge.

Proof. Condition (320) gives $R_j/L_j \rightarrow 0$ and $B_j/L_j \rightarrow 0$. Therefore the three elementary assertions about J_j and B_j hold. Since $\|V_j\|_\infty \leq \|V\|_\infty$ and $g_{J_j, R_j} \sim L_j/R_j \rightarrow \infty$, apply Proposition E.5. $\square$

The most dangerous explicit way to enter the active block is the wrapped azimuthal phase from Proposition F.18. Allowing its pulse area to grow makes the conversion arbitrarily broadband, but the exact scale ledger shows that it cannot be both cheap and clean at its own centrifugal turning radius.

Proposition E.7 (Broadband wrapped masks miss the evanescent scale). *Fix $s_* > 0$ and $0 < \delta < 1/2$, put $S_\delta = [s_*(1-\delta), s_*(1+\delta)]$, and let $\Theta_L = \text{Arg}(e^{iL\phi})$. Given $\tau \geq 1$ and $\epsilon > 0$, set*

$$T = \frac{\tau}{s_* \epsilon}, \quad v(r, \omega) = \epsilon \Theta_L(\omega) \mathbf{1}_{[R, R+T]}(r). \quad (329)$$

After deleting A_3/r^2 , the terminal state from e_0 is $\psi_s = e^{-i(s/s_*)\tau\Theta_L} e_0$. If

$$M_{\tau, L} = L \lfloor (1 - 2\delta)\tau \rfloor, \quad (330)$$

then, for all sufficiently large τ ,

$$\sup_{s \in S_\delta} \|P_{< M_{\tau, L}} \psi_s\|_2^2 \leq \frac{C_\delta}{\tau}. \quad (331)$$

The charged cost and the centrifugal phase scale at that target degree are

$$\mathcal{C}_{\tau, L} = \int_R^{R+T} \|M_{v(r)} e_0\|_2^2 dr = \frac{\pi^2 \tau^2}{3s_*^2 T}, \quad (332)$$

$$\mathcal{D}_{\tau, L} = M_{\tau, L} (M_{\tau, L} + 1) \frac{T}{R(R+T)}. \quad (333)$$

Along any sequence with $\tau \rightarrow \infty$ and $\mathcal{C}_{\tau, L} \rightarrow 0$, the turning-scale condition $R \leq CM_{\tau, L}$ forces $\mathcal{D}_{\tau, L} \rightarrow \infty$. Consequently, if both the charged cost and the centrifugal error tend to zero, then

$$\frac{R}{M_{\tau, L}} \rightarrow \infty; \quad (334)$$

the modes created in (331) are then far inside their propagating region and cannot supply an exterior centrifugal barrier in series with the mask.

Proof. Write $a = s/s_*$ and $\beta = a\tau$. The azimuthal Fourier coefficients of $e^{-i\beta\Theta_L}$ occur only at frequencies qL and satisfy

$$|c_q(\beta)| = \left| \frac{\sin \pi(\beta - q)}{\pi(\beta - q)} \right|. \quad (335)$$

For $|q| < (1 - 2\delta)\tau$ and $\beta \in [(1 - \delta)\tau, (1 + \delta)\tau]$, summing the squared right-hand side gives

$$\sum_{|q| < (1-2\delta)\tau} |c_q(\beta)|^2 \leq \frac{C_\delta}{\tau}. \quad (336)$$

A spherical harmonic with azimuthal frequency qL has degree at least $|q|L$, proving (331). Since $(4\pi)^{-1}\|\Theta_L\|_2^2 = \pi^2/3$, substitution of $\epsilon = \tau/(s_*T)$ proves (332); the second identity is direct.

Vanishing cost implies $T/\tau^2 \rightarrow \infty$. Suppose $R \leq CM_{\tau,L}$. If $T \leq R$, then

$$\mathcal{D}_{\tau,L} \geq \frac{M_{\tau,L}^2 T}{2R^2} \geq \frac{T}{2C^2} \rightarrow \infty. \quad (337)$$

If $T \geq R$, then

$$\mathcal{D}_{\tau,L} \geq \frac{M_{\tau,L}^2}{2R} \geq \frac{M_{\tau,L}}{2C} \rightarrow \infty. \quad (338)$$

This proves the turning-scale assertion. If (334) failed, a bounded-ratio subsequence would contradict that assertion. $\square$

The last sentence has an exact algebraic counterpart. A clean converter placed outside an inner amplifier is a right unitary factor of the boundary Jost matrix, and the scalar Plücker quotient does not see it at all.

Proposition E.8 (Outer unitary converters are Plücker-invisible). *Let $P_0 = e_0 e_0^*$ and $\Gamma_0(J) = \|J^{-1}e_0\|^{-1}$. If a radially inner system has boundary matrix J_{in} and a reflectionless outer converter contributes a right factor U , so that*

$$J = J_{\text{in}}U, \quad (339)$$

then

$$U^*U = \mathbf{I} \implies \Gamma_0(J) = \Gamma_0(J_{\text{in}}). \quad (340)$$

More generally, if $\|U^*U - \mathbf{I}\| \leq \eta < 1$, then

$$|\log \Gamma_0(J) - \log \Gamma_0(J_{\text{in}})| \leq -\frac{1}{2} \log(1 - \eta). \quad (341)$$

Thus a far-field phase mask may populate arbitrarily high propagating modes, but it cannot exploit the centrifugal amplification already accumulated to its inside. To increase Γ_0 , conversion must overlap the nonunitary turning dynamics or occur on its inner side.

Proof. Put $y = J_{\text{in}}^{-1}e_0$. Equation (339) gives $J^{-1}e_0 = U^{-1}y$. This proves (340). Under the approximate hypothesis, the singular values of U lie between $\sqrt{1 - \eta}$ and $\sqrt{1 + \eta}$. Hence

$$\frac{\|y\|}{\sqrt{1 + \eta}} \leq \|U^{-1}y\| \leq \frac{\|y\|}{\sqrt{1 - \eta}}, \quad (342)$$

and (341) follows. The final statement is the radial ordering encoded by the right factor in (339). $\square$

The weakest entropy hypothesis needed on this diagonal uses a liminf, not a supremum over all cutoffs. Let $\Gamma_{P_N; j}$ be the finite-block quotient for $H_{R_j, L_j}(V_j)$.

Theorem E.9 (Completion from buffered diagonal entropy). *Suppose that, for every bounded real $V \in X_3$, every fixed N , and every $K \in (0, \infty)$, one can choose (316) and (320) so that*

$$\liminf_{j \rightarrow \infty} \int_K \log^+ \Gamma_{P_N; j}(k) dk \leq C_{K,N}(1 + \mathcal{E}_N(V)). \quad (343)$$

Then the exterior spectral density has rank at least N for almost every positive energy. If (343) holds for every N , then Simon's conclusion (4) follows.

Proof. Choose a subsequence realizing the finite liminf. The Plücker identity and the compact-probe boundary factor convert (343) into a uniform upper bound for the negative logarithmic determinant of the fixed N -probe densities. Proposition E.2 gives vague convergence of their measures, and the entropy lower-semicontinuity estimate (20) passes the bound to the limiting exterior density. Its determinant is therefore positive almost everywhere. The rank/countability argument and the free-collar equivalence in Proposition 3.3 finish exactly as in Theorem 12.1. $\square$

There is also a positive result for the dimension-normalized route. It shows that matrix size itself is not the obstruction.

Proposition E.10 (Zero-background dimension-normalized matrix sum rule). *Let $H_m = -\partial_r^2 + W(r)$ be a half-line Schrödinger operator with componentwise Neumann or Dirichlet boundary condition and a compactly supported real symmetric $m \times m$ potential. If h_m and $h_{0,m}$ are its matrix density and the matching free density, then for every $I \in (0, \infty)$,*

$$\frac{1}{m} \int_I \log^- \det(h_{0,m}^{-1/2} h_m h_{0,m}^{-1/2}) dE \leq C_I \left(1 + \frac{1}{m} \int_0^\infty \text{tr } W(r)^2 dr \right), \quad (344)$$

with C_I independent of m . Consequently, for $W = P_L M_V P_L$ and $m = m_L$ the right-hand side is bounded by $C_I(1 + (4\pi)^{-1} \|V\|_{X_3}^2)$ uniformly in L .

Proof. For smooth compactly supported W , extend it by zero to the negative half-line and let $S(E)$ and $P(E)$ be the Jost transmission and reflection matrices. The direct and reciprocal flux identities are

$$S^* S - P^* P = I, \quad SS^* - \bar{P} P^T = I. \quad (345)$$

They follow from Wronskian conservation for the real symmetric equation. In particular

$$A = S^{-1} \bar{P}, \quad AA^* = I - S^{-1} S^{-*} \preceq I. \quad (346)$$

For Neumann boundary condition the incoming coefficient of the regular solution is $B_N = (S + \bar{P})/2 = S(I + A)/2$; for Dirichlet boundary condition its ratio to the free incoming coefficient is $S - \bar{P} = S(I - A)$. Since all singular values of A are at most one,

$$|\det(I \pm A)| \leq 2^m, \quad |\det B_N| \leq |\det S|. \quad (347)$$

The $m = 1$ trace identity of the matrix L^2 sum rule [7, Theorem 7] has quadratic coefficient $\int \text{tr } W^2$ and a favorable nonpositive bound-state term. On a compact positive interval it yields

$$\int_I \log |\det S(E)| dE \leq C_I \int_0^\infty \text{tr } W(r)^2 dr. \quad (348)$$

The density formula $h_m = c(E) B^{-1} B^{-*}$, divided by the matching free formula, and (347) give pointwise

$$\log^- \det(h_{0,m}^{-1/2} h_m h_{0,m}^{-1/2}) \leq 2m \log 2 + 2 \log |\det S|. \quad (349)$$

Equations (348) and (349) prove (344). Approximation gives the stated L^2 class. Finally,

$$\frac{1}{m_L} \text{tr}(P_L M_V P_L)^2 = \frac{1}{m_L} \|P_L M_V P_L\|_{\mathfrak{S}_2}^2 \leq \frac{1}{4\pi} \|V(r, \cdot)\|_2^2 \quad (350)$$

by Proposition 5.1. $\square$

The obstruction to inserting the centrifugal background into this proof is visible already in the first quadratic trace coefficient.

Proposition E.11 (The naive Bessel subtraction has an L^2 -uncontrolled linear term). *In dimension three let $D_L Y_{nm} = n(n+1)Y_{nm}$ and $m_L = (L+1)^2$. For every real $v \in L^2(\mathbb{S}^2)$,*

$$\frac{1}{m_L} \operatorname{tr}(D_L P_L M_v P_L) = \frac{L(L+2)}{8\pi} \int_{\mathbb{S}^2} v(\omega) d\omega. \quad (351)$$

Consequently, subtracting the free quadratic invariant from the invariant for $D_L/r^2 + P_L M_V P_L$ leaves the cross term

$$\frac{2}{m_L} \int_1^\infty \frac{1}{r^2} \operatorname{tr}(D_L P_L M_{V(r)} P_L) dr = \frac{L(L+2)}{4\pi} \int_1^\infty \frac{1}{r^2} \int_{\mathbb{S}^2} V(r, \omega) d\omega dr. \quad (352)$$

Its coefficient grows quadratically in L and is not bounded by the scalar X_3 norm uniformly in L . Thus a proof of (133) must use a genuinely Bessel-relative determinant or an explicit channelwise phase renormalization; subtracting two zero-background trace identities is insufficient.

Proof. The degree- n addition theorem gives

$$\begin{aligned} \operatorname{tr}(D_L P_L M_v P_L) &= \int_{\mathbb{S}^2} v(\omega) \sum_{n=0}^L n(n+1) \frac{2n+1}{4\pi} d\omega \\ &= \frac{L(L+1)^2(L+2)}{8\pi} \int_{\mathbb{S}^2} v(\omega) d\omega. \end{aligned} \quad (353)$$

Divide by $m_L = (L+1)^2$ and expand $\operatorname{tr}(D_L/r^2 + P_L M_V P_L)^2 - \operatorname{tr}(D_L/r^2)^2$. $\square$

The divergent term in Proposition E.11 has a precise origin: it is entirely the radial angular mean. Splitting that mean into the reference operator removes the term exactly and leaves the original quadratic cost.

Proposition E.12 (Radial-mean reference cancels every Bessel cross term). *For a real scalar potential on $\mathbb{S}^2$, write*

$$q(r) = \frac{1}{4\pi} \int_{\mathbb{S}^2} V(r, \omega) d\omega, \quad V^\circ(r, \omega) = V(r, \omega) - q(r), \quad W_L^\circ = P_L M_{V^\circ} P_L. \quad (354)$$

Then

$$\|V(r, \cdot)\|_2^2 = 4\pi |q(r)|^2 + \|V^\circ(r, \cdot)\|_2^2, \quad (355)$$

$$\operatorname{tr} W_L^\circ = 0, \quad \operatorname{tr}(D_L W_L^\circ) = 0, \quad (356)$$

and hence, pointwise in r ,

$$\frac{1}{m_L} \operatorname{tr} \left[\left(\frac{D_L}{r^2} + q\mathbf{I} + W_L^\circ \right)^2 - \left(\frac{D_L}{r^2} + q\mathbf{I} \right)^2 \right] = \frac{1}{m_L} \operatorname{tr}(W_L^\circ)^2 \leq \frac{1}{4\pi} \|V^\circ(r, \cdot)\|_2^2. \quad (357)$$

Thus the correct comparison operator for a Bessel-relative sum rule is

$$H_{q,L} = -\partial_r^2 + \frac{D_L}{r^2} + q(r)\mathbf{I}, \quad (358)$$

not the bare Bessel operator.

Proof. The first identity is angular orthogonality. The addition theorem gives

$$\operatorname{tr}(P_L M_f P_L) = \frac{m_L}{4\pi} \int_{\mathbb{S}^2} f, \quad \operatorname{tr}(D_L P_L M_f P_L) = \frac{L(L+1)^2(L+2)}{8\pi} \int_{\mathbb{S}^2} f. \quad (359)$$

Apply these formulas to $f = V^\circ$. Expanding the two squares leaves only $\operatorname{tr}(W_L^\circ)^2$, and Proposition 5.1 proves the last inequality. $\square$

Proposition E.13 (Signed radial-mean-relative trace identity). *Assume first that q and V° are smooth and compactly supported in $(1, \infty)$. Put*

$$U_{V,L} = \frac{D_L}{r^2} + qI + W_L^\circ, \quad U_{q,L} = \frac{D_L}{r^2} + qI, \quad (360)$$

and let $S_{V,L}(E)$ and $S_{q,L}(E)$ be their matrix Jost transmission matrices relative to the flat half-line reference. If $\{\lambda_a(V, L)\}$ and $\{\lambda_a(q, L)\}$ are the negative eigenvalues, then

$$\begin{aligned} & \frac{1}{\pi m_L} \int_0^\infty E^{1/2} \log \left| \frac{\det S_{V,L}(E)}{\det S_{q,L}(E)} \right| dE \\ &= \frac{1}{m_L} \int_1^\infty \text{tr}(W_L^\circ(r))^2 dr - \frac{2}{3m_L} \left(\sum_a |\lambda_a(V, L)|^{3/2} - \sum_a |\lambda_a(q, L)|^{3/2} \right). \end{aligned} \quad (361)$$

Moreover, its right-hand side is bounded in absolute value by

$$C \left(\|V\|_{X_3}^2 + \|q\|_{L^2(1,\infty)}^2 \right) \leq C' \|V\|_{X_3}^2, \quad (362)$$

with constants independent of L . The identity and bound extend to bounded X_3 data at the level needed here: every smooth compact approximation obeys the same bound, uniformly in its support and angular cutoff. No full-space transmission determinant is asserted by this statement.

Proof. Apply the $m = 1$ trace identity of Molchanov–Vainberg [7, Theorem 7] separately to $U_{V,L}$ and $U_{q,L}$ and subtract. One first truncates the common D_L/r^2 tail; for fixed L it lies in $L^1 \cap L^2$, so standard Jost and negative-spectrum convergence passes the identity to the untruncated tail. Proposition E.12 gives

$$\int \text{tr}(U_{V,L}^2 - U_{q,L}^2) = \int \text{tr}(W_L^\circ)^2, \quad (363)$$

which proves (361). The matrix Lieb–Thirring inequality and positivity of D_L/r^2 give

$$\sum_a |\lambda_a(V, L)|^{3/2} \leq C \int_1^\infty \text{tr}(P_L M_V P_L)_-^2 dr \leq C \int_1^\infty \text{tr}(P_L M_V P_L)^2 dr, \quad (364)$$

$$\sum_a |\lambda_a(q, L)|^{3/2} \leq C m_L \int_1^\infty |q(r)|^2 dr. \quad (365)$$

Divide by m_L and use Proposition 5.1 and (355). $\square$

Remark E.14 (The remaining one-sided gap). Identity (361) controls a signed logarithm. It does not control

$$\int_I \log^+ \left| \frac{\det S_{V,L}}{\det S_{q,L}} \right|, \quad (366)$$

because an arbitrarily large positive part could cancel an equally large negative part. Thus the divergent Bessel coefficient has been removed, but the fixed-coupling theorem still requires a one-sided relative determinant estimate using scalar locality, or the fixed-probe estimate (368) directly.

For fixed low channels the radial-mean reference already has the required entropy. Indeed, it is a finite direct sum of scalar half-line operators with potentials $q(r) + n(n+1)/r^2 \in L^2(1, \infty)$, so the one-dimensional L^2 sum rule applies channel by channel.

Proposition E.15 (Exact reduction to radial-mean-relative entropy). *Let V_j be the double-buffered approximants and let $h_{V,j,N}$ and $h_{q,j,N}$ be the fixed P_N probe densities for $H_{R_j,L_j}(V_j)$ and its radial-mean reference (358), respectively. For every fixed N and $I \Subset (0, \infty)$,*

$$\sup_j \int_I \log^- \det h_{q,j,N}(E) dE < \infty, \quad (367)$$

after any fixed smooth positive normalization of the probe density. Therefore the cutoff-compatible estimate

$$\liminf_{j \rightarrow \infty} \int_I \log^- \det(h_{q,j,N}^{-1/2} h_{V,j,N} h_{q,j,N}^{-1/2}) dE \leq C_{I,N} (1 + \|V^\circ\|_{X_3}^2) \quad (368)$$

implies rank at least N for the limiting exterior density. If (368) holds for every N , it implies Simon's conclusion (4).

Proof. For fixed N , the finitely many centrifugal potentials $n(n+1)/r^2$ have bounded $L^2(1, \infty)$ norm, while $\|q\|_2^2 \leq (4\pi)^{-1} \|V\|_{X_3}^2$. The scalar physical-coupling entropy bound, applied to the radial truncations of each channel, proves (367). Determinants give the scalar identity

$$\det h_{V,j,N} = \det h_{q,j,N} \det(h_{q,j,N}^{-1/2} h_{V,j,N} h_{q,j,N}^{-1/2}). \quad (369)$$

Using $(x+y)^+ \leq x^+ + y^+$ for the negative logarithms, equations (367) and (368) give a uniform subsequential entropy bound for $h_{V,j,N}$. Strong-resolvent convergence, entropy lower semicontinuity (20), and the rank/countability finish prove the claim. $\square$

For $N = 1$ one can remove still more uncharged structure than in the radial-mean determinant. Keep the entire high-high block in the reference operator. The perturbation is then a star with one charged radial column.

Proposition E.16 (Exact star-Feshbach reduction and charged resolvent bound). *Fix a finite angular model containing $e_0 = (4\pi)^{-1/2}$ and decompose it as $P_0 \oplus Q_0$. With the notation of (354), identify $P_0 L^2$ with $L^2(1, \infty)$ and write*

$$H_{V,L} = \begin{pmatrix} h_q & B_L^* \\ B_L & H_{Q,L} \end{pmatrix}, \quad h_q = -\partial_r^2 + q(r), \quad (B_L f)(r) = b_L(r) f(r), \quad (370)$$

where

$$b_L(r) = Q_0 P_L M_{V^\circ(r)} e_0. \quad (371)$$

Then

$$\int_1^\infty \|b_L(r)\|^2 dr \leq \frac{1}{4\pi} \|V^\circ\|_{X_3}^2, \quad (372)$$

with equality whenever $P_L(V^\circ e_0) = V^\circ e_0$, as happens eventually for each buffered finite-band approximant. For $z \in \mathbb{C} \setminus \mathbb{R}$ the exact compressed resolvent is

$$P_0(H_{V,L} - z)^{-1} P_0 = [h_q - z - B_L^*(H_{Q,L} - z)^{-1} B_L]^{-1}. \quad (373)$$

Assume $\|V\|_\infty \leq M$ and choose $\kappa^2 > M + 1$. Put

$$R_q = (h_q + \kappa^2)^{-1}, \quad R_Q = (H_{Q,L} + \kappa^2)^{-1}, \quad R_{00} = P_0(H_{V,L} + \kappa^2)^{-1} P_0. \quad (374)$$

There is a constant $C_{\kappa,M}$, independent of L and of the radial support, such that

$$0 \leq R_{00} - R_q \in \mathfrak{S}_1, \quad (375)$$

$$\|R_{00} - R_q\|_{\mathfrak{S}_1} \leq C_{\kappa,M} \int_1^\infty \|b_L(r)\|^2 dr \leq \frac{C_{\kappa,M}}{4\pi} \|V^\circ\|_{X_3}^2. \quad (376)$$

Finally, if $J_{V,L}$ is the physical Jost matrix and j_q is the scalar Jost factor of h_q , then the canonical boundary densities obey

$$\frac{h_{V,L;0}(k^2)}{h_q(k^2)} = \|\mathfrak{r}_L(k)\|^2, \quad \mathfrak{r}_L(k) := j_q(k) J_{V,L}(k)^{-1} e_0. \quad (377)$$

Consequently, the $N = 1$ case of (368) is exactly the one-sided boundary estimate

$$\liminf_j \int_I \log^- \|\mathfrak{r}_{L_j}(\sqrt{E})\|^2 dE \leq C_I (1 + \|V^\circ\|_{X_3}^2). \quad (378)$$

Proof. The mean-zero identity gives $P_0 M_{V^\circ} P_0 = 0$, so the block decomposition and (373) are the ordinary Schur complement. Also

$$\|b_L(r)\|_2^2 \leq \|M_{V^\circ(r)} e_0\|_2^2 = \frac{1}{4\pi} \|V^\circ(r, \cdot)\|_2^2, \quad (379)$$

which proves (372) and its equality statement.

At $z = -\kappa^2$, both diagonal blocks in (370) are positive. The Schur complement is no larger than $h_q + \kappa^2$, hence inverse monotonicity gives (375). Moreover,

$$R_{00} - R_q = R_{00} B_L^* R_Q B_L R_q. \quad (380)$$

The vector-valued heat semigroup is dominated in pointwise norm by e^{Mt} times the scalar Dirichlet heat semigroup. Therefore the diagonal kernels of R_Q^2 and R_q^2 are bounded by a constant depending only on κ and M . It follows that

$$\|R_Q B_L\|_{\mathfrak{S}_2}^2 = \int_1^\infty \langle b_L(r), R_Q^2(r, r) b_L(r) \rangle dr \leq C_{\kappa, M} \int \|b_L(r)\|^2 dr, \quad (381)$$

$$\|B_L R_q\|_{\mathfrak{S}_2}^2 = \int_1^\infty R_q^2(r, r) \|b_L(r)\|^2 dr \leq C_{\kappa, M} \int \|b_L(r)\|^2 dr. \quad (382)$$

Thus $B_L^* R_Q B_L R_q = (R_Q B_L)^* (B_L R_q)$ is trace class with the claimed charged bound. Since $\|R_{00}\| \leq (\kappa^2 - M)^{-1}$, (380) proves (376).

Finally, (13) and its scalar q analogue give

$$h_{V, L; 0}(k^2) = \frac{k}{\pi} \|J_{V, L}(k)^{-1} e_0\|^2, \quad h_q(k^2) = \frac{k}{\pi |j_q(k)|^2}, \quad (383)$$

which proves (377) and the final exact reformulation. $\square$

Proposition E.17 (The star-relative bound-state divisor has the favorable sign). *Assume first that the finite-model coefficients are smooth and compactly supported. In the star decomposition (370), write*

$$H_{\star, L} = \begin{pmatrix} h_q & 0 \\ 0 & H_{Q, L} \end{pmatrix}, \quad \mathbf{K}_L = \begin{pmatrix} 0 & B_L^* \\ B_L & 0 \end{pmatrix}, \quad H_{V, L} = H_{\star, L} + \mathbf{K}_L. \quad (384)$$

Let $\mathbf{S}_{V, L}$ and $\mathbf{S}_{\star, L}$ be the finite matrix transmission matrices relative to the same flattened reference, and define

$$\Delta_{3/2, L} := \sum_a |\lambda_a(H_{V, L})|^{3/2} - \sum_a |\lambda_a(H_{\star, L})|^{3/2}. \quad (385)$$

Then

$$\Delta_{3/2, L} \geq 0 \quad (386)$$

and the signed second trace identity has the dimension-free upper bound

$$\begin{aligned} \frac{1}{\pi} \int_0^\infty E^{1/2} \log \left| \frac{\det \mathbf{S}_{V, L}(E)}{\det \mathbf{S}_{\star, L}(E)} \right| dE &= 2 \int_1^\infty \|b_L(r)\|^2 dr - \frac{2}{3} \Delta_{3/2, L} \\ &\leq 2 \int_1^\infty \|b_L(r)\|^2 dr \leq \frac{1}{2\pi} \|V^\circ\|_{X_3}^2. \end{aligned} \quad (387)$$

Thus neither the uncharged high-high potential nor the self-adjoint negative spectrum causes the remaining one-sided $N = 1$ gap.

Proof. Let $\mathcal{Z} = P_0 - Q_0$ on the angular channel space. It commutes with $H_{*,L}$ and changes the sign of K_L , so

$$\mathcal{Z}(H_{*,L} + K_L)\mathcal{Z} = H_{*,L} - K_L. \quad (388)$$

The trace functional $A \mapsto \text{Tr}(-A)_+^{3/2}$ is convex. Since $H_{*,L}$ is the midpoint of the two unitarily equivalent operators $H_{*,L} \pm K_L$, convexity gives (386). The assertion follows first on a finite radial interval and then by monotone exhaustion; the compact support matrix Lieb–Thirring bound makes all displayed moments finite.

At the potential level the star background is block diagonal and K_L is block off-diagonal. Hence every cross trace vanishes and

$$\text{tr}(U_{V,L}^2 - U_{*,L}^2) = \text{tr} K_L^2 = 2\|b_L(r)\|^2. \quad (389)$$

Apply the same finite-matrix second trace identity used in Proposition E.13 to the two operators and subtract. This proves the equality in (387); its inequalities follow from (386) and (372). $\square$

Remark E.18 (Why the favorable divisor does not yet prove the entropy bound). Proposition E.17 removes a tempting but false diagnosis: the uncontrolled term is not the relative self-adjoint negative spectrum. The scalar Feshbach transmission coefficient is a Schur quotient of the hybrid Jost matrix. By the Jacobi identity its denominator is one chosen high-channel minor. Zeros of that minor create the artificial meromorphic pole divisor already warned about after (84). In a second trace formula its complex poles carry cubic terms with no fixed sign; this is the destructive term in the Feshbach–Szegő route of Laptev–Naboko–Safronov [8].

The pole-free Plücker norm, equivalently the full radiation column $\|J_{V,L}^{-1}e_0\|$, sums all complementary minors and retains the beneficial denominator in (89). But its positive boundary part does not follow from the signed determinant bound (387). The reciprocal identity (409) is precisely a pole-free real-axis replacement: it identifies the missing term as positive state-correlated refocusing work rather than as an unbounded negative-eigenvalue contribution.

Remark E.19 (The boundary-value bottleneck). The trace-class statement (376) is genuinely dimension-free and deletes the whole uncharged high–high block. It is still an off-axis estimate. Trace-class control at one negative spectral point does not by itself bound the negative logarithm of the boundary density, and the general Hilbert–Schmidt perturbation determinant can even lack nontangential boundary values [12]. The remaining theorem is now precise: use the local radial Schrödinger structure and centrifugal ordering to upgrade (376) to (378). No determinant averaged over the high block is required for $N = 1$.

There is an exact real-axis factorization of the remaining boundary quantity. It removes every forward phase, including the long weak phases which destroyed the ungauged Hardy normalization, and leaves one directional reflection coefficient.

Proposition E.20 (Exact radial-mean two-port gauge and directional reflection). *Assume first that the finite-model coefficients are smooth and compactly supported. Let $F_{q,L}(r, k)$ be the diagonal outgoing Jost matrix for $H_{q,L}$ and let $F_{V,L}(r, k)$ be the outgoing Jost matrix for $H_{V,L}$, both normalized to have Wronskian $2ikI$. For real $k > 0$, define coefficient matrices $T_L(r, k)$ and $R_L(r, k)$ by*

$$\begin{pmatrix} F_{V,L} \\ F'_{V,L} \end{pmatrix} = \begin{pmatrix} F_{q,L} & \overline{F_{q,L}} \\ F'_{q,L} & \overline{F'_{q,L}} \end{pmatrix} \begin{pmatrix} T_L \\ R_L \end{pmatrix}, \quad (390)$$

with terminal data $T_L = I$, $R_L = 0$ beyond the support. Then

$$T_L^* T_L - R_L^* R_L = I \quad (391)$$

at every radius.

Put $F = F_{q,L}$ and $W = W_L^\circ$. The coefficient evolution is exactly

$$\frac{d}{dr} \begin{pmatrix} \mathbf{T}_L \\ \mathbf{R}_L \end{pmatrix} = \frac{1}{2ik} \begin{pmatrix} \overline{F}WF & \overline{F}W\overline{F} \\ -FWF & -FW\overline{F} \end{pmatrix} \begin{pmatrix} \mathbf{T}_L \\ \mathbf{R}_L \end{pmatrix}. \quad (392)$$

On the real axis the two diagonal blocks in (392) are anti-Hermitian. Consequently, if

$$U'_+ = \frac{1}{2ik} \overline{F}WFU_+, \quad U'_- = -\frac{1}{2ik} FW\overline{F}U_-, \quad U_+(R) = U_-(R) = \mathbf{I}, \quad (393)$$

$$\tilde{\mathbf{T}}_L = U_+^{-1} \mathbf{T}_L, \quad \tilde{\mathbf{R}}_L = U_-^{-1} \mathbf{R}_L, \quad (394)$$

then $U_\pm$ are unitary and

$$\frac{d}{dr} \begin{pmatrix} \tilde{\mathbf{T}}_L \\ \tilde{\mathbf{R}}_L \end{pmatrix} = \begin{pmatrix} 0 & \mathbf{C}_L \\ \mathbf{C}_L^* & 0 \end{pmatrix} \begin{pmatrix} \tilde{\mathbf{T}}_L \\ \tilde{\mathbf{R}}_L \end{pmatrix}, \quad \mathbf{C}_L = \frac{1}{2ik} U_+^{-1} \overline{F}WFU_-. \quad (395)$$

Thus all real-axis norm growth is carried by the reflection-only off-diagonal system.

At $r = 1$, set

$$\Theta_{q,L}(k) = J_{q,L}(k)^{-1} \overline{J_{q,L}(k)}, \quad J_{q,L}(k) = F_{q,L}(1, k). \quad (396)$$

The matrix $\Theta_{q,L}$ is diagonal unitary. With

$$x_L(k) = (\mathbf{T}_L(k) + \Theta_{q,L}(k) \mathbf{R}_L(k))^{-1} e_0 = j_q(k) J_{V,L}(k)^{-1} e_0, \quad (397)$$

$$\xi_L(k) = \frac{x_L(k)}{\|x_L(k)\|}, \quad y_L(k) = \|\mathbf{R}_L(k) \xi_L(k)\|, \quad (398)$$

one has the pointwise sharp estimate

$$\|x_L(k)\|^{-1} \leq \sqrt{1 + y_L(k)^2} + y_L(k), \quad (399)$$

$$\log^- \|x_L(k)\|^2 \leq 2 \operatorname{arsinh} y_L(k). \quad (400)$$

Moreover $y_L = \|\tilde{\mathbf{R}}_L \xi_L\|$, so the right-hand side is unchanged by the exact forward gauges.

Proof. Variation of constants in the reference basis gives (392); the Wronskian of F and $\overline{F}$ is $-2ik\mathbf{I}$ with the displayed convention. Conservation of the full outgoing flux and the vanishing mixed reference Wronskians give (391). Since $W = W^*$ and F is diagonal, $\overline{F}WF$ and $FW\overline{F}$ are Hermitian. This proves unitarity of $U_\pm$. Direct differentiation gives (395); its lower-left block is the adjoint of its upper-right block on the real axis.

At the boundary,

$$J_{V,L} = J_{q,L} \mathbf{T}_L + \overline{J_{q,L}} \mathbf{R}_L = J_{q,L} (\mathbf{T}_L + \Theta_{q,L} \mathbf{R}_L). \quad (401)$$

The reference is a direct sum of scalar channels, hence every diagonal entry of $\Theta_{q,L}$ has modulus one. Since $J_{q,L} e_0 = j_q e_0$, this proves (397). Now

$$\begin{aligned} 1 &= \|(\mathbf{T}_L + \Theta_{q,L} \mathbf{R}_L) x_L\| \\ &\leq \|x_L\| (\|\mathbf{T}_L \xi_L\| + \|\mathbf{R}_L \xi_L\|) = \|x_L\| \left(\sqrt{1 + y_L^2} + y_L \right), \end{aligned} \quad (402)$$

where the last equality is (391). This proves (399); take logarithms and use $\operatorname{arsinh} y = \log(\sqrt{1 + y^2} + y)$ to obtain (400). Finally U_- is unitary, so it does not change y_L . $\square$

Proposition E.21 (Exact reciprocal-refocusing identity). *In the setting of Proposition E.20, define at every radius*

$$\Theta_{q,L}(r, k) = F_{q,L}(r, k)^{-1} \overline{F_{q,L}(r, k)}, \quad \mathbf{A}_L(r, k) = \mathbf{T}_L(r, k) + \Theta_{q,L}(r, k) \mathbf{R}_L(r, k). \quad (403)$$

Then $\mathbf{A}_L = F_{q,L}^{-1} F_{V,L}$ and, on the real axis,

$$\Theta'_{q,L}(r, k) = -2ik F_{q,L}(r, k)^{-2}, \quad (404)$$

$$\mathbf{A}'_L(r, k) = \Theta'_{q,L}(r, k) \mathbf{R}_L(r, k), \quad \mathbf{R}'_L(r, k) = -\frac{1}{2ik} F_{q,L} W_L^\circ F_{q,L} \mathbf{A}_L. \quad (405)$$

Thus a large coefficient reflection changes the physical radiation column only after multiplication by the reciprocal phase velocity $\Theta'_{q,L}$. The flux law also becomes the exact interference identity

$$\mathbf{A}_L^* \mathbf{A}_L - \mathbf{I} = \mathbf{A}_L^* \Theta_{q,L} \mathbf{R}_L + \mathbf{R}_L^* \Theta_{q,L}^* \mathbf{A}_L. \quad (406)$$

In particular, the quadratic size $\mathbf{R}_L^* \mathbf{R}_L$ cancels: only its phase-correlated interference with the physical column can change the density.

Fix the boundary-selected unit vector $\xi_L(k)$ from (398), and put

$$a_L(r, k) = \mathbf{A}_L(r, k) \xi_L(k), \quad b_L(r, k) = \mathbf{R}_L(r, k) \xi_L(k). \quad (407)$$

If R lies beyond the support, then $a_L(R, k) = \xi_L(k)$, $b_L(R, k) = 0$, and the exact state-correlated reciprocal work

$$\mathfrak{W}_L^{\text{ref}}(k) := - \int_1^R \Re \frac{\langle a_L(r, k), \Theta'_{q,L}(r, k) b_L(r, k) \rangle}{\|a_L(r, k)\|^2} dr \quad (408)$$

satisfies

$$\mathfrak{W}_L^{\text{ref}}(k) = \log \|\mathbf{A}_L(1, k) \xi_L(k)\| = -\log \|x_L(k)\|, \quad \log^- \|x_L(k)\|^2 = 2[\mathfrak{W}_L^{\text{ref}}(k)]_+. \quad (409)$$

There is also the exact vector refocusing formula

$$\frac{e_0}{\|x_L(k)\|} = \xi_L(k) - \int_1^R \Theta'_{q,L}(r, k) \mathbf{R}_L(r, k) \xi_L(k) dr. \quad (410)$$

Consequently, with

$$\mathfrak{Q}_L(k) := \int_1^R \|\Theta'_{q,L}(r, k) \mathbf{R}_L(r, k) \xi_L(k)\| dr, \quad (411)$$

one has the phase-sensitive improvement

$$\log^- \|x_L(k)\|^2 \leq 2 \log(1 + \mathfrak{Q}_L(k)). \quad (412)$$

Proof. Since $F = F_{q,L}$ is diagonal, its entries never vanish on the positive real axis: a zero would contradict the nonzero Wronskian of $f_{q,n}$ and $\overline{f_{q,n}}$. The identity $F \mathbf{A}_L = F_{V,L}$ follows directly from (390). The scalar Wronskian convention $f_{q,n} \overline{f_{q,n}'} - f_{q,n}' \overline{f_{q,n}} = -2ik$ gives (404). Differentiate $\mathbf{A}_L = \mathbf{T}_L + \Theta_{q,L} \mathbf{R}_L$ and use (392). The two terms multiplying $\mathbf{T}_L$ cancel, as do the two terms multiplying $\mathbf{R}_L$; only $\Theta'_{q,L} \mathbf{R}_L$ remains. The lower row of (392), together with $F \mathbf{A}_L = F \mathbf{T}_L + \overline{F} \mathbf{R}_L$, gives the second equation in (405).

Substitute $\mathbf{T}_L = \mathbf{A}_L - \Theta_{q,L} \mathbf{R}_L$ into (391). Since $\Theta_{q,L}$ is unitary, the two copies of $\mathbf{R}_L^* \mathbf{R}_L$ cancel, which proves (406).

The vector $a_L(r, k)$ cannot vanish: otherwise $F_{V,L}(r, k) \xi_L = 0$, which would make its outgoing flux zero at that radius, contradicting the positive outgoing flux of the nonzero coefficient vector ξ_L . Hence

$$\partial_r \log \|a_L(r, k)\| = \Re \frac{\langle a_L(r, k), \Theta'_{q,L}(r, k) b_L(r, k) \rangle}{\|a_L(r, k)\|^2}. \quad (413)$$

Integrate from 1 to R . The terminal norm is one, while (397) gives $a_L(1, k) = e_0 / \|x_L(k)\|$. This proves (409). Integrating $\mathbf{A}'_L = \Theta'_{q,L} \mathbf{R}_L$ and using $\mathbf{A}_L(R) = \mathbf{I}$ gives (410). Its triangle inequality and the first equality in (409) prove (412). $\square$

Proposition E.22 (Bessel balancing recovers the scalar charged generator). *Keep the notation of Proposition E.21 and put*

$$\Omega_{q,L}(r, k) = 2k(F_{q,L}(r, k)^* F_{q,L}(r, k))^{-1}, \quad \mathcal{U}_{q,L}(r, k) = F_{q,L}(r, k)(F_{q,L}^* F_{q,L})^{-1/2}. \quad (414)$$

Thus $\Omega_{q,L} > 0$ is diagonal and $\mathcal{U}_{q,L}$ is diagonal unitary. Define

$$\mathbf{C}_L^{\text{ph}} = \Theta_{q,L} \mathbf{R}_L, \quad \mathbf{H}_L = \frac{1}{2k} \overline{F_{q,L}} W_L^\circ F_{q,L}. \quad (415)$$

Then $\mathbf{H}_L = \mathbf{H}_L^*$ and the exact refocusing system becomes

$$\mathbf{A}'_L = -i\Omega_{q,L} \mathbf{C}_L^{\text{ph}}, \quad (\mathbf{C}_L^{\text{ph}})' = -i\Omega_{q,L} \mathbf{C}_L^{\text{ph}} + i\mathbf{H}_L \mathbf{A}_L, \quad (416)$$

$$\mathbf{A}_L(R) = \mathbf{I}, \quad \mathbf{C}_L^{\text{ph}}(R) = 0. \quad (417)$$

Most importantly, its geometrically balanced interaction is exactly

$$\boxed{\Omega_{q,L}^{1/2} \mathbf{H}_L \Omega_{q,L}^{1/2} = \mathcal{U}_{q,L}^* W_L^\circ \mathcal{U}_{q,L}.} \quad (418)$$

Consequently every unitarily invariant norm is preserved. Since $\mathcal{U}_{q,L}$ commutes with every angular-degree projection, for every fixed $N \leq L$,

$$\|\Omega_{q,L}^{1/2} \mathbf{H}_L \Omega_{q,L}^{1/2} P_N\|_{\text{HS}} = \|W_L^\circ P_N\|_{\text{HS}}, \quad (419)$$

and in particular

$$\|\Omega_{q,L}^{1/2} \mathbf{H}_L \Omega_{q,L}^{1/2} e_0\|^2 = \|W_L^\circ e_0\|^2 = \frac{1}{4\pi} \|P_L V^\circ(r, \cdot)\|_2^2. \quad (420)$$

If $V^\circ(r, \cdot)$ has angular bandwidth at most B , the balanced generator has the same degree-coupling bandwidth: its matrix element between degrees n and m vanishes whenever $|n - m| > B$.

Proof. Equation (404) gives $\Theta'_{q,L} \Theta_{q,L}^* = -i\Omega_{q,L}$. Multiply the second equation in (405) by $\Theta_{q,L}$ and differentiate the product $\Theta_{q,L} \mathbf{R}_L$ to obtain (416). Diagonality of $F_{q,L}$ and self-adjointness of W_L° give $\mathbf{H}_L^* = \mathbf{H}_L$. Finally,

$$\begin{aligned} \Omega_{q,L}^{1/2} \mathbf{H}_L \Omega_{q,L}^{1/2} &= (F_{q,L}^* F_{q,L})^{-1/2} F_{q,L}^* W_L^\circ F_{q,L} (F_{q,L}^* F_{q,L})^{-1/2} \\ &= \mathcal{U}_{q,L}^* W_L^\circ \mathcal{U}_{q,L}, \end{aligned} \quad (421)$$

which proves (418). The phase matrix is constant on each spherical-harmonic eigenspace of a fixed degree. It therefore commutes with P_N and only changes phases of the matrix elements of W_L° . This proves (419), preserves the degree-coupling zero pattern, and gives (420) because $W_L^\circ e_0 = (4\pi)^{-1/2} P_L V^\circ$. $\square$

Proposition E.23 (Signed pole-free work is sufficient for the scalar endpoint). *Fix $\kappa > 0$ and put*

$$\varpi_\kappa(E) = \frac{\sqrt{E}}{(E + \kappa^2)^3}, \quad E > 0. \quad (422)$$

For a finite radial-mean model define

$$u_L(E) = \log \|\mathbf{r}_L(\sqrt{E})\|^2, \quad \mathfrak{S}_{\kappa,L} = - \int_0^\infty \varpi_\kappa(E) u_L(E) \, dE. \quad (423)$$

Assume $\|V\|_\infty \leq M$ and take $\kappa^2 > M + 1$. Along any double-buffered fixed-potential diagonal, the following statements hold. At stage j , the symbols q and h_q below abbreviate the truncated radial mean $q_j = \mathbf{1}_{[1, R_j]} q$ and its density; in particular $\|q_j\|_2 \leq \|q\|_2$, so every displayed scalar bound is uniform in j .

First, density amplification has the uniform bound

$$\sup_j \int_0^\infty \varpi_\kappa(E) u_{L_j}(E)^+ dE \leq C_{\kappa,M} + C_\kappa(1 + \|q\|_2^2). \quad (424)$$

Second, if the signed pole-free estimate

$$\liminf_{j \rightarrow \infty} \mathfrak{S}_{\kappa,L_j} \leq C_{\kappa,M} \left(1 + \|V^\circ\|_{X_3}^2\right) \quad (425)$$

holds, then for every $I \Subset (0, \infty)$,

$$\liminf_{j \rightarrow \infty} \int_I \log^- \|\mathfrak{r}_{L_j}(\sqrt{E})\|^2 dE \leq C_{I,\kappa,M} \left(1 + \|V\|_{X_3}^2\right). \quad (426)$$

Consequently (425), which is a signed estimate, is sufficient for the $N = 1$ fixed-potential entropy theorem and hence for rank at least one in the limiting exterior density.

In the notation of Proposition E.21,

$$\mathfrak{S}_{\kappa,L} = 2 \int_0^\infty \varpi_\kappa(E) \mathfrak{M}_L^{\text{ref}}(\sqrt{E}) dE. \quad (427)$$

Thus the remaining scalar endpoint does not require a direct estimate of $[\mathfrak{M}_L^{\text{ref}}]_+$: its negative part is already paid for by the ordinary spectral-mass budget.

Proof. The density identity (377) gives

$$e^{u_L(E)} = \frac{h_{V,L;0}(E)}{h_q(E)}. \quad (428)$$

For arbitrary positive numbers a, b ,

$$\log^+(a/b) \leq a + \log^- b. \quad (429)$$

Hence

$$u_L(E)^+ \leq h_{V,L;0}(E) + \log^- h_q(E). \quad (430)$$

The harmless choice of a fixed positive normalization for both densities only changes the right-hand side by an integrable function depending on κ .

Let $\rho_{V,L;0}$ be the constant-boundary-channel spectral measure. The one-dimensional Dirichlet heat-kernel domination estimate, uniform in the channel dimension because $D_L/r^2 \geq 0$ and $\|V\|_\infty \leq M$, gives

$$\int_{\mathbb{R}} \frac{d\rho_{V,L;0}(E)}{(E + \kappa^2)^2} \leq C_{\kappa,M}. \quad (431)$$

For smooth coefficients this follows by taking both inward boundary derivatives in

$$\|e^{-tH_{V,L}}(r, s)\| \leq e^{Mt} e_D^{t\partial_r^2}(r, s) \quad (432)$$

and using $\partial_r \partial_s e_D^{t\partial_r^2}(1, 1) \asymp t^{-3/2}$ in the Laplace representation of $(H_{V,L} + \kappa^2)^{-2}$; approximation gives the stated bounded-potential version. Since $\varpi_\kappa(E) \leq C_\kappa(E + \kappa^2)^{-2}$ for $E \geq 0$, this proves

$$\sup_L \int_0^\infty \varpi_\kappa(E) h_{V,L;0}(E) dE \leq C_{\kappa,M}. \quad (433)$$

The scalar L^2 sum rule for h_q , in its rationally localized form, gives

$$\int_0^\infty \varpi_\kappa(E) \log^- h_q(E) dE \leq C_\kappa(1 + \|q\|_2^2). \quad (434)$$

Equations (430)–(434) prove (424).

Now $u_L^- = u_L^+ - u_L$. If $c_{I,\kappa} = \min_{E \in I} \varpi_\kappa(E) > 0$, then

$$\begin{aligned} c_{I,\kappa} \int_I u_L(E)^- dE &\leq \int_0^\infty \varpi_\kappa(E) u_L(E)^- dE \\ &= \int_0^\infty \varpi_\kappa(E) u_L(E)^+ dE + \mathfrak{S}_{\kappa,L}. \end{aligned} \quad (435)$$

Take a subsequence realizing the liminf in (425), use (424), and recall $4\pi\|q\|_2^2 + \|V^\circ\|_{X_3}^2 = \|V\|_{X_3}^2$. This proves (426). Finally, (409) gives $u_L = -2\mathfrak{W}_L^{\text{ref}}$, which is (427). $\square$

Proposition E.24 (A fixed forward row majorizes the signed selected loss). *Let $J_{\star,L}(k)$ be the Jost matrix of the block-diagonal star reference $H_{\star,L} = h_q \oplus H_{Q,L}$ from (384), and put*

$$\mathbf{M}_{\star,L}(k) = J_{\star,L}(k)^{-1} J_{V,L}(k). \quad (436)$$

Because $J_{\star,L}e_0 = j_q e_0$, for real $k > 0$,

$$x = \mathbf{M}_{\star,L}^{-1} e_0 = \mathfrak{r}_L(k). \quad (437)$$

Equivalently, if $h_{V,L}(k^2)$ is the full canonical boundary-density matrix, then the star-normalized relative density is

$$\mathbf{W}_{\star,L}^{\text{rel}}(k^2) := \frac{\pi}{k} J_{\star,L}(k)^* h_{V,L}(k^2) J_{\star,L}(k) = \mathbf{M}_{\star,L}^{-*} \mathbf{M}_{\star,L}^{-1}, \quad (438)$$

and

$$\langle e_0, \mathbf{W}_{\star,L}^{\text{rel}} e_0 \rangle = \|\mathfrak{r}_L(k)\|^2. \quad (439)$$

Then

$$-\log \|x\|^2 \leq \langle e_0, \log(\mathbf{M}_{\star,L} \mathbf{M}_{\star,L}^*) e_0 \rangle \leq \log \|\mathbf{M}_{\star,L}^* e_0\|^2. \quad (440)$$

Consequently the fixed-diagonal logarithmic estimate

$$\liminf_{j \rightarrow \infty} \int_0^\infty \varpi_\kappa(E) \langle e_0, \log(\mathbf{M}_{\star,L_j}(\sqrt{E}) \mathbf{M}_{\star,L_j}(\sqrt{E})^*) e_0 \rangle dE \leq C_{\kappa,M} (1 + \|V^\circ\|_{X_3}^2) \quad (441)$$

implies (425) and hence the $N = 1$ fixed-potential entropy theorem. Either of the successively stronger bounds obtained by replacing the integrand in (441) by

$$\log \|\mathbf{M}_{\star,L_j}^* e_0\|^2 \quad \text{or} \quad \log^+ \|\mathbf{M}_{\star,L_j}^* e_0\|^2 \quad (442)$$

is also sufficient. The first estimate is the preferred forward factorization: the coefficient difference between $H_{V,L}$ and $H_{\star,L}$ is only $\mathbf{K}_L$, and

$$\int_1^\infty \text{tr } \mathbf{K}_L(r)^2 dr = 2 \int_1^\infty \|b_L(r)\|^2 dr \leq \frac{1}{2\pi} \|V^\circ\|_{X_3}^2. \quad (443)$$

Thus the unresolved strengthening of Proposition E.17 is a fixed diagonal entry of the noncommutative logarithm, not another determinant estimate. In the relative-density notation it is

$$\langle e_0, \log(\mathbf{M}_{\star,L} \mathbf{M}_{\star,L}^*) e_0 \rangle = - \langle e_0, \log \mathbf{W}_{\star,L}^{\text{rel}} e_0 \rangle. \quad (444)$$

The same harmonic–arithmetic majorant also holds with the radial-mean relative matrix $\mathbf{A}_L(1, k) = J_{q,L}^{-1} J_{V,L}$ in place of $\mathbf{M}_{\star,L}$. For that choice the forward row obeys an exact fixed-state system. Put

$$p_L(r, k) = \mathbf{A}_L(r, k)^* e_0, \quad z_L(r, k) = (\mathbf{C}_L^{\text{ph}}(r, k))^* e_0, \quad \omega_0(r, k) = \langle e_0, \Omega_{q,L}(r, k) e_0 \rangle. \quad (445)$$

Then

$$p'_L = i\omega_0 z_L, \quad p_L(R, k) = e_0, \quad (446)$$

$$z'_L = i\omega_0 z_L - iA_L^* H_L e_0, \quad z_L(R, k) = 0. \quad (447)$$

If

$$G_L = \Omega_{q,L}^{1/2} H_L \Omega_{q,L}^{1/2}, \quad (448)$$

then its only forcing column has the exact representation

$$H_L e_0 = \omega_0^{-1/2} \Omega_{q,L}^{-1/2} G_L e_0, \quad \|G_L e_0\|^2 = \frac{1}{4\pi} \|P_L V^\circ(r, \cdot)\|_2^2. \quad (449)$$

Thus the radial-mean version using either row integrand in (442) is a fixed-state nonlinear Fourier estimate whose bare forcing is exactly the charged scalar column. The boundary-selected vector has disappeared. The remaining nonlinearity is the dressing of that column by the full factor A_L^* in (447); no estimate of this dressing is claimed here.

Proof. Apply the following argument first with $C = M_{*,L}$ and then, for the asserted radial-mean variant, with $C = A_L(1, k)$. Set $B = CC^* > 0$. Since

$$\|x\|^2 = \langle e_0, B^{-1} e_0 \rangle, \quad (450)$$

operator concavity of the logarithm, applied first to B^{-1} and then to B , gives

$$-\log \langle e_0, B^{-1} e_0 \rangle \leq \langle e_0, \log B e_0 \rangle \quad (451)$$

$$\leq \log \langle e_0, B e_0 \rangle = \log \|C^* e_0\|^2. \quad (452)$$

This proves (440). Integrating its upper bound against ϖ_κ proves the sufficient implication.

Take adjoints in (416). Since $\Omega_{q,L} e_0 = \omega_0 e_0$, evaluation on e_0 gives (446)–(447) and their terminal data. Finally, $\Omega_{q,L}^{-1/2} e_0 = \omega_0^{-1/2} e_0$, so the definition of G_L gives the first identity in (449); the second is (420). $\square$

Remark E.25 (The row theorem must use scalar closure). No estimate of the form (441) can hold for an arbitrary high-channel background using only the off-diagonal cost (443): the matrix rotator–barrier counterexample already disproves that abstract statement. For a scalar multiplier, however, the high block is not independent of the charged column. Identifying $b_L(r)$ with its mean-zero angular function and assuming the buffered finite-band model contains $V^\circ(r, \cdot)$, one has the exact closure relation

$$V^\circ(r, \omega) = \sqrt{4\pi} b_L(r, \omega), \quad Q_0 W_L^\circ(r) Q_0 = \sqrt{4\pi} Q_0 M_{b_L(r)} Q_0. \quad (453)$$

Thus the true remaining theorem is a scalar, cutoff-buffered *self-background* diagonal-log inequality. Treating $H_{Q,L}$ as an arbitrary uncharged bath discards exactly the structure which distinguishes Simon's conjecture from the false matrix analogue.

Proposition E.26 (The negative-energy Poisson shortcut is false). *Delete the angular kinetic term, but retain scalar multiplication on $L^2(\mathbb{S}^2)$. Fix $\kappa > 0$ and a nonzero real spherical polynomial w such that*

$$\int_{\mathbb{S}^2} w \, d\omega = 0. \quad (454)$$

There are compactly supported slabs v_n of this one fixed angular bandwidth for which

$$\int_0^\infty \|M_{v_n(r)} e_0\|_2^2 \, dr \longrightarrow 0, \quad \log \|J_{v_n}(i\kappa)^* e_0\|_2^2 \longrightarrow +\infty, \quad (455)$$

where the Jost row is normalized by the free decaying solution. On every $K \Subset (0, \infty)$ the same slabs satisfy

$$\sup_{k \in K} \log^+ \|J_{v_n}(k)^* e_0\|_2^2 \longrightarrow 0. \quad (456)$$

Moreover, one can concatenate long weak slabs to obtain a single fixed mean-zero, fixed-band scalar potential v with

$$\int_0^\infty \|M_{v(r)} e_0\|_2^2 dr < \infty \quad (457)$$

whose compact radial truncations $v^{(N)}$ obey

$$\log \|J_{v^{(N)}}(ik)^* e_0\|_2^2 \longrightarrow +\infty. \quad (458)$$

Consequently no estimate of a negative-energy forward row by the charged L^2 cost, even with an additive constant, can replace the real-axis rational second-trace estimate (441).

Proof. Work first with one slab

$$v_n(r, \omega) = \epsilon_n w(\omega) \mathbf{1}_{[0, T_n]}(r), \quad \epsilon_n = n^{-3}, \quad T_n = n^4. \quad (459)$$

At the angular point ω , put $\lambda_n(\omega) = (\kappa^2 + \epsilon_n w(\omega))^{1/2}$. For all sufficiently large n , the free-normalized value at the left end of the solution which equals $e^{-\kappa r}$ to the right of the slab is

$$G_n(\omega) = e^{-\kappa T_n} \left(\cosh(\lambda_n T_n) + \frac{\kappa}{\lambda_n} \sinh(\lambda_n T_n) \right). \quad (460)$$

Choose $a > 0$ so that $A = \{\omega : w(\omega) \geq a\}$ has positive surface measure. On A , for all large n ,

$$\lambda_n - \kappa = \frac{\epsilon_n w}{\lambda_n + \kappa} \geq \frac{a \epsilon_n}{3\kappa}, \quad G_n \geq \frac{1}{2} \exp\left(\frac{a \epsilon_n T_n}{3\kappa}\right). \quad (461)$$

Since the forward Jost operator in the decomposable model is M_{G_n} ,

$$\log \|M_{G_n} e_0\|_2^2 \geq \frac{2an}{3\kappa} + \log \frac{|A|}{16\pi}, \quad (462)$$

$$\int_0^{T_n} \|M_{v_n(r)} e_0\|_2^2 dr = \frac{\epsilon_n^2 T_n}{4\pi} \|w\|_2^2 = \frac{\|w\|_2^2}{4\pi n^2}. \quad (463)$$

This proves (455).

For real $k \in K$, let $q_n(\omega, k) = (k^2 - \epsilon_n w(\omega))^{1/2}$. Direct propagation through the constant slab gives

$$F_n(\omega, k) = e^{ikT_n} \left(\cos(q_n T_n) - i \frac{k}{q_n} \sin(q_n T_n) \right). \quad (464)$$

Uniformly on $K \times \mathbb{S}^2$,

$$|F_n|^2 = \cos^2(q_n T_n) + \frac{k^2}{q_n^2} \sin^2(q_n T_n) \leq 1 + C_{K,w} \epsilon_n. \quad (465)$$

The multiplier row therefore proves (456). Notice that the large pulse area $\epsilon_n T_n = n$ is a real-axis phase, but becomes exponential growth at $i\kappa$.

For the fixed-potential assertion, choose $\epsilon_m = \delta 2^{-m}$ and successive intervals I_m of length τ/ϵ_m , and set

$$v(r, \omega) = \epsilon_m w(\omega), \quad r \in I_m. \quad (466)$$

Then

$$\int_0^\infty \|M_{v(r)} e_0\|_2^2 dr = \frac{\tau \|w\|_2^2}{4\pi} \sum_{m=1}^\infty \epsilon_m < \infty. \quad (467)$$

It remains only to verify that the imaginary-energy growth persists through the interfaces. For a scalar nonnegative compactly supported function u , let f be its decaying Jost solution at $-\kappa^2$ and put $g(r) = e^{\kappa r} f(r)$. The Volterra equation is

$$g(r) = 1 + \int_r^\infty \frac{1 - e^{-2\kappa(s-r)}}{2\kappa} u(s) g(s) ds. \quad (468)$$

Its Picard series is positive. Partition the half-line into intervals of length $d = \kappa^{-1}$ and retain the parity class carrying at least half of $\int u$. Points chosen from distinct retained intervals are separated by d , so the Volterra kernel is at least $c_\kappa = (1 - e^{-2})/(2\kappa)$. Summing the terms which use at most one point from each retained interval gives, whenever $d\|u\|_\infty$ is bounded by a sufficiently small fixed constant,

$$g(0) \geq C_\kappa \prod_j \left(1 + c_\kappa \int_{I_{2j}} u(s) ds \right) \geq C_\kappa \exp \left(c'_\kappa \int_0^\infty u(s) ds \right). \quad (469)$$

Changing the retained parity only changes C_κ .

For $\omega \in A$, the first N slabs give a nonnegative scalar potential $u_N(r) = v^{(N)}(r, \omega)$ with

$$\int_0^\infty u_N(r) dr = \tau N w(\omega) \geq a\tau N. \quad (470)$$

Choose δ small enough for the hypothesis of (469). Integration over A now gives (458). The construction has the angular bandwidth of the single fixed polynomial w throughout. $\square$

Proposition E.27 (The transport-free scalar endpoint is true). *Delete A_3/r^2 and let $v \in L^2((0, \infty) \times \mathbb{S}^2; \mathbb{R})$ be bounded, initially with compact radial support. For a.e. ω , let $J_\omega(k)$ be the scalar half-line Jost factor for $v(\cdot, \omega)$, and put*

$$H_v(k^2) = \frac{1}{4\pi} \int_{\mathbb{S}^2} |J_\omega(k)|^{-2} d\omega. \quad (471)$$

This is exactly the free-normalized density of the constant angular vector. For every $I = [E_0, E_1] \Subset (0, \infty)$,

$$\int_I \log^- H_v(E) dE \leq 2\sqrt{E_1} |K_I| \log 2 + \frac{\sqrt{E_1}}{16E_0} \int_0^\infty \int_{\mathbb{S}^2} |v(r, \omega)|^2 d\omega dr, \quad K_I = [\sqrt{E_0}, \sqrt{E_1}]. \quad (472)$$

The constant is independent of the radial support and the angular bandwidth. Consequently the estimate extends by L^2 approximation to every bounded v in the stated space.

Proof. The function $\Phi(t) = \log^- t = \max\{0, -\log t\}$ is convex on $(0, \infty)$. Jensen's inequality applied to (471) gives, for a.e. $k > 0$,

$$\log^- H_v(k^2) \leq \frac{1}{4\pi} \int_{\mathbb{S}^2} \log^+ |J_\omega(k)|^2 d\omega. \quad (473)$$

Apply Theorem 8.5 in every angular fiber and then use Fubini. Since $k \leq \sqrt{E_1}$ and $dE = 2k dk$,

$$\begin{aligned} \int_I \log^- H_v(E) dE &\leq \frac{2\sqrt{E_1}}{4\pi} \int_{\mathbb{S}^2} \int_{K_I} \log^+ |J_\omega(k)|^2 dk d\omega \\ &\leq 2\sqrt{E_1} |K_I| \log 2 + \frac{\sqrt{E_1}}{16E_0} \int_0^\infty \int_{\mathbb{S}^2} |v(r, \omega)|^2 d\omega dr. \end{aligned} \quad (474)$$

This is (472). The scalar spectral measures converge under compactly supported L^2 approximation, and the entropy lower-semicontinuity argument in (20) passes the same bound to the limit. $\square$

Remark E.28 (The exact location of the remaining obstruction). Propositions E.26 and E.27 separate two facts which a Poisson argument conflates. Long weak scalar phases can make an imaginary-energy row arbitrarily large at negligible cost, but on the real axis the complete transport-free selected density still obeys the desired entropy estimate. Thus neither scalar multiplication, angular concentration by itself, nor the long radial support is the missing mechanism. The only term absent from Proposition E.27 is A_3/r^2 ; the remaining theorem must quantify its noncommutative angular transport together with the reciprocal return in (405).

Proposition E.29 (Exact signed centrifugal interpolation). *Fix a finite cutoff L , a compact radial support $[1, R]$, and real $k > 0$. For $0 \leq \lambda \leq 1$, replace A_3/r^2 by $\lambda A_3/r^2$ in both the physical operator and the radial-mean reference. Write*

$$F_{\lambda,L}(r, k) = \text{diag}(f_{\lambda,n}(r, k))_{0 \leq n \leq L}, \quad d_n = n(n+1), \quad (475)$$

with the usual multiplicities, and define

$$\Omega_\lambda = 2k(F_{\lambda,L}^* F_{\lambda,L})^{-1}, \quad H_\lambda = \frac{1}{2k} \overline{F_{\lambda,L}} W_L^\circ F_{\lambda,L}, \quad (476)$$

$$U_\lambda = F_{\lambda,L}(F_{\lambda,L}^* F_{\lambda,L})^{-1/2}, \quad G_\lambda = \Omega_\lambda^{1/2} H_\lambda \Omega_\lambda^{1/2} = U_\lambda^* W_L^\circ U_\lambda. \quad (477)$$

Put

$$\zeta_\lambda = F_{\lambda,L}^{-1} \partial_\lambda F_{\lambda,L}, \quad Z_\lambda = -i U_\lambda^* \partial_\lambda U_\lambda = \Im \zeta_\lambda. \quad (478)$$

Then

$$\partial_\lambda \Omega_\lambda = -2\Omega_\lambda \Re \zeta_\lambda, \quad (479)$$

$$\partial_\lambda H_\lambda = \zeta_\lambda^* H_\lambda + H_\lambda \zeta_\lambda, \quad (480)$$

$$\boxed{\partial_\lambda G_\lambda = i[G_\lambda, Z_\lambda]}, \quad Z_\lambda e_0 = 0. \quad (481)$$

Thus the Bessel-balanced scalar interaction changes only by a diagonal phase commutator; the exponentially large variation of evanescent Bessel amplitudes has disappeared.

For the exact refocusing solution, first set

$$Y_\lambda = \begin{pmatrix} \mathbf{A}_\lambda \\ \mathbf{C}_\lambda^{\text{ph}} \end{pmatrix}, \quad \mathcal{K}_\lambda = \begin{pmatrix} 0 & -i\Omega_\lambda \\ iH_\lambda & -i\Omega_\lambda \end{pmatrix}, \quad Y'_\lambda = \mathcal{K}_\lambda Y_\lambda, \quad Y_\lambda(R) = \begin{pmatrix} \mathbf{I} \\ 0 \end{pmatrix}. \quad (482)$$

Extend this system through the free tail $r > R$ and introduce the reciprocal amplitude balance

$$\hat{\mathbf{A}}_\lambda = \Omega_\lambda^{-1/2} \mathbf{A}_\lambda, \quad \hat{\mathbf{C}}_\lambda = \Omega_\lambda^{1/2} \mathbf{C}_\lambda^{\text{ph}}, \quad B_\lambda = \Omega_\lambda^{-1/2} (\Omega_\lambda^{1/2})' = \frac{1}{2} \Omega_\lambda^{-1} \Omega'_\lambda. \quad (483)$$

Then, with $\hat{Y}_\lambda = (\hat{\mathbf{A}}_\lambda, \hat{\mathbf{C}}_\lambda)^\top$,

$$\hat{Y}'_\lambda = \hat{\mathcal{K}}_\lambda \hat{Y}_\lambda, \quad \hat{\mathcal{K}}_\lambda = \begin{pmatrix} -B_\lambda & -i\mathbf{I} \\ iG_\lambda & B_\lambda - i\Omega_\lambda \end{pmatrix}, \quad \hat{Y}_\lambda(\infty) = \begin{pmatrix} (2k)^{-1/2} \mathbf{I} \\ 0 \end{pmatrix}. \quad (484)$$

The terminal vector is independent of λ , and

$$\boxed{\partial_\lambda \hat{\mathcal{K}}_\lambda = \begin{pmatrix} -\partial_\lambda B_\lambda & 0 \\ -[G_\lambda, Z_\lambda] & \partial_\lambda B_\lambda - i\partial_\lambda \Omega_\lambda \end{pmatrix}}. \quad (485)$$

Thus no derivative of the amplitude-weighted interaction H_λ remains. More precisely, the two diagonal reference terms obey

$$B_\lambda = -\Re(F_{\lambda,L}^{-1} F'_{\lambda,L}), \quad \partial_\lambda B_\lambda = -(\Re \zeta_\lambda)', \quad \partial_\lambda \Omega_\lambda = -2\Omega_\lambda \Re \zeta_\lambda. \quad (486)$$

In particular, both $\partial_\lambda B_\lambda$ and $\partial_\lambda \Omega_\lambda$ annihilate e_0 .

Let $\widehat{\mathcal{P}}_\lambda(r, s)$ be the propagator of (484) and let $\Pi_+ = (\mathbf{I}, 0)$. Differentiation in λ gives the exact Duhamel formula

$$\partial_\lambda \widehat{\mathbf{A}}_\lambda(1, k) = -\Pi_+ \int_1^R \widehat{\mathcal{P}}_\lambda(1, r)(\partial_\lambda \widehat{\mathcal{K}}_\lambda(r))\widehat{Y}_\lambda(r) dr - \Pi_+ \int_R^\infty \widehat{\mathcal{P}}_\lambda(1, r)(\partial_\lambda \widehat{\mathcal{K}}_\lambda(r))\widehat{Y}_\lambda(r) dr. \quad (487)$$

Equivalently the two integrals form one integral over $[1, \infty)$; the split only displays that the diagonal inverse-square reference tail remains present after the potential has vanished.

There is a further exact cancellation. Put

$$P_\lambda^{\text{ph}}(r, k) = e^{-ikr} U_\lambda(r, k), \quad \widetilde{Y}_\lambda = \begin{pmatrix} P_\lambda^{\text{ph}} & 0 \\ 0 & P_\lambda^{\text{ph}} \end{pmatrix} \widehat{Y}_\lambda. \quad (488)$$

Writing

$$\Gamma_\lambda = (P_\lambda^{\text{ph}})'(P_\lambda^{\text{ph}})^* = i \left(\frac{1}{2} \Omega_\lambda - k\mathbf{I} \right), \quad (489)$$

the phase-gauged system is

$$\widetilde{Y}'_\lambda = \widetilde{\mathcal{K}}_\lambda \widetilde{Y}_\lambda, \quad \widetilde{\mathcal{K}}_\lambda = \begin{pmatrix} \Gamma_\lambda - B_\lambda & -i\mathbf{I} \\ iW_L^\circ & \Gamma_\lambda + B_\lambda - i\Omega_\lambda \end{pmatrix}, \quad \widetilde{Y}_\lambda(\infty) = \begin{pmatrix} (2k)^{-1/2}\mathbf{I} \\ 0 \end{pmatrix}. \quad (490)$$

In particular the interaction W_L° is now independent of λ . The Wronskian identity gives

$$(\Im \zeta_\lambda)' = \frac{1}{2} \partial_\lambda \Omega_\lambda = -\Omega_\lambda \Re \zeta_\lambda. \quad (491)$$

Together with (486), this proves the pure radial-gauge formula

$$\boxed{\partial_\lambda \widetilde{\mathcal{K}}_\lambda = \partial_r \mathcal{Q}_\lambda, \quad \mathcal{Q}_\lambda = \begin{pmatrix} \zeta_\lambda & 0 \\ 0 & -\zeta_\lambda \end{pmatrix}.} \quad (492)$$

Let $\widetilde{\mathcal{P}}_\lambda(r, s)$ be the propagator of (490). Since $\mathcal{Q}_\lambda(\infty) = 0$, integration by parts in the exact Duhamel formula yields

$$\partial_\lambda \widetilde{Y}_\lambda(1) = \mathcal{Q}_\lambda(1) \widetilde{Y}_\lambda(1) + \int_1^\infty \widetilde{\mathcal{P}}_\lambda(1, r) [\mathcal{Q}_\lambda(r), \widetilde{\mathcal{K}}_\lambda(r)] \widetilde{Y}_\lambda(r) dr, \quad (493)$$

$$[\mathcal{Q}_\lambda, \widetilde{\mathcal{K}}_\lambda] = \begin{pmatrix} 0 & -2i\zeta_\lambda \\ -i(\zeta_\lambda W_L^\circ + W_L^\circ \zeta_\lambda) & 0 \end{pmatrix}. \quad (494)$$

Thus all bare centrifugal variation is a removable endpoint gauge. The only interior term is the charged off-diagonal curvature in (494); it records conversion and reciprocal return in the same integral. If

$$x_\lambda(k) = \mathbf{A}_\lambda(1, k)^{-1} e_0, \quad \ell_\lambda(k) = -\log \|x_\lambda(k)\|^2, \quad (495)$$

and $\widehat{x}_\lambda = \widehat{\mathbf{A}}_\lambda(1, k)^{-1} e_0$, then

$$x_\lambda = \omega_0(1, k)^{-1/2} \widehat{x}_\lambda, \quad \ell_\lambda(k) = \log \omega_0(1, k) - \log \|\widehat{x}_\lambda(k)\|^2, \quad (496)$$

where $\Omega_\lambda e_0 = \omega_0 e_0$ and ω_0 is independent of λ . If also

$$\widetilde{x}_\lambda = \widetilde{\mathbf{A}}_\lambda(1, k)^{-1} e_0, \quad \mathcal{R}_\lambda(k) = \Pi_+ \int_1^\infty \widetilde{\mathcal{P}}_\lambda(1, r) [\mathcal{Q}_\lambda(r), \widetilde{\mathcal{K}}_\lambda(r)] \widetilde{Y}_\lambda(r) dr, \quad (497)$$

then $\|\widetilde{x}_\lambda\| = \|\widehat{x}_\lambda\|$. Moreover the endpoint term in (493) contributes zero to the selected logarithmic derivative, because $\zeta_\lambda e_0 = 0$ and $\widetilde{\mathbf{A}}_\lambda \widetilde{x}_\lambda = e_0$. Hence, if

$$\mathfrak{S}_{\kappa, L}^{(\lambda)} := \int_0^\infty \varpi_\kappa(E) \ell_\lambda(\sqrt{E}) dE. \quad (498)$$

then

$$\partial_\lambda \ell_\lambda(k) = 2\Re \frac{\langle \tilde{x}_\lambda, \tilde{\mathbf{A}}_\lambda(1, k)^{-1} \mathcal{R}_\lambda(k) \tilde{x}_\lambda \rangle}{\|\tilde{x}_\lambda\|^2}. \quad (499)$$

Equivalently, define the right dressed trajectory and the normalized left refocusing trajectory by

$$\begin{pmatrix} a_\lambda(r, k) \\ c_\lambda(r, k) \end{pmatrix} := \tilde{Y}_\lambda(r, k) \tilde{x}_\lambda(k), \quad (500)$$

$$\begin{pmatrix} p_\lambda(r, k) \\ q_\lambda(r, k) \end{pmatrix} := \frac{1}{\|\tilde{x}_\lambda(k)\|^2} \tilde{\mathcal{P}}_\lambda(1, r)^* \Pi_+^* \tilde{\mathbf{A}}_\lambda(1, k)^{-*} \tilde{x}_\lambda(k). \quad (501)$$

They solve the reciprocal equations $v'_\lambda = \tilde{\mathcal{K}}_\lambda v_\lambda$ and $w'_\lambda = -\tilde{\mathcal{K}}_\lambda^* w_\lambda$. Formula (499) is therefore exactly

$$\partial_\lambda \ell_\lambda(k) = 2\Re \int_1^\infty [-2i \langle p_\lambda, \zeta_\lambda c_\lambda \rangle - i \langle q_\lambda, (\zeta_\lambda W_L^\circ + W_L^\circ \zeta_\lambda) a_\lambda \rangle] dr. \quad (502)$$

No undifferentiated centrifugal term or derivative of W_L° occurs. The first integrand is charged through the return state c_λ , which vanishes when $W_L^\circ = 0$; the second displays the charged scalar multiplier explicitly. Both must remain paired with the reciprocal left trajectory. In particular, for the rational weight in (422),

$$\mathfrak{S}_{\kappa, L}^{(1)} = \mathfrak{S}_{\kappa, L}^{(0)} + \int_0^1 \int_0^\infty \varpi_\kappa(E) \partial_\lambda \ell_\lambda(\sqrt{E}) dE d\lambda. \quad (503)$$

The transport-free estimate gives

$$\mathfrak{S}_{\kappa, L}^{(0)} \leq C_{\kappa, M} \left(1 + \frac{1}{4\pi} \int_1^R \|V(r, \cdot)\|_2^2 dr \right), \quad (504)$$

uniformly in L , R , and the angular bandwidth. Consequently the sole remaining $N = 1$ estimate is the signed centrifugal interpolation condition SCI_1 :

$$\liminf_{j \rightarrow \infty} \int_0^1 \int_0^\infty \varpi_\kappa(E) \partial_\lambda \ell_{\lambda, L_j}(\sqrt{E}) dE d\lambda \leq C_{\kappa, M} (1 + \|V\|_{X_3}^2). \quad (505)$$

Along a double-buffered fixed-potential diagonal, (505) implies (425), and hence the fixed-coupling $N = 1$ spectral-density lower bound.

Proof. All diagonal entries of $F_{\lambda, L}$ are nonzero on the positive real axis by their Wronskians, so the displayed derivatives are legitimate. Differentiating the first two definitions in (476) gives (479) and (480). Since $U_\lambda^* \partial_\lambda U_\lambda = iZ_\lambda$, differentiation of $G_\lambda = U_\lambda^* W_L^\circ U_\lambda$ gives (481). The degree-zero reference equation does not contain λ , hence $\zeta_\lambda e_0 = Z_\lambda e_0 = 0$.

Equation (482) is (416) with its λ dependence shown explicitly. Substitution of (483) gives (484). Because $\Omega_\lambda(r, k) \rightarrow 2k\mathbf{I}$ as $r \rightarrow \infty$ for fixed L , its terminal vector is common to all λ . Differentiating its generator and using (481) gives (485). The first identity in (486) follows from $\Omega_\lambda = 2k|F_{\lambda, L}|^{-2}$; the other two follow by differentiation. Variation of constants from infinity gives (487); convergence on the free tail follows from the inverse-square reference coefficients. Since $\mathbf{A}_\lambda = \Omega_\lambda^{1/2} \tilde{\mathbf{A}}_\lambda$ and the degree-zero entry of Ω_λ is independent of λ , we obtain (496).

For (489), write each diagonal reference solution as $f_{\lambda, n} = |f_{\lambda, n}| e^{i\theta_{\lambda, n}}$. Its outgoing Wronskian gives $\theta'_{\lambda, n} = k|f_{\lambda, n}|^{-2} = \frac{1}{2}(\Omega_\lambda)_{nn}$, proving both (489) and (491). Direct substitution then gives (490) and (492). Differentiating the terminal-value problem, integrating by parts, and using

$$\partial_r (\tilde{\mathcal{P}}_\lambda(1, r) \mathcal{Q}_\lambda(r) \tilde{Y}_\lambda(r)) = \tilde{\mathcal{P}}_\lambda(1, r) (\mathcal{Q}'_\lambda + [\mathcal{Q}_\lambda, \tilde{\mathcal{K}}_\lambda]) \tilde{Y}_\lambda(r)$$

proves (493); block multiplication proves (494). Finally, $\partial_\lambda \tilde{x}_\lambda = -\tilde{\mathbf{A}}_\lambda^{-1}(\partial_\lambda \tilde{\mathbf{A}}_\lambda)\tilde{x}_\lambda$. The top endpoint block is $\zeta_\lambda(1)\tilde{\mathbf{A}}_\lambda(1)$, and its contribution is zero because it sends $\tilde{x}_\lambda$ to $\zeta_\lambda(1)e_0 = 0$. Differentiating $-\log \|\tilde{x}_\lambda\|^2$ proves (499). Move the propagator in (497) to the first slot of the inner product and insert (494); this gives (502). Integration in λ proves (503).

At $\lambda = 0$ the reference is scalar in angle, so the operator is the direct integral used in Proposition E.27. Jensen's inequality followed by the global Deift–Killip transmission bound, now multiplied by $2k^2/(k^2 + \kappa^2)^3 \leq 2\kappa^{-6}k^2$, proves (504). Finally combine (503), (504), and Proposition E.23. $\square$

Remark E.30 (Absolute centrifugal variation is the wrong target). The commutator in (481) must remain inside the signed state-correlated integral. Indeed, already on $\mathbb{C}^2$, take

$$U_N(\lambda) = \text{diag}(1, e^{iN\lambda}), \quad W_N = \delta_N \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad G_N = U_N^* W_N U_N. \quad (506)$$

Then

$$\|G_N e_0\|^2 = \delta_N^2, \quad \int_0^1 \|\partial_\lambda G_N(\lambda) e_0\| d\lambda = N\delta_N. \quad (507)$$

With $\delta_N = N^{-1}$ the charged quadratic column tends to zero while the absolute variation stays equal to one. This is an algebraic diagnostic, not a scalar Schrödinger counterexample.

The stronger absolute-curvature failure does have a scalar realization. Take zero radial mean and let

$$p_n(\theta, \phi) = \sin^n \theta \cos(n\phi), \quad W_n^\circ(r, \omega) = \chi(r)p_n(\omega), \quad (508)$$

where $\chi = 1$ on a fixed nontrivial shell. Then p_n is a pure real harmonic of degree n , $\|p_n\|_\infty = 1$, and $\|p_n\|_2^2 \asymp n^{-1/2}$. At any fixed $k > 0$ and on that shell, the degree- n diagonal entry of ζ_λ satisfies

$$\int_0^1 |\zeta_{\lambda,n}(r, k)| d\lambda \geq |\log |f_{1,n}(r, k)| - \log |f_{0,n}(r, k)|| \gtrsim_{r,k} n \log n. \quad (509)$$

Indeed $f_{0,n} = e^{ikr}$, whereas the fixed-argument spherical Hankel asymptotic gives $|f_{1,n}(r, k)| \asymp (2n-1)!!(kr)^{-n}$. Since $\zeta_\lambda e_0 = 0$ and $W_n^\circ e_0 = (4\pi)^{-1/2} \chi p_n$, the lower-left block of (494) gives

$$\int_0^1 \int_{\text{supp } \chi} \left\| [\mathcal{Q}_\lambda, \tilde{\mathcal{K}}_\lambda] \begin{pmatrix} e_0 \\ 0 \end{pmatrix} \right\| dr d\lambda \gtrsim n^{3/4} \log n, \quad \int \|W_n^\circ e_0\|^2 dr \asymp n^{-1/2}. \quad (510)$$

Thus even for uniformly bounded compactly supported real scalar multipliers, no absolute curvature estimate at the charged scale is possible. Taking an absolute value before the signed, dressed λ and energy integrations destroys exactly the high-phase and reciprocal-refocusing cancellation which (505) must retain.

Proposition E.31 (Charged negative-energy centrifugal action). *In the finite $N = 1$ model of Proposition E.29, decompose $P_0 \oplus Q_0$ and write*

$$H_{\lambda,L} = \begin{pmatrix} h_q & B_L^* \\ B_L & H_{Q,\lambda,L} \end{pmatrix}, \quad H_{Q,\lambda,L} = H_{Q,0,L} + \lambda \frac{D_{Q,L}}{r^2}. \quad (511)$$

Assume $\|V\|_\infty \leq M$, choose $\kappa^2 > M + 1$, and put

$$R_{Q,\lambda} = (H_{Q,\lambda,L} + \kappa^2)^{-1}, \quad R_{00}^{(\lambda)} = P_0(H_{\lambda,L} + \kappa^2)^{-1}P_0, \quad R_q = (h_q + \kappa^2)^{-1}. \quad (512)$$

Then

$$R_{00}^{(\lambda)} = \left(h_q + \kappa^2 - B_L^* R_{Q,\lambda} B_L \right)^{-1} \quad (513)$$

is decreasing in λ and satisfies the exact derivative identity

$$-\partial_\lambda R_{00}^{(\lambda)} = T_\lambda^* \frac{D_{Q,L}}{r^2} T_\lambda \succeq 0, \quad T_\lambda = R_{Q,\lambda} B_L R_{00}^{(\lambda)}. \quad (514)$$

Moreover,

$$0 \preceq R_{00}^{(0)} - R_{00}^{(1)} \preceq R_{00}^{(0)} - R_q, \quad (515)$$

$$\int_0^1 \left\| \left(\frac{D_{Q,L}}{r^2} \right)^{1/2} T_\lambda \right\|_{\mathfrak{S}_2}^2 d\lambda = \text{Tr}(R_{00}^{(0)} - R_{00}^{(1)}) \leq C_{\kappa,M} \int_1^\infty \|b_L(r)\|^2 dr \leq \frac{C_{\kappa,M}}{4\pi} \|V^\circ\|_{X_3}^2. \quad (516)$$

The constants are independent of L , the angular bandwidth, and the radial support.

Proof. For finite L , $D_{Q,L}/r^2$ is bounded on the exterior half-line, so the resolvent is norm differentiable in λ and

$$\partial_\lambda R_{Q,\lambda} = -R_{Q,\lambda} \frac{D_{Q,L}}{r^2} R_{Q,\lambda}. \quad (517)$$

Differentiate the Schur complement in (513). This proves (514) and hence monotonicity. Because $B_L^* R_{Q,\lambda} B_L \succeq 0$, inverse monotonicity also gives $R_{00}^{(\lambda)} \succeq R_q$. Consequently (515) follows.

The right side of (515) is positive trace class, and Proposition E.16, applied at $\lambda = 0$, bounds its trace by the charged column. Integrating (514), taking traces, and using monotone convergence proves (516). Approximation removes the temporary smoothness needed to justify pointwise kernel manipulations. $\square$

Remark E.32 (Resolvent order alone cannot yield the entropy). Proposition E.31 is a genuine charged estimate, but it is not by itself a proof of SCI_1 . The failure is already visible for abstract scalar measures. On $[0, 1]$, let the reference density be $h_0 = 1$ and set, with $\epsilon_n = e^{-n}$,

$$h_n(E) = \begin{cases} \epsilon_n, & 0 < E < \frac{1}{2}, \\ 2 - \epsilon_n, & \frac{1}{2} < E < 1. \end{cases} \quad (518)$$

The masses agree. For every $\kappa > 0$,

$$\int_0^1 \frac{h_0(E) - h_n(E)}{E + \kappa^2} dE = (1 - \epsilon_n) \left(\int_0^{1/2} \frac{dE}{E + \kappa^2} - \int_{1/2}^1 \frac{dE}{E + \kappa^2} \right) > 0, \quad (519)$$

and the left side is uniformly bounded. The affine path $h_{n,\lambda} = (1 - \lambda)h_0 + \lambda h_n$ therefore has decreasing Stieltjes transform at every negative energy. Nevertheless,

$$\int_0^1 \log^- h_n(E) dE = \frac{n}{2}, \quad \int_0^1 \varpi_\kappa(E) \log^- h_n(E) dE \geq n \int_0^{1/2} \varpi_\kappa(E) dE \longrightarrow \infty. \quad (520)$$

Thus even monotonicity at every negative energy plus a uniformly bounded total resolvent action does not control boundary entropy for general Herglotz data. Any implication from (516) to SCI_1 must use the local radial Schrödinger evolution, not only operator order or abstract spectral-measure positivity.

Proposition E.33 (Exact angular heat-flow interpolation). *Let $A_3 = -\Delta_\omega$, write $V = q + V^\circ$ with zero angular mean V° , and define*

$$V_t(r, \omega) = q(r) + e^{-tA_3/r^2} V^\circ(r, \cdot)(\omega), \quad t \geq 0. \quad (521)$$

For smooth compact finite-band data this path is real, has the same radial support, obeys $\|V_t\|_\infty \leq \|V\|_\infty$, and satisfies the exact heat budget

$$\int_0^\infty \int_1^R \frac{1}{r^2} \|\nabla_\omega V_t(r, \cdot)\|_2^2 dr dt = \frac{1}{2} \int_1^R \|V^\circ(r, \cdot)\|_2^2 dr. \quad (522)$$

Let $F_{t,L}(r, k)$ be the outgoing Jost matrix for

$$H_{t,L} = -\partial_r^2 + \frac{D_L}{r^2} + P_L M_{V_t} P_L, \quad J_{t,L}(k) = F_{t,L}(1, k), \quad (523)$$

and let $j_q(k)$ be the degree-zero radial-mean Jost factor. Put

$$x_t(k) = J_{t,L}(k)^{-1} e_0, \quad \ell_t^h(k) = -\log(|j_q(k)|^2 \|x_t(k)\|^2). \quad (524)$$

Then $\ell_\infty^h = 0$. To state its derivative, write the phase-space system for $(F_{t,L}, F'_{t,L})^\top$ as $\mathbb{Y}'_t = \mathbb{K}_t \mathbb{Y}_t$, let $\mathbb{P}_t(r, s)$ be its propagator, and set $\Pi_1 = (I, 0)$. Define right and left reciprocal trajectories by

$$\begin{pmatrix} \alpha_t(r, k) \\ \delta_t(r, k) \end{pmatrix} = \mathbb{Y}_t(r, k) x_t(k), \quad (525)$$

$$\begin{pmatrix} \gamma_t(r, k) \\ \beta_t(r, k) \end{pmatrix} = \frac{1}{\|x_t(k)\|^2} \mathbb{P}_t(1, r)^* \Pi_1^* J_{t,L}(k)^{-*} x_t(k). \quad (526)$$

One has the exact identities

$$\partial_t \ell_t^h(k) = 2\Re \int_1^R \langle \beta_t(r, k), M_{r^{-2}A_3V_t^\circ(r)} \alpha_t(r, k) \rangle dr \quad (527)$$

$$= 2\Re \int_1^R \frac{1}{r^2} \int_{\mathbb{S}^2} \nabla_\omega V_t^\circ \cdot \nabla_\omega (\overline{\beta_t} \alpha_t) d\omega dr, \quad (528)$$

and consequently

$$\ell_0^h(k) = -2\Re \int_0^\infty \int_1^R \frac{1}{r^2} \int_{\mathbb{S}^2} \nabla_\omega V_t^\circ \cdot \nabla_\omega (\overline{\beta_t} \alpha_t) d\omega dr dt. \quad (529)$$

For the rational weight (422), define the heat-flow reciprocal Fisher action

$$\mathfrak{D}_{\kappa,L}^h := \int_0^\infty \int_0^\infty \int_1^R \frac{\varpi_\kappa(E)}{r^2} \|\nabla_\omega (\overline{\beta_t(r, \sqrt{E})} \alpha_t(r, \sqrt{E}))\|_2^2 dr dE dt. \quad (530)$$

Pointwise on the sphere,

$$\|\nabla_\omega (\overline{\beta} \alpha)\|_2^2 \leq 2 \int_{\mathbb{S}^2} (|\beta|^2 |\nabla_\omega \alpha|^2 + |\alpha|^2 |\nabla_\omega \beta|^2) d\omega. \quad (531)$$

Thus $\mathfrak{D}_{\kappa,L}^h$ is controlled by the sum of the two reciprocal density actions, with exactly the physical centrifugal weight r^{-2} and no support-radius loss. The trajectories have additional structure not visible in (531). Set

$$\mathfrak{j}_t(r, k, \omega) = \overline{\beta_t} \partial_r \alpha_t - \overline{\partial_r \beta_t} \alpha_t = \overline{\beta_t} \delta_t + \overline{\gamma_t} \alpha_t, \quad (532)$$

$$\mathfrak{J}_t(r, k, \omega) = \overline{\beta_t} \nabla_\omega \alpha_t - \alpha_t \nabla_\omega \overline{\beta_t}. \quad (533)$$

Hamiltonian duality and scalarity of V_t give the exact local conservation law

$$\partial_r \mathfrak{j}_t + \frac{1}{r^2} \operatorname{div}_\omega \mathfrak{J}_t = 0, \quad \int_{\mathbb{S}^2} \mathfrak{j}_t(r, k, \omega) d\omega = 1. \quad (534)$$

Thus the product gradient in (530) is the symmetric partner of a divergence-controlled reciprocal current; this is the compensation absent from the arbitrary H^1 diagnostic below. Since $\int_0^\infty \varpi_\kappa(E) dE = \pi/(8\kappa^3)$, Cauchy-Schwarz and (522) give

$$\left| \int_0^\infty \varpi_\kappa(E) \ell_0^h(\sqrt{E}) dE \right| \leq C_\kappa \|V^\circ\|_{X_3} (\mathfrak{D}_{\kappa,L}^h)^{1/2}. \quad (535)$$

Therefore the cutoff-compatible estimate

$$\liminf_{j \rightarrow \infty} \mathfrak{D}_{\kappa, L_j}^h \leq C_{\kappa, M} (1 + \|V^\circ\|_{X_3}^2) \quad (536)$$

along the double-buffered fixed-potential diagonal implies the $N = 1$ relative entropy bound and hence the fixed-coupling spectral-density lower bound.

Proof. The spherical heat semigroup is Markov and self-adjoint, proving the reality and L^∞ assertions. Expanding V° in nonconstant spherical harmonics gives

$$\int_0^\infty \frac{1}{r^2} \|A_3^{1/2} e^{-tA_3/r^2} V^\circ(r)\|_2^2 dt = \frac{1}{2} \|V^\circ(r)\|_2^2; \quad (537)$$

integration in r proves (522).

The terminal Jost data beyond R are independent of t . Since $\partial_t V_t^\circ = -r^{-2} A_3 V_t^\circ$, variation of constants in the phase-space system gives

$$\partial_t \mathbb{Y}_t(1) = - \int_1^R \mathbb{P}_t(1, r) \begin{pmatrix} 0 & 0 \\ -M_{r^{-2} A_3 V_t^\circ(r)} & 0 \end{pmatrix} \mathbb{Y}_t(r) dr. \quad (538)$$

Differentiate (524), move the propagator to the first slot, and use (525)–(526); this proves (527). Angular integration by parts proves (528). As $t \rightarrow \infty$, $V_t \rightarrow q$ and $J_{t,L}^{-1} e_0 = j_q^{-1} e_0$, so $\ell_\infty^h = 0$ and integration in t proves (529). Finally apply Cauchy–Schwarz in (t, E, r, ω) and use the heat budget. Estimate (536) then gives a uniform signed relative entropy bound; Proposition E.15 and the diagonal completion theorem E.9 finish the asserted implication. $\square$

Remark E.34 (Why the heat endpoint is genuinely cutoff-compatible). For each buffered approximant V_j , the whole heat path has angular bandwidth at most that of V_j , while (522) is uniform in j . Thus the degree- $2L_j$ cutoff-edge lift cannot reappear during the interpolation. A bound such as (536) is false with uniform quantifiers over arbitrary unbuffered multiplier families, because together with (535) it would imply the already false PSR_1 . Its live content is precisely the fixed-potential, bandwidth-before-cutoff diagonal order.

Remark E.35 (Separate angular-energy bounds do not control the heat action). The reciprocal relation in (526) is essential. Indeed $H^1(\mathbb{S}^2)$ is not an algebra. In a fixed polar coordinate disk, put $L_\epsilon = \log(1/\epsilon)$ and, up to harmless smooth cutoffs, let

$$f_\epsilon(r) = \begin{cases} \sqrt{L_\epsilon}, & 0 < r \leq \epsilon, \\ \frac{\log(1/r)}{\sqrt{L_\epsilon}}, & \epsilon < r < 1, \\ 0, & r \geq 1. \end{cases} \quad (539)$$

Then $\|f_\epsilon\|_2 = O(L_\epsilon^{-1/2})$ and

$$\int |\nabla f_\epsilon|^2 = 2\pi + o(1), \quad \int |\nabla(f_\epsilon^2)|^2 = \frac{8\pi}{3} L_\epsilon + O(1). \quad (540)$$

After adding a constant and normalizing, one obtains smooth $\alpha_\epsilon = \beta_\epsilon$ with bounded L^2 norms and bounded separate angular energies, but $\|\nabla(\beta_\epsilon \alpha_\epsilon)\|_2 \rightarrow \infty$. Consequently (536) cannot follow from independent centrifugal-energy bounds for α_t and β_t . A proof must use their exact right/left Schrödinger reciprocity, or establish additional integrability generated by the scalar multiplication dynamics. This functional diagnostic is not a Schrödinger counterexample to (536).

Remark E.36 (What the literature trace formulas actually supply). The established finite-dimensional matrix Szegő sum rules are formulated through log det of the matrix spectral weight, equivalently the trace of its operator logarithm; see, for example, the matrix orthogonal-polynomial survey of Damanik–Pushnitski–Simon [9]. Laptev–Naboko–Safronov instead select the scalar radial Weyl measure, but their second trace develops the complex high-minor pole contribution described above [8]. Neither result contains the fixed-vector inequality (441). Proposition E.26 also shows why that missing inequality cannot be recovered by evaluating a forward row at one point of the upper half-plane: the uncontrolled exponential type is already present in a commutative fixed-band scalar model.

Remark E.37 (What the signed reduction does and does not remove). Estimate (425) is strictly weaker than the original positive-work endpoint, but it is not a consequence of the star-relative determinant trace. At the algebraic two-port level, take

$$\mathbf{T} = \cosh \rho \mathbf{I}, \quad \mathbf{R} = \sinh \rho \operatorname{diag}(1, -1), \quad \mathbf{\Theta} = \mathbf{I}. \quad (541)$$

Then $\mathbf{T}^* \mathbf{T} - \mathbf{R}^* \mathbf{R} = \mathbf{I}$ and $\mathbf{A} = \mathbf{T} + \mathbf{\Theta} \mathbf{R} = \operatorname{diag}(e^\rho, e^{-\rho})$. Thus $\det \mathbf{A} = 1$, while for the first coordinate $x = \mathbf{A}^{-1} e_0 = e^{-\rho} e_0$ and $-\log \|x\| = \rho$. A determinant can therefore hide the selected loss by an opposite high-channel gain even under the exact flux law.

Likewise, positivity and trace control at one negative resolvent point are not enough at the level of abstract measures. On $[0, 1]$, compare the reference density 1 with

$$h_n(E) = \begin{cases} 2 - e^{-n}, & 0 < E < 1/2, \\ e^{-n}, & 1/2 < E < 1. \end{cases} \quad (542)$$

The masses are equal and, for every decreasing Stieltjes weight $(E + \kappa^2)^{-1}$, the weighted difference is positive and uniformly bounded, but $\int_0^1 \log^- h_n \, dE = n/2$. This is only an abstract diagnostic, not a Schrödinger counterexample; it proves that the local pole-free propagation structure must enter the passage from (376) to (425).

Corollary E.38 (A cutoff-compatible refocusing endpoint is sufficient). *For the double-buffered approximants, suppose that for every $I \Subset (0, \infty)$,*

$$\liminf_{j \rightarrow \infty} \int_I \log(1 + \mathfrak{Q}_{L_j}(\sqrt{E})) \, dE \leq C_I (1 + \|V^\circ\|_{X_3}^2). \quad (543)$$

Then (378) holds for $N = 1$, and the limiting exterior density has rank at least one almost everywhere on $(0, \infty)$. Equivalently, the desired relative entropy itself is exactly twice the positive energy integral of the work in (408). Proposition E.23 shows that this sufficient condition is stronger than necessary: after rational energy localization, a charged bound on the signed work already suffices.

Proof. Combine (412), (377), and Proposition E.15. □

Corollary E.39 (A positive directional-reflection endpoint is sufficient). *For the double-buffered approximants, suppose that for every $I \Subset (0, \infty)$,*

$$\liminf_{j \rightarrow \infty} \int_I \operatorname{arsinh}(\|\mathbf{R}_{L_j}(\sqrt{E}) \xi_{L_j}(\sqrt{E})\|) \, dE \leq C_I (1 + \|V^\circ\|_{X_3}^2). \quad (544)$$

Then (378) holds for $N = 1$, and the limiting exterior density has rank at least one almost everywhere on $(0, \infty)$.

Proof. Combine (400), (377), and Proposition E.15. □

Proposition E.40 (Exact hyperbolic-angle identity and bilinear density target). *Fix real $k > 0$ and keep the unit vector $\xi_L(k)$ from (398) fixed as r varies. Put*

$$t(r, k) = \tilde{T}_L(r, k)\xi_L(k), \quad s(r, k) = \tilde{R}_L(r, k)\xi_L(k), \quad \rho(r, k) = \operatorname{arsinh} \|s(r, k)\|. \quad (545)$$

Then $\|t\|^2 - \|s\|^2 = 1$. At points where $s \neq 0$, define

$$\eta_+(r, k) = \frac{t(r, k)}{\|t(r, k)\|}, \quad \eta_-(r, k) = \frac{s(r, k)}{\|s(r, k)\|}. \quad (546)$$

The reflection-only system (395) gives the exact identity

$$\partial_r \rho(r, k) = \Re \langle \eta_+(r, k), C_L(r, k) \eta_-(r, k) \rangle. \quad (547)$$

Consequently, with R beyond the support, the terminal condition gives the exact state-correlated return work

$$\rho(1, k) = - \int_1^R \Re \langle \eta_+(r, k), C_L(r, k) \eta_-(r, k) \rangle dr \geq 0. \quad (548)$$

In particular,

$$\rho(1, k) \leq \frac{1}{2k} \int_1^R \|V^\circ(r, \cdot)\|_2 \|\Psi_+(r, k) \Psi_-(r, k)\|_2 dr, \quad (549)$$

where

$$\Psi_+(r, k) = F_{q,L}(r, k) U_+(r, k) \eta_+(r, k), \quad \Psi_-(r, k) = \overline{F_{q,L}(r, k)} U_-(r, k) \eta_-(r, k). \quad (550)$$

The product in (549) is pointwise on $\mathbb{S}^2$. Values of η_- on the set $s = 0$ may be chosen arbitrarily; they do not contribute to the integrated identity.

In particular, define along the buffered diagonal

$$\mathfrak{B}_{I,j} := \int_I \int_1^{R_j} \int_{\mathbb{S}^2} |\Psi_{+,j}(r, \sqrt{E}, \omega)|^2 |\Psi_{-,j}(r, \sqrt{E}, \omega)|^2 d\omega dr dE. \quad (551)$$

If

$$\limsup_{j \rightarrow \infty} \mathfrak{B}_{I,j} \leq C_I \left(1 + \|V^\circ\|_{X_3}^2\right), \quad (552)$$

then (544) holds and the limiting exterior density has rank at least one almost everywhere.

Proof. The flux identity gives $\|t\|^2 - \|s\|^2 = 1$. Using $s' = C_L^* t$ and $\|t\| = \sqrt{1 + \|s\|^2}$ gives

$$\partial_r \operatorname{arsinh} \|s\| = \Re \langle \eta_-, C_L^* \eta_+ \rangle, \quad (553)$$

which is (547). Since $s(R, k) = 0$, integrating inward proves (548); taking the absolute value inside the integral gives an upper bound. From (395), scalar multiplication gives

$$\begin{aligned} |\langle \eta_+, C_L \eta_- \rangle| &= \frac{1}{2k} \left| \int_{\mathbb{S}^2} V^\circ(r, \omega) \overline{\Psi_+(r, k, \omega)} \Psi_-(r, k, \omega) d\omega \right| \\ &\leq \frac{1}{2k} \|V^\circ(r, \cdot)\|_2 \|\Psi_+(r, k) \Psi_-(r, k)\|_2. \end{aligned} \quad (554)$$

This proves (549). Finally integrate over $E \in I$, use $k \geq \sqrt{\inf I}$, and apply Cauchy–Schwarz on $I \times (1, R_j) \times \mathbb{S}^2$:

$$\int_I \rho(1, \sqrt{E}) dE \leq C_I \sqrt{|I|} \|V^\circ\|_{X_3} \mathfrak{B}_{I,j}^{1/2}. \quad (555)$$

Under (552), the right-hand side is bounded by $C_I(1 + \|V^\circ\|_{X_3}^2)$. Apply Corollary E.39. $\square$

Proposition E.41 (An ideal scalar shell exposes the evanescent-coordinate artifact). *Fix an observation radius $a > 1$ and a real $k > 0$. In a finite diagonal reference, insert the ideal shell $W(r) = g\delta_a(r)W_0$, where $W_0 = W_0^*$, and normalize the coefficient matrices to $(T, R) = (I, 0)$ just outside the shell. Then the coefficients just inside it are exactly*

$$T(a^-) = I - \frac{g}{2ik} \overline{F(a)} W_0 F(a), \quad R(a^-) = \frac{g}{2ik} F(a) W_0 F(a), \quad (556)$$

and therefore

$$A(a^-) = T(a^-) + \Theta(a)R(a^-) = I. \quad (557)$$

Below the shell the coefficient matrices are constant and

$$A(r) = I + (\Theta(r) - \Theta(a))R(a^-), \quad 1 \leq r < a. \quad (558)$$

This cancellation occurs for scalar angular multiplication. Indeed, take the free reference, let $e_0 = Y_{00}$, choose a real normalized spherical harmonic $Y_{\ell 0}$, and set $w_\ell = \sqrt{4\pi}Y_{\ell 0}$ and $W_0 = P_L M_{w_\ell} P_L$ with $L \geq \ell$. Then $W_0 e_0 = Y_{\ell 0}$ and

$$\|R(a^-)e_0\| = \frac{|g|}{2k} |f_0(a, k)f_\ell(a, k)| \longrightarrow \infty \quad (\ell \rightarrow \infty), \quad (559)$$

whereas at the observation plane $x = e_0$ and the relative density ratio is exactly one. Thus raw directional reflection, and any estimate that replaces the exact boundary interference by its hyperbolic angle, can arbitrarily overcount a harmless evanescent coefficient.

Proof. Continuity across the shell and the derivative jump $F'_{V,L}(a^+) - F'_{V,L}(a^-) = gW_0 F_{V,L}(a)$ give (556); equivalently, integrate the coefficient system across the atom. Its atomic generator squares to zero because $(F, \overline{F})(\overline{F}, -F)^T = 0$. Since $\Theta(a)F(a) = \overline{F(a)}$, the two shell terms cancel exactly in (557). Equation (558) follows because T and R are constant where $W = 0$.

For the scalar example, $w_\ell e_0 = Y_{\ell 0}$. Hence the only component of $R(a^-)e_0$ is the displayed degree- ℓ component. With $f_n(r, k) = krh_n^{(1)}(kr)$, the positive-coefficient identity (325) gives

$$|f_\ell(a, k)|^2 \geq \frac{((2\ell - 1)!!)^2}{(ka)^{2\ell}}, \quad (560)$$

which diverges for fixed ka . On the other hand, (557) gives $x = e_0$ at the observation plane. This proves the claim. $\square$

Remark E.42 (Scope of the shell diagnostic). The atom $g\delta_a w_\ell$ is not an admissible X_3 potential, so Proposition E.41 is neither a counterexample to (543) nor a disproof of Simon's conjecture. It is an exact conditioning test. It proves that the unweighted endpoint (544) is not the intrinsic quantity to attack. A proof may first eliminate elliptic channels by the outgoing Feshbach estimate, or work directly with the exact reciprocal factor $\Theta' = -2ikF^{-2}$ in (408). Either route retains the cancellation in (557).

Remark E.43 (What remains after the exact gauge). The directional endpoint (544) remains a valid sufficient condition, and its hyperbolic derivative contains neither a signed logarithm nor a forward WKB phase. It is nevertheless too strong to be the canonical target. Proposition E.41 shows that it can count the conditioning of an elliptic Bessel basis rather than physical loss. The same warning applies to the bilinear density action (551).

The exact cutoff-compatible target is instead (409): density loss is the positive part of one state-correlated reciprocal work. Its factor $|\Theta'_n| = 2k/|f_{q,n}|^2$ automatically suppresses channels below their turning radius and preserves boundary refocusing. The buffered high-channel coercivity still removes the Galerkin edge. Proposition E.23 makes the remaining theorem strictly sharper: prove a scalar charged bound on the rationally weighted *signed* refocusing work inside the growing active block. Density amplification pays for its negative part automatically. Bounding the two factors in the work separately would again destroy the correlation which the exact identity retains.

Proposition E.10 does *not* prove (133) for the physical finite models: their reference operator is $-\partial_r^2 + D_L/r^2$, and Proposition E.11 shows that ordinary background subtraction still destroys dimension uniformity. Proposition E.12 removes that algebraic divergence. For $N = 1$, Proposition E.16 sharpens the next positive target to the relative-radiation-column estimate (378). For general N the target remains the radial-mean-relative estimate (368), or its dimension-normalized version followed by fixed-subspace tightness (136). None of these conclusions follows from strong resolvent convergence, one off-axis trace-class bound, or the signed trace coefficient alone.

F The first full-space converter obstruction

The preceding corollary leaves open whether angular dynamics repairs the non-invariance of the compressed lift. It is useful first to remove that dynamics completely. The resulting model has an exact obstruction which is independent of the radial profile and of its cost.

Before doing so, one can quantify exactly what the minimum lift loses into the complement. This already rejects the most literal adiabatic use of the compressed rotator.

Proposition F.1 (Complementary leakage of the minimum lift). *Let $S = 4\pi$, let h_ℓ be (268), and let Π_ℓ be the projection onto $\text{span}\{e_0, h_\ell\}$. For $v_{a,b,d}$ in (258),*

$$\|(I - \Pi_\ell)M_{v_{a,b,d}}e_0\|_2^2 = \frac{(d-a)^2}{SG_\ell}, \quad (561)$$

$$\|(I - \Pi_\ell)M_{v_{a,b,d}}h_\ell\|_2^2 = Sb^2G_\ell + \frac{(d-a)^2}{G_\ell^2}(H_{6,\ell} - H_{4,\ell}^2). \quad (562)$$

Moreover,

$$\frac{H_{6,\ell} - H_{4,\ell}^2}{G_\ell^2} \rightarrow \frac{20}{9\sqrt{3}} - 1. \quad (563)$$

For the lifted rotator in Corollary C.5, on every fixed interior portion of the mixing region on which $|\sin \theta \cos \theta| \geq c > 0$,

$$\|(I - \Pi_\ell)M_{v_{a,b,d}}h_\ell\|_2 \geq c'\delta_\Lambda \ell^{1/4}. \quad (564)$$

Thus the complementary coupling divided by the intended two-channel gap δ_Λ diverges like $\ell^{1/4}$.

Proof. Write $g_\ell = h_\ell^2 - S^{-1}$ and

$$w_\ell = g_\ell h_\ell - G_\ell h_\ell. \quad (565)$$

Direct substitution in (258) gives

$$(I - \Pi_\ell)M_{v_{a,b,d}}e_0 = \frac{d-a}{\sqrt{S}G_\ell}g_\ell, \quad (566)$$

$$(I - \Pi_\ell)M_{v_{a,b,d}}h_\ell = \sqrt{S}bg_\ell + \frac{d-a}{G_\ell}w_\ell. \quad (567)$$

The azimuthal frequencies of g_ℓ are 0 and 2ℓ , while those of w_ℓ are ℓ and 3ℓ , so the two terms in (567) are orthogonal. Also

$$\|w_\ell\|_2^2 = \|g_\ell h_\ell\|_2^2 - G_\ell^2 = H_{6,\ell} - H_{4,\ell}^2. \quad (568)$$

This proves (561) and (562). The asymptotics in (270)–(271) give (563). Finally, $G_\ell \asymp \sqrt{\ell}$ and $|b| \geq c\delta_\Lambda$ on the stated region, so the first term of (562) proves (564). $\square$

The preceding growth is specific to the minimum lift, but non-negligible leakage is forced for every bounded scalar lift. The reason is pointwise: near the large nodal region of a concentrated highest-weight state, multiplication cannot turn h_ℓ into a nonzero constant component.

Proposition F.2 (Nodal-set obstruction to a two-state scalar lift). *Let (X, μ) have measure S , let $e_0 = S^{-1/2}$, let h be real with $\|h\|_2 = 1$ and $h \perp e_0$, and let Π project onto $\text{span}\{e_0, h\}$. For every real $v \in L^\infty(X)$, put*

$$b = \langle e_0, M_v h \rangle, \quad d = \langle h, M_v h \rangle, \quad B = \|v\|_\infty. \quad (569)$$

Then, for every $t > 0$,

$$\|(I - \Pi)M_v h\|_2^2 \geq \mu\{|h| \leq t\} \left(\frac{|b|}{\sqrt{S}} - (B + |d|)t \right)_+^2. \quad (570)$$

If $b \neq 0$ and

$$t_b = \frac{|b|}{2\sqrt{S}(B + |d|)}, \quad (571)$$

then

$$\|(I - \Pi)M_v h\|_2^2 \geq \frac{b^2}{4S} \mu\{|h| \leq t_b\}. \quad (572)$$

For $h = h_\ell$, if

$$\log \frac{B_\ell + |d_\ell|}{|b_\ell|} = o(\ell), \quad (573)$$

then $\mu\{|h_\ell| \leq t_{b_\ell}\} \rightarrow 4\pi$ and hence

$$\|(I - \Pi_\ell)M_{v_\ell} h_\ell\|_2 \geq \left(\frac{1}{2} - o(1) \right) |b_\ell|. \quad (574)$$

Proof. Since

$$(I - \Pi)M_v h = vh - be_0 - dh, \quad (575)$$

the reverse triangle inequality gives, on $\{|h| \leq t\}$,

$$|vh - be_0 - dh| \geq \frac{|b|}{\sqrt{S}} - (B + |d|)t. \quad (576)$$

Integration proves (570), and the choice (571) proves (572).

For the highest-weight state, fix $\eta > 0$. On $\{|\theta - \pi/2| \geq \eta\}$,

$$|h_\ell(\theta, \phi)| \leq C\ell^{1/4}(\cos \eta)^\ell. \quad (577)$$

Condition (573) says $t_{b_\ell} = e^{-o(\ell)}$ up to fixed factors. Thus the displayed region is contained in $\{|h_\ell| \leq t_{b_\ell}\}$ for all large ℓ . Its area tends to 4π as $\eta \downarrow 0$, proving the measure assertion and (574). $\square$

For the lifted rotating block, $|b| \asymp \delta_\Lambda$ during a fixed fraction of the quarter-turn. The minimum lift has $B = O(\delta_\Lambda \ell^{1/4})$, so (573) holds. It also holds for any uniformly bounded alternative lift when $\ell = (\log \Lambda)^p$ with $p > 1$, because $\log(B/|b|) = O(\log \Lambda) = o(\ell)$. Consequently no such scalar realization makes $\text{span}\{e_0, h_\ell\}$ asymptotically invariant during the mixing stage. A successful converter, if it exists, must deliberately enter the complementary angular sector and later refocus it.

Proposition F.3 (Decomposable scalar converters cannot concentrate). *Let (X, μ) have finite measure S , let $e_0 = S^{-1/2}$, and let $h \in L^2(X)$ have norm one. Suppose that a lossless two-port scatterer is decomposable over X :*

$$(Sf)(x) = S(x)f(x), \quad S(x) \in U(2) \quad \text{for a.e. } x. \quad (578)$$

If $\mathbf{e}_+$ denotes incidence in the first port and the same symbol denotes the desired transmitted port, then, for every constant phase γ ,

$$\left\| \mathcal{S}(e_0 \mathbf{e}_+) - e^{i\gamma} h \mathbf{e}_+ \right\|_{L^2(X; \mathbb{C}^2)}^2 \geq 2 - \frac{2}{\sqrt{S}} \|h\|_{L^1(X)}. \quad (579)$$

In particular, deleting A_3/r^2 from a real scalar angular Schrödinger converter makes (578) hold with $X = \mathbb{S}^2$: every angular point is then an independent one-dimensional lossless scattering problem.

Proof. Both vectors in (579) have norm one. If $s_{++}(x)$ is the desired output component of $\mathcal{S}(x)\mathbf{e}_+$, unitarity gives $|s_{++}(x)| \leq 1$. Hence

$$\begin{aligned} \left| \left\langle e^{i\gamma} h \mathbf{e}_+, \mathcal{S}(e_0 \mathbf{e}_+) \right\rangle \right| &= \left| S^{-1/2} \int_X e^{-i\gamma} \overline{h(x)} s_{++}(x) d\mu(x) \right| \\ &\leq S^{-1/2} \|h\|_1. \end{aligned} \quad (580)$$

Expanding the squared distance proves the claim. If A_3/r^2 is absent, the real scalar radial equation contains no angular derivatives. Its direction-resolved scattering matrix is therefore the direct integral of the unitary scalar matrices $\mathcal{S}(x)$. $\square$

For the real highest-weight harmonic (268),

$$\|h_\ell\|_1 = 4C_\ell \sqrt{\pi} \frac{\Gamma((\ell+2)/2)}{\Gamma((\ell+3)/2)} = 4\sqrt{2} \pi^{-1/4} \ell^{-1/4} (1 + o(1)). \quad (581)$$

Indeed, the azimuthal integral of $|\cos(\ell\phi)|$ is 4, and the polar integral is $\int_0^\pi \sin^{\ell+1} \theta d\theta = \sqrt{\pi} \Gamma((\ell+2)/2) / \Gamma((\ell+3)/2)$. Consequently Proposition F.3 gives

$$\inf_{S, \gamma} \left\| \mathcal{S}(e_0 \mathbf{e}_+) - e^{i\gamma} h_\ell \mathbf{e}_+ \right\|^2 \geq 2 - 4\sqrt{2} \pi^{-3/4} \ell^{-1/4} (1 + o(1)). \quad (582)$$

Thus the best decomposable modal error tends to $\sqrt{2}$, not to zero. If $\mathcal{S}_\ell^{\text{full}}$ is a proposed full converter and $\mathcal{S}_\ell^{\text{dec}}$ is the scatterer obtained by deleting angular transport, the triangle inequality yields the useful perturbative test

$$\left\| (\mathcal{S}_\ell^{\text{full}} - \mathcal{S}_\ell^{\text{dec}}) e_0 \mathbf{e}_+ \right\| \geq \sqrt{2 - \frac{\|h_\ell\|_1}{\sqrt{\pi}}} - \text{err}_{\text{conv}}(\ell). \quad (583)$$

Every successful converter must therefore use the centrifugal term at order one; a comparison which makes it $o(1)$ disproves the candidate construction.

The same necessity has a local conservation-law formulation. It also identifies the geometric transport which a positive estimate must control.

Proposition F.4 (Exact angular-current requirement). *Let v be real and let u solve on $[R_1, R_2] \times \mathbb{S}^2$*

$$-\partial_r^2 u - \frac{1}{r^2} \Delta_{\mathbb{S}^2} u + vu = k^2 u. \quad (584)$$

Define the radial and angular currents

$$j_r = \Im(\bar{u} \partial_r u), \quad J_\omega = \frac{1}{r^2} \Im(\bar{u} \nabla_\omega u). \quad (585)$$

Then

$$\partial_r j_r + \text{div}_\omega J_\omega = 0. \quad (586)$$

If the boundary data are exact forward modes e_0^+ at R_1 and h_ℓ^+ at R_2 , then

$$\int_{R_1}^{R_2} \frac{1}{r^2} \int_{\mathbb{S}^2} |u| |\nabla_\omega u| d\omega dr \geq k \left(\frac{\pi}{2} - 1 - C\ell^{-1/2} \right). \quad (587)$$

The same conclusion holds with an $O_K(\varepsilon)$ loss when the two directional boundary modes have phase-space error at most ε .

Proof. Taking the imaginary part of $\bar{u}$ times (584) proves (586); the real scalar potential disappears identically. Integrate in r and test against the one-Lipschitz function

$$\varphi(\theta, \phi) = |\theta - \pi/2|. \quad (588)$$

Since a forward mode $e^{ikr} f(\omega)$ has radial current $k|f|^2$, integration by parts on the sphere gives

$$\begin{aligned} k \left| \int_{\mathbb{S}^2} \varphi \left(h_\ell^2 - \frac{1}{4\pi} \right) d\omega \right| &\leq \int_{R_1}^{R_2} \int_{\mathbb{S}^2} |J_\omega| d\omega dr \\ &\leq \int_{R_1}^{R_2} \frac{1}{r^2} \int_{\mathbb{S}^2} |u| |\nabla_\omega u| d\omega dr. \end{aligned} \quad (589)$$

The uniform spherical measure satisfies

$$\frac{1}{4\pi} \int_{\mathbb{S}^2} |\theta - \pi/2| d\omega = \int_0^{\pi/2} (\pi/2 - \theta) \sin \theta d\theta = \frac{\pi}{2} - 1. \quad (590)$$

The polar marginal of $h_\ell^2 d\omega$ is $J_\ell^{-1} \sin^{2\ell+1} \theta d\theta$, whose first absolute moment about the equator is $O(\ell^{-1/2})$. Substitution in (589) proves (587). The phase-space stability statement follows by expanding the boundary currents and applying Cauchy–Schwarz. $\square$

For example, if along the converter $\|u(r)\|_2 \leq C_0$ and $\|\nabla_\omega u(r)\|_2 \leq C_1 L_{\text{eff}}$, then

$$\int_{R_1}^{R_2} \frac{1}{r^2} \int |u| |\nabla_\omega u| \leq C_0 C_1 L_{\text{eff}} \left(\frac{1}{R_1} - \frac{1}{R_2} \right). \quad (591)$$

Therefore a converter supported far beyond its effective angular bandwidth is impossible. A construction must instead operate near the centrifugal scale $R_1 = O(L_{\text{eff}})$ or create a genuine cascade to angular frequencies much larger than the target degree. This is the sharpened scalar fork: the compressed rotator cannot simply be moved to a region where A_3/r^2 is negligible and then justified perturbatively.

The scalar cost alone does not bound L_{eff} . A long weak slab can create an arbitrarily oscillatory angular phase at vanishing cost. This is a second necessary falsification test for any proposed positive converter bound.

Proposition F.5 (Cheap scalar phase masks). *Let $K \Subset (0, \infty)$, let $\tau > 0$, and let $w \in L^\infty(\mathbb{S}^2; \mathbb{R})$ with $\|w\|_\infty \leq 1$. Delete A_3/r^2 and put the scalar potential*

$$v_\epsilon(r, \omega) = \epsilon w(\omega) \mathbf{1}_{[R, R+\tau/\epsilon]}(r). \quad (592)$$

Its full constant-column cost is

$$\int \|M_{v_\epsilon(r)} e_0\|_2^2 dr = \frac{\tau\epsilon}{4\pi} \|w\|_2^2 \longrightarrow 0. \quad (593)$$

Nevertheless, after removal of the free propagation phase, its two-port scattering matrix has reflection $O_K(\epsilon)$ and transmission multiplier

$$t_\epsilon(k, \omega) = \exp\left(-\frac{i\tau}{2k} w(\omega)\right) + O_K(\epsilon), \quad (594)$$

uniformly for $k \in K$ and almost every ω . The same statement holds in the reverse direction.

In particular, for the smooth bounded masks

$$w_L(\theta, \phi) = \cos(L\theta) = T_L(\cos \theta), \quad (595)$$

one has $\|w_L\|_2 \asymp 1$ but $\|\nabla_\omega w_L\|_2 \asymp L$. The limiting transmitted wave from e_0 therefore has angular gradient comparable to L , while the cost in (593) can tend to zero independently of L .

Proof. For each fixed ω , the slab has internal wave number

$$p_\epsilon(k, \omega) = \sqrt{k^2 - \epsilon w(\omega)} = k - \frac{\epsilon}{2k} w(\omega) + O_K(\epsilon^2). \quad (596)$$

The reflection coefficient at either interface is $O_K(\epsilon)$, uniformly in ω . The exact rectangular-slab formula therefore gives total reflection $O_K(\epsilon)$ and, relative to free propagation over the same length,

$$t_\epsilon(k, \omega) = \exp\left(i(p_\epsilon - k)\frac{\tau}{\epsilon}\right) (1 + O_K(\epsilon)), \quad (597)$$

which proves (594). Reciprocity gives the reverse statement. The cost is immediate. Finally,

$$\|w_L\|_2^2 = 2\pi \left(1 - \frac{1}{4L^2 - 1}\right), \quad (598)$$

$$\|\nabla_\omega w_L\|_2^2 = 2\pi L^2 \left(1 + \frac{1}{4L^2 - 1}\right), \quad (599)$$

so differentiating the phase in (594) proves the last assertion. $\square$

Proposition F.5 shows why (587) is a necessary condition but not yet a converter lower bound. Vanishing column cost can seed arbitrarily large angular frequency, after which the centrifugal propagator may transport angular density. The remaining issue is not generation of a cascade but its broadband refocusing into one prescribed highest-weight mode.

There is, however, a radial incompatibility hidden by the idealized product notation. A weak slab is a clean phase mask only where the centrifugal evolution during that slab is negligible. That is incompatible with making the same slab cheap at the angular-transport scale.

Proposition F.6 (Phase-mask/transport incompatibility). *Let w_L be real with $0 < c_0 \leq \|w_L\|_2 \leq C_0$ and $\|w_L\|_\infty \leq C_0$. Put*

$$v_L(r, \omega) = \epsilon_L w_L(\omega) \mathbf{1}_{[R_L, R_L + T_L]}(r), \quad \tau_L = \epsilon_L T_L. \quad (600)$$

Assume $0 < c_1 \leq \tau_L \leq C_1$. Its constant-column cost and the maximal centrifugal phase scale on degrees at most L are

$$\mathcal{C}_L := \int \|M_{v_L(r)} e_0\|_2^2 dr = \frac{\tau_L^2}{4\pi T_L} \|w_L\|_2^2, \quad (601)$$

$$\mathcal{D}_L := L(L+1) \int_{R_L}^{R_L + T_L} \frac{dr}{r^2} = \frac{L(L+1)T_L}{R_L(R_L + T_L)}. \quad (602)$$

If $R_L \leq CL$ and $\mathcal{C}_L \rightarrow 0$, then $T_L \rightarrow \infty$ and $\mathcal{D}_L \rightarrow \infty$. Conversely, under the same radius condition, $\mathcal{D}_L \rightarrow 0$ forces $T_L \rightarrow 0$ and $\mathcal{C}_L \rightarrow \infty$. Hence a nontrivial phase mask cannot be both cheap and centrifugally clean at the scale where degree- L angular transport is available.

Proof. Equations (601) and (602) are direct calculations. Vanishing cost with τ_L and $\|w_L\|_2$ bounded below implies $T_L \rightarrow \infty$. If $T_L \leq R_L$, then

$$\mathcal{D}_L \geq \frac{L(L+1)}{2R_L^2} T_L \geq cT_L. \quad (603)$$

If $T_L \geq R_L$, then

$$\mathcal{D}_L \geq \frac{L(L+1)}{2R_L} \geq cL. \quad (604)$$

This proves the first assertion. If $\mathcal{D}_L \rightarrow 0$, the second case is impossible and the first estimate forces $T_L \rightarrow 0$; then (601) diverges. $\square$

Thus the ideal masks in (594) are available cheaply only in a far-field regime relative to the frequencies which remain active during the slab. But Proposition F.4 and (591) show that such a regime cannot change the angular density unless the slab simultaneously creates frequencies much larger than its radius. The putative converter must therefore use a coupled scalar–centrifugal cascade during the weak slabs themselves; it is not a literal succession of independent optical elements.

The linearized Fourier budget has an exact nonlinear counterpart for the standard pure-detuning two-level ensemble. Recording it separates a genuine nonlinear obstruction from the additional energy scaling in the radial Schrödinger problem.

Proposition F.7 (Nonlinear Plancherel budget for a pure-detuning ensemble). *Let $q \in C_c^\infty(\mathbb{R}; \mathbb{C})$ and, for $\xi \in \mathbb{R}$, let $G(\xi, x)$ solve*

$$\partial_x G(\xi, x) = \begin{pmatrix} 0 & q(x)e^{2i\xi x} \\ -\overline{q(x)}e^{-2i\xi x} & 0 \end{pmatrix} G(\xi, x), \quad G(\xi, -\infty) = \mathbf{I}. \quad (605)$$

Write

$$G(\xi, +\infty) = \begin{pmatrix} a(\xi) & b(\xi) \\ -\overline{b(\xi)} & \overline{a(\xi)} \end{pmatrix}. \quad (606)$$

Then $|a|^2 + |b|^2 = 1$ and

$$-\int_{\mathbb{R}} \log(1 - |b(\xi)|^2) d\xi \leq \pi \int_{\mathbb{R}} |q(x)|^2 dx. \quad (607)$$

More precisely, if z_j are the zeros, counted with multiplicity, of the analytic continuation of a to $\mathbb{C}_+$, then

$$-\int_{\mathbb{R}} \log(1 - |b(\xi)|^2) d\xi = \pi \|q\|_2^2 - 4\pi \sum_j \Im z_j. \quad (608)$$

Consequently, if $|b(\xi)| \geq \eta$ on an interval J , then

$$\|q\|_2^2 \geq \frac{|J|}{\pi} \log \frac{1}{1 - \eta^2}. \quad (609)$$

Proof. For real ξ , the generator in (605) is skew-Hermitian and traceless, so $G(\xi, +\infty) \in \mathrm{SU}(2)$. The Volterra series gives an analytic continuation of a to $\mathbb{C}_+$ and the high-imaginary-energy expansion

$$\log a(z) = -\frac{i}{2z} \|q\|_2^2 + O(z^{-2}). \quad (610)$$

Factor $a = Ba_{\mathrm{out}}$ into its Blaschke and outer parts. With

$$B(z) = \prod_j \frac{z - z_j}{z - \overline{z_j}}, \quad \log a_{\mathrm{out}}(z) = \frac{1}{\pi i} \int_{\mathbb{R}} \frac{\log |a(\xi)|}{\xi - z} d\xi, \quad (611)$$

their expansions at infinity are

$$\log B(z) = -\frac{2i}{z} \sum_j \Im z_j + O(z^{-2}), \quad \log a_{\mathrm{out}}(z) = \frac{i}{\pi z} \int_{\mathbb{R}} \log |a(\xi)| d\xi + O(z^{-2}). \quad (612)$$

Comparison with (610), followed by $2 \log |a| = \log(1 - |b|^2)$, proves (608). Dropping the nonnegative zero sum proves (607); restricting its left side to J proves (609). This is the continuous nonlinear-Fourier analogue of the nonlinear Plancherel identities discussed in [16]. $\square$

This proposition does not yet settle the physical frozen converter. After subtracting a scalar phase, that two-state forward model has the form

$$i\partial_x \psi_s = s \begin{pmatrix} 0 & \beta_\ell u(x) \\ \beta_\ell u(x) & c_\ell \end{pmatrix} \psi_s, \quad s = (2k)^{-1}, \quad c_\ell = \frac{\ell(\ell+1)}{R_\ell^2}. \quad (613)$$

In the interaction picture the spectral parameter is $\xi = sc_\ell/2$, but the nonlinear-Fourier potential is $q_s = -is\beta_\ell u$ and therefore varies with the same parameter. The data occupy a diagonal curve in (ξ, q) , whereas Proposition F.7 integrates ξ with one fixed q . Thus (609) is the correct exact model theorem, but applying it to (613) requires an additional scaled-potential or second-order Schrödinger argument.

That missing step is not formal: the corresponding scaled-ensemble inequality is false for arbitrary matrix Hamiltonians.

Proposition F.8 (Scaled nonlinear Plancherel fails for matrix controls). *Fix $0 < s_0 < s_1$. There are real symmetric 2×2 Hamiltonians $H_\Lambda(x)$ supported on $[0, \Lambda]$ such that*

$$\int_0^\Lambda \|H_\Lambda(x)e_1\|^2 dx = \frac{1}{2}\Lambda^{-1/3} \longrightarrow 0, \quad (614)$$

while the solutions of

$$i\partial_x \psi_s = sH_\Lambda(x)\psi_s, \quad \psi_s(0) = e_1, \quad s \in [s_0, s_1], \quad (615)$$

satisfy

$$\sup_{s \in [s_0, s_1]} \inf_{\gamma \in \mathbb{R}} \|\psi_s(\Lambda) - e^{i\gamma} e_2\| = O(\Lambda^{-1/3}). \quad (616)$$

Consequently the fixed-band transition entropy diverges although the charged cost vanishes:

$$\int_{s_0}^{s_1} -\log(1 - |\langle e_2, \psi_s(\Lambda) \rangle|^2) ds \geq c \log \Lambda - C. \quad (617)$$

Proof. Use the first-order version of the rotator in Appendix B:

$$H_\Lambda(x) = \delta_\Lambda R(\omega_\Lambda x) \text{diag}(0, 1) R(\omega_\Lambda x)^*, \quad \delta_\Lambda = \Lambda^{-2/3}, \quad \omega_\Lambda = \frac{\pi}{2\Lambda}. \quad (618)$$

The cost is $\delta_\Lambda^2 \int_0^\Lambda \sin^2(\omega_\Lambda x) dx$, which is (614). Writing $\psi_s = R(\omega_\Lambda x)y_s$ gives the constant equation

$$i\partial_x y_s = [s\delta_\Lambda \text{diag}(0, 1) - i\omega_\Lambda \mathbf{K}] y_s. \quad (619)$$

Its lower spectral projection differs from $e_1 e_1^*$ by $O(\omega_\Lambda/\delta_\Lambda) = O(\Lambda^{-1/3})$, uniformly in s . Since $R(\pi/2)e_1 = e_2$, this proves (616). The complementary transition probability is $O(\Lambda^{-2/3})$, and taking its negative logarithm proves (617). $\square$

The principal nonlinear escape from a Fourier budget would be robust adiabatic passage. Such passage is effective in finite quantum ladders and qubit ensembles [17, 18]. Scalar angular Schrödinger operators, however, possess an order structure absent from an arbitrary matrix Hamiltonian.

Proposition F.9 (Ground-state positivity excludes a scalar adiabatic swap). *Let*

$$\mathcal{L}(r) = r^{-2}A_3 + M_{v(r)}, \quad v(r, \cdot) \in L^\infty(\mathbb{S}^2; \mathbb{R}), \quad (620)$$

with v continuous in r in the L^∞ norm. Its lowest eigenvalue is simple and has a strictly positive normalized eigenfunction g_r . Hence any continuous rank-one isolated eigenprojection of

$\mathcal{L}(r)$ which equals $e_0 e_0^*$ at a free left endpoint remains the ground-state projection. If the right endpoint is also free, it equals $e_0 e_0^*$ there and cannot converge to $h_\ell h_\ell^*$.

Quantitatively, if at some radius R one has $\|v(R, \cdot)\|_\infty \leq \epsilon$ and g_R is the normalized ground state, then

$$|\langle h_\ell, g_R \rangle|^2 \leq \frac{2\epsilon R^2}{\ell(\ell+1)}. \quad (621)$$

In particular, for $R \leq C\ell$ and $\epsilon = o(1)$, an adiabatically followed ground mode has $o(1)$ overlap with h_ℓ .

Proof. The heat semigroup of $r^{-2}A_3 + M_v$ is positivity improving on the connected sphere. Equivalently, the strong maximum principle and the variational characterization show that the ground eigenvalue is simple and its eigenfunction may be chosen strictly positive. A continuous eigenvalue branch which starts at the bottom cannot leave the ground branch without meeting a second eigenvalue; simplicity of the ground state forbids that meeting. At a free endpoint the ground eigenspace is $\mathbb{C}e_0$.

For the quantitative assertion, let E_R be the ground energy. Testing on e_0 gives $E_R \leq \epsilon$, while

$$E_R \geq R^{-2} \langle g_R, A_3 g_R \rangle - \epsilon. \quad (622)$$

Thus $\langle g_R, A_3 g_R \rangle \leq 2\epsilon R^2$. Since $A_3 h_\ell = \ell(\ell+1)h_\ell$ and $A_3 \geq 0$, the spectral theorem gives (621). $\square$

Proposition F.9 rules out the direct scalar analogue of the slowly rotating lowest-eigenvector construction in Appendix B, including a vanishing endpoint-matching potential at the centrifugal scale. It does not exclude a chirped rotating-frame branch, an excited-state path with deliberate transitions, or a genuinely nonadiabatic multimode cascade. Those mechanisms must leave the positive ground bundle; the problem is now to charge that departure on a fixed energy band.

The most direct rotating-frame escape is a monotone chirped highest-weight pulse. The continuum of energies makes its usual adiabatic scaling expensive.

Proposition F.10 (A monotone chirped scalar passage is not cheap on a band). *Let φ_ℓ be the normalized complex highest-weight harmonic, $L_z \varphi_\ell = \ell \varphi_\ell$, and put*

$$M_\ell^c = \|\varphi_\ell\|_\infty \asymp \ell^{1/4}, \quad w_{\ell, \Phi}(\omega) = \frac{\Re(e^{-i\Phi} \varphi_\ell(\omega))}{M_\ell^c}. \quad (623)$$

Consider the forward angular equation $i\partial_r U_{\ell, s} = s(R_\ell^{-2} A_3 + M_{v_\ell(r)}) U_{\ell, s}$ at $R_\ell = C\ell$ with

$$v_\ell(r, \omega) = \epsilon_\ell w_{\ell, \Phi_\ell(r)}(\omega), \quad g_\ell = \frac{\epsilon_\ell}{2\sqrt{4\pi} M_\ell^c}. \quad (624)$$

Let $\nu_\ell = \Phi'_\ell$ be monotone and sweep the complete resonance interval

$$I_\ell = c_\ell [s_0, s_1], \quad c_\ell = \frac{\ell(\ell+1)}{R_\ell^2} \asymp 1. \quad (625)$$

If the standard uniform adiabatic parameter

$$\eta_\ell := \sup_{\{r: \nu_\ell(r) \in I_\ell\}} \frac{|\nu'_\ell(r)|}{g_\ell^2} \longrightarrow 0, \quad (626)$$

then the charged cost on the sweep obeys

$$\int \|M_{v_\ell(r)} e_0\|_2^2 dr \geq \frac{2|I_\ell|}{\eta_\ell} \longrightarrow \infty. \quad (627)$$

In particular, a uniformly adiabatic monotone chirp cannot produce a vanishing-cost converter on a fixed nondegenerate energy band.

Proof. Rotation by $e^{-i\Phi_\ell L_z/\ell}$ fixes the angular mask and adds $-\nu_\ell L_z/\ell$ to the transformed generator. On $\text{span}\{e_0, \varphi_\ell\}$ the detuning is therefore $sc_\ell - \nu_\ell(r)$ and the off-diagonal coupling has size sg_ℓ . Every $s \in [s_0, s_1]$ crosses resonance, so uniform adiabaticity requires (626) throughout the complete interval I_ℓ . Monotonicity gives a sweep duration

$$T_\ell \geq \frac{|I_\ell|}{\eta_\ell g_\ell^2}. \quad (628)$$

Since $\|w_{\ell,\Phi}\|_2^2 = (2(M_\ell^c)^2)^{-1}$, the cost density is

$$\|M_{\nu_\ell} e_0\|_2^2 = \frac{\epsilon_\ell^2}{8\pi(M_\ell^c)^2} = 2g_\ell^2. \quad (629)$$

Multiplication by T_ℓ proves (627). $\square$

This proposition is deliberately restricted to a single monotone chirp and the usual uniform adiabatic regime. It does not exclude nonmonotone composite pulses, rotating frames involving several angular momenta, or nonadiabatic multimode transfer. It proves that the most immediate ensemble-control construction cannot be the missing cheap converter.

A seemingly cheaper alternative is to use a rotating low-degree scalar field and climb the highest-weight ladder one unit at a time. The centrifugal spacing between adjacent rungs is only $O(\ell^{-2})$ at $R \asymp \ell$, and a complete sweep through ℓ rungs has width $O(\ell^{-1})$. The fixed energy band nevertheless prevents an adiabatic ground ladder from ending on one common rung.

Proposition F.11 (Rotating-frame ground ladders fan out on an energy band). *Let $L_z = -i\partial_\phi$, let $\varphi_n = Y_n^n$, and consider any rotating-frame scalar angular generator whose control vanishes at its right endpoint and whose endpoint operator is*

$$\mathcal{H}_{s,\Omega}^{\text{end}} = sR^{-2}A_3 - \Omega L_z, \quad s \in [s_0, s_1]. \quad (630)$$

For $\Omega \geq 0$, the vector φ_ℓ can be a ground state of (630) only if

$$\frac{2s\ell}{R^2} \leq \Omega \leq \frac{2s(\ell+1)}{R^2}. \quad (631)$$

Consequently no single $\Omega = \Omega_\ell$ makes φ_ℓ a ground state for every $s \in [s_0, s_1]$ once

$$\ell > \frac{s_0}{s_1 - s_0}. \quad (632)$$

Thus a uniformly adiabatic construction which starts from the ground state e_0 , follows an isolated rotating-frame ground branch, turns off the scalar control, and aims at one common highest-weight mode cannot work on a fixed nondegenerate energy band.

Proof. At the free endpoint,

$$\mathcal{H}_{s,\Omega}^{\text{end}} Y_n^m = \left(\frac{sn(n+1)}{R^2} - \Omega m \right) Y_n^m. \quad (633)$$

For $\Omega \geq 0$ and fixed n , the lowest value occurs at $m = n$. Comparing the energies of φ_ℓ with $\varphi_{\ell-1}$ and $\varphi_{\ell+1}$ gives exactly (631). A common endpoint value would therefore have to satisfy

$$\frac{2s_1\ell}{R^2} \leq \Omega_\ell \leq \frac{2s_0(\ell+1)}{R^2}. \quad (634)$$

The interval is empty precisely when $s_1\ell > s_0(\ell+1)$, which is (632). An isolated ground branch remains the ground branch under uniform adiabatic following, so it cannot terminate at φ_ℓ for the complete band. $\square$

For $\Omega < 0$ the same argument selects Y_n^{-n} . A real highest-weight packet, which combines the two signs of m , is even less compatible with a nonzero endpoint rotation rate. Proposition F.11 does not exclude an excited-branch pump or a deliberately nonadiabatic ladder, but it closes the most economical adiabatic supercascade suggested by the shrinking adjacent-level spacings.

The natural next attempt is to keep the e_0 - h_ℓ centrifugal gap cancelled and replace the chirp by a finite composite sequence. The scalar cost and the exact angular current are incompatible with every such bounded-depth construction. More precisely, gap cancellation can be cheap only for a very short time, while changing the angular density in that time requires frequencies far above the target frequency.

Proposition F.12 (A gap-cancelled converter must supercascade). *Let $S = 4\pi$, $R_\ell \asymp \ell$, and $c_\ell = \ell(\ell + 1)/R_\ell^2$. Fix $s \in [s_0, s_1] \Subset (0, \infty)$, and let the normalized solution of*

$$i\partial_x \psi_{\ell,s} = s(R_\ell^{-2} A_3 + M_{v_\ell(x)}) \psi_{\ell,s}, \quad \psi_{\ell,s}(0) = e_0, \quad (635)$$

be defined on $[0, T_\ell]$. Suppose that, for almost every x ,

$$\left| c_\ell + \langle h_\ell, M_{v_\ell(x)} h_\ell \rangle - \langle e_0, M_{v_\ell(x)} e_0 \rangle \right| \leq \frac{c_\ell}{2}. \quad (636)$$

There are constants $\varepsilon_, C, c > 0$, independent of ℓ , such that if*

$$\inf_{\gamma \in \mathbb{R}} \|\psi_{\ell,s}(T_\ell) - e^{i\gamma} h_\ell\|_2 \leq \varepsilon_*, \quad (637)$$

and

$$\mathcal{E}_\ell := \int_0^{T_\ell} \|M_{v_\ell(x)} e_0\|_2^2 dx, \quad (638)$$

then

$$\mathcal{E}_\ell \geq c T_\ell \ell^{-1/2}, \quad (639)$$

$$\int_0^{T_\ell} \|A_3^{1/2} \psi_{\ell,s}(x)\|_2 dx \geq c R_\ell^2, \quad (640)$$

$$\sup_{0 \leq x \leq T_\ell} \|A_3^{1/2} \psi_{\ell,s}(x)\|_2 \geq \frac{c \ell^{3/2}}{\mathcal{E}_\ell}. \quad (641)$$

In particular, vanishing charged cost forces angular frequency $\gg \ell^{3/2}$ somewhere during the converter.

Proof. Put $a_\ell(x) = \langle e_0, M_{v_\ell(x)} e_0 \rangle$ and $d_\ell(x) = \langle h_\ell, M_{v_\ell(x)} h_\ell \rangle$. Since

$$d_\ell - a_\ell = \int_{\mathbb{S}^2} v_\ell \left(h_\ell^2 - \frac{1}{S} \right) d\omega, \quad (642)$$

Cauchy–Schwarz, (636), and $G_\ell = \|h_\ell^2 - S^{-1}\|_2^2 \asymp \sqrt{\ell}$ give

$$\|M_{v_\ell(x)} e_0\|_2^2 = \frac{1}{S} \|v_\ell(x)\|_2^2 \geq \frac{|d_\ell(x) - a_\ell(x)|^2}{S G_\ell} \geq c \ell^{-1/2}. \quad (643)$$

Integration proves (639).

The density $\rho = |\psi_{\ell,s}|^2$ satisfies the exact continuity equation

$$\partial_x \rho + \frac{2s}{R_\ell^2} \operatorname{div}_\omega \Im(\overline{\psi_{\ell,s}} \nabla_\omega \psi_{\ell,s}) = 0; \quad (644)$$

the real scalar potential again disappears. Test against $\varphi(\theta, \phi) = |\theta - \pi/2|$. The uniform density has φ -moment $\pi/2 - 1$, whereas the h_ℓ^2 moment is $O(\ell^{-1/2})$. Also

$$\| |\psi|^2 - h_\ell^2 \|_1 \leq 2 \inf_\gamma \|\psi - e^{i\gamma} h_\ell\|_2. \quad (645)$$

Choosing ε_* small and integrating (644) therefore yields

$$c \leq \frac{2s}{R_\ell^2} \int_0^{T_\ell} \int_{\mathbb{S}^2} |\psi_{\ell,s}| |\nabla_\omega \psi_{\ell,s}| d\omega dx \leq \frac{C}{R_\ell^2} \int_0^{T_\ell} \|A_3^{1/2} \psi_{\ell,s}\|_2 dx, \quad (646)$$

which proves (640). Finally, (639) gives $T_\ell \leq C\mathcal{E}_\ell \sqrt{\ell}$. Combining this with (640) and $R_\ell \asymp \ell$ proves (641). $\square$

This has a direct finite-depth consequence once the layers have the usual bounded phase-gradient budget. A quadratic-form calculation in (635) gives

$$\|A_3^{1/2} \psi_{\ell,s}(x)\|_2 \leq s \int_0^x \|\nabla_\omega v_\ell(t)\|_\infty dt. \quad (647)$$

The discrete counterpart is that free angular propagation preserves the H^1 seminorm, while a phase mask satisfies

$$\|A_3^{1/2}(e^{-isw}f)\|_2 \leq \|A_3^{1/2}f\|_2 + s\|\nabla_\omega w\|_\infty \|f\|_2. \quad (648)$$

Hence a gap-cancelled implementation with m_ℓ layers satisfying

$$\int_{I_{j,\ell}} \|\nabla_\omega v_\ell(t)\|_\infty dt \leq C\ell \quad (649)$$

for each layer—or its ideal mask/free analogue with $\|\nabla w_{j,\ell}\|_\infty \leq C\ell$ —has intermediate frequency at most $C(1+m_\ell)\ell$. Under the hypotheses of Proposition F.12, it must satisfy

$$(1+m_\ell)\mathcal{E}_\ell \geq c\sqrt{\ell}. \quad (650)$$

Thus fixed-depth designs fail, and if $\mathcal{E}_\ell \rightarrow 0$ then necessarily $m_\ell/\sqrt{\ell} \rightarrow \infty$. The surviving counterexample mechanism must be a genuinely deep super-target cascade, not an ordinary composite pulse.

The same conclusion explains why a norm-only argument cannot close the problem. The exact block dynamics has a useful normal form which isolates the remaining high-sector rotation.

Proposition F.13 (Exact scalar star-frame reduction). *Let $Q = I - e_0 e_0^*$ and write*

$$\begin{aligned} m_0(x) &= \langle e_0, M_{v(x)} e_0 \rangle, & q(x) &= Q M_{v(x)} e_0, \\ K(x) &= Q(D + M_{v(x)})Q. \end{aligned} \quad (651)$$

Let W_s solve $i\partial_x W_s = sK(x)W_s$, $W_s(0) = I_Q \mathcal{K}$, and decompose a solution as $\psi_s = z_s e_0 + \chi_s$. Set

$$\phi_0(x) = \int_0^x m_0(t) dt, \quad \alpha_s = e^{is\phi_0} z_s, \quad \eta_s = W_s^* \chi_s, \quad r_s = e^{-is\phi_0} W_s^* q. \quad (652)$$

Then

$$i\partial_x \begin{pmatrix} \alpha_s \\ \eta_s \end{pmatrix} = s \begin{pmatrix} 0 & r_s^* \\ r_s & 0 \end{pmatrix} \begin{pmatrix} \alpha_s \\ \eta_s \end{pmatrix}, \quad \|r_s(x)\|_2 = \|q(x)\|_2, \quad (653)$$

and the charged density is exactly

$$\|M_{v(x)} e_0\|_2^2 = |m_0(x)|^2 + \|r_s(x)\|_2^2. \quad (654)$$

Proof. Projection of the equation onto $\mathbb{C}e_0 \oplus Q\mathcal{K}$ gives

$$iz_s' = s(m_0 z_s + q^* \chi_s), \quad (655)$$

$$i\chi_s' = s(qz_s + K\chi_s). \quad (656)$$

Substitution of (652) proves (653). Unitarity of W_s gives the norm identity, and orthogonality of $m_0 e_0$ and q gives (654). $\square$

The star frame also upgrades the scale obstruction. Reaching an orthogonal state requires an order-one pulse area in the charged vector, irrespective of how violently its direction rotates in the complementary sector.

Corollary F.14 (Pulse-area and scalar-roughness toll). *Under the hypotheses of Proposition F.12, put*

$$q_\ell(x) = QM_{v_\ell(x)}e_0, \quad \mathcal{G}_\ell = \int_0^{T_\ell} \|\nabla_\omega v_\ell(x)\|_\infty dx \quad (657)$$

whenever the second quantity is finite. Then

$$\int_0^{T_\ell} \|q_\ell(x)\|_2 dx \geq c, \quad (658)$$

$$T_\ell \mathcal{E}_\ell \geq c, \quad (659)$$

$$\mathcal{E}_\ell \geq c\ell^{-1/4}. \quad (660)$$

If in addition $v_\ell(x, \cdot) \in W^{1,\infty}(\mathbb{S}^2)$ for almost every x , then

$$\mathcal{G}_\ell \geq \frac{c\ell^2}{T_\ell} \geq \frac{c\ell^{3/2}}{\mathcal{E}_\ell}, \quad (661)$$

$$\frac{\mathcal{G}_\ell}{T_\ell} \geq \frac{c\ell}{\mathcal{E}_\ell^2}. \quad (662)$$

Consequently, if $\text{ess sup}_x \|\nabla_\omega v_\ell(x)\|_\infty \leq B_\ell$, then

$$\mathcal{E}_\ell \geq c\sqrt{\frac{\ell}{B_\ell}}. \quad (663)$$

In particular, uniformly bounded scalar controls whose angular Lipschitz scale is $B_\ell = O(\ell)$ have a strictly positive cost floor. If each $v_\ell(x, \cdot)$ is a spherical polynomial of degree at most L_ℓ and $\|v_\ell\|_\infty \leq B$, the spherical Bernstein inequality gives

$$\mathcal{E}_\ell \geq c\sqrt{\frac{\ell}{BL_\ell}}. \quad (664)$$

Thus a bounded band-limited converter with $\mathcal{E}_\ell \rightarrow 0$ must satisfy $L_\ell \gtrsim \ell/\mathcal{E}_\ell^2 \gg \ell$.

Proof. Use the variables of Proposition F.13. Since $|\alpha_s| = |\langle e_0, \psi_{\ell,s} \rangle|$, set $\vartheta_s = \arccos |\alpha_s|$. Equation (653) and $\|\eta_s\|_2^2 = 1 - |\alpha_s|^2$ imply, in the a.e. metric-derivative sense,

$$|\vartheta'_s(x)| \leq s\|r_s(x)\|_2 = s\|q_\ell(x)\|_2. \quad (665)$$

Initially $\vartheta_s(0) = 0$, while (637) and $h_\ell \perp e_0$ give $|\alpha_s(T_\ell)| \leq \varepsilon_*$. After decreasing the fixed ε_* if necessary, integration proves (658). Cauchy–Schwarz and (654) give

$$c \leq \left(\int_0^{T_\ell} \|q_\ell\|_2 dx \right)^2 \leq T_\ell \mathcal{E}_\ell, \quad (666)$$

which is (659). On the other hand, (639) says $T_\ell \leq C\mathcal{E}_\ell\sqrt{\ell}$. Combining the last two inequalities proves (660).

Write $g_\ell(x) = \|\nabla_\omega v_\ell(x)\|_\infty$. Integrating (647) in x and reversing the order of integration gives

$$\int_0^{T_\ell} \|A_3^{1/2}\psi_{\ell,s}(x)\|_2 dx \leq s \int_0^{T_\ell} (T_\ell - t)g_\ell(t) dt \leq sT_\ell \mathcal{G}_\ell. \quad (667)$$

Together with (640), this yields the first bound in (661); the cost–time estimate yields the second and then (662). The bound (663) follows by comparing the last display with $\mathcal{G}_\ell/T_\ell \leq B_\ell$. Finally, a spherical polynomial of degree L_ℓ restricts to a trigonometric polynomial of that degree on each great circle, so Bernstein’s inequality gives $\|\nabla_\omega v_\ell\|_\infty \leq CL_\ell\|v_\ell\|_\infty$. This proves (664). $\square$

The conclusion is stronger than a state-frequency warning. A cheap gap-cancelled converter cannot be assembled from bounded scalar profiles at the target angular scale: the scalar control itself must contain a super-target rough component. At the extremal pulse-area scale $\mathcal{E}_\ell \asymp \ell^{-1/4}$, its degree must already be at least of order $\ell^{3/2}$. The remaining question is whether such a rough component can be populated and refocused uniformly in energy.

Thus the magnitude of the off-diagonal star coupling is fully charged. The only uncharged freedom is its direction $W_s(x) * q(x) / \|q(x)\|$, which is rotated by the complementary scalar dynamics and depends on the same ensemble parameter s . Generic bilinear quantum systems can transfer eigenstates with arbitrarily small L^p control for $p > 1$, and related ensemble systems exploit averaging and noncommuting brackets [19, 20]. Those theorems do not apply directly to the common scaling and scalar-multiplication geometry here, but they rule out any argument based only on the smallness of (654). A proof of SSNP_ℓ must charge the energy-dependent directional rotation and its reciprocal refocusing.

Remark F.15 (Resonant-drive scaling and the first broadband barrier). There is a natural coupled mechanism which survives the preceding incompatibility at one energy. Put

$$M_\ell = \|h_\ell\|_\infty \asymp \ell^{1/4}, \quad w_\ell = M_\ell^{-1} h_\ell, \quad \beta_\ell = (\sqrt{4\pi} M_\ell)^{-1} \asymp \ell^{-1/4}. \quad (668)$$

Then $\|w_\ell\|_\infty = 1$ and, for every real radial envelope u ,

$$\langle h_\ell, M_{uw_\ell} e_0 \rangle = \beta_\ell u, \quad \|M_{uw_\ell} e_0\|_2^2 = \beta_\ell^2 |u|^2. \quad (669)$$

Freeze the forward angular evolution on a slab beginning at $R_\ell = C\ell$. Its e_0 - h_ℓ gap is

$$\Delta_\ell(k) = \frac{\ell(\ell+1)}{2kR_\ell^2} \longrightarrow \frac{1}{2C^2k}. \quad (670)$$

At a single k_0 , the resonant choice $u(x) = \epsilon \cos(\Delta_\ell(k_0)x) \mathbf{1}_{[0,T]}(x)$ with $T \asymp (\beta_\ell \epsilon)^{-1}$ has order-one first-order transition amplitude but charged cost

$$\beta_\ell^2 \int_0^T |u(x)|^2 dx \asymp \beta_\ell \epsilon \longrightarrow 0. \quad (671)$$

For fixed $\epsilon > 0$, one also has $T = O(\ell^{1/4}) = o(R_\ell)$, so the centrifugal gap is nearly frozen. This scaling is therefore a serious candidate mechanism, not a clean far-field phase mask.

The same calculation exposes its first obstruction. In the linearized frozen model the transition amplitude is

$$c_\ell(\Delta) = -i\beta_\ell \hat{u}(\Delta), \quad \hat{u}(\Delta) = \int_{\mathbb{R}} u(x) e^{i\Delta x} dx. \quad (672)$$

If $|c_\ell(\Delta)| \geq \eta$ throughout an interval J , Plancherel gives the exact Born-level lower bound

$$\int_{\mathbb{R}} \|M_{u(x)w_\ell} e_0\|_2^2 dx = \beta_\ell^2 \|u\|_2^2 \geq \frac{\eta^2 |J|}{2\pi}. \quad (673)$$

For any fixed nondegenerate K , the interval $J_\ell = \Delta_\ell(K)$ has length bounded below as $\ell \rightarrow \infty$. Consequently the cheap resonant pulse is intrinsically single-energy at first order. A full converter must either prove a nonlinear composite-pulse evasion of (673) through complementary modes, or a positive proof must upgrade this Fourier budget to the exact reciprocal radial evolution. The statement here is only a rigorous diagnostic for the linearized frozen model, not such an upgrade.

The Fourier obstruction becomes stronger, rather than weaker, if the rough control tries to launch the supercascade in one step. Let $P_{\geq L}$ denote the projection onto spherical harmonics of degrees at least L .

Proposition F.16 (No broadband one-jump supercascade at Born order). *In the frozen laboratory (737), let $\psi_s^{(1)}(T)$ be the first Born correction to the free evolution from e_0 . For every integer $L \geq 1$,*

$$\int_{s_0}^{s_1} \|P_{\geq L} \psi_s^{(1)}(T)\|_2^2 ds \leq C_S \frac{R^2}{L(L+1)} \int_0^T \|M_{v(x)} e_0\|_2^2 dx. \quad (674)$$

Consequently, if the first Born term has norm at least η in degrees $\geq L$ for every $s \in S$, then

$$\int_0^T \|M_{v(x)} e_0\|_2^2 dx \geq c_S \eta^2 \frac{L^2}{R^2}. \quad (675)$$

For $R \asymp \ell$, a direct broadband launch to the roughness scale $L \gtrsim \ell/\mathcal{E}_\ell^2$ forced by Corollary F.14 is therefore incompatible with $\mathcal{E}_\ell \rightarrow 0$ at Born order.

Proof. Let $\{Y_{n,m}\}$ be an orthonormal spherical-harmonic basis and set

$$q_{n,m}(x) = \langle Y_{n,m}, M_{v(x)} e_0 \rangle, \quad \lambda_n = \frac{n(n+1)}{R^2}. \quad (676)$$

Up to an irrelevant terminal free phase, the first Born coefficient is

$$b_{n,m}^{(1)}(s) = -is \int_0^T q_{n,m}(x) e^{is\lambda_n x} dx. \quad (677)$$

With the Fourier convention $\hat{q}(\xi) = \int q(x) e^{i\xi x} dx$, the change of variables $\xi = s\lambda_n$ and Plancherel give

$$\int_{s_0}^{s_1} |b_{n,m}^{(1)}(s)|^2 ds \leq \frac{2\pi s_1^2}{\lambda_n} \int_0^T |q_{n,m}(x)|^2 dx. \quad (678)$$

Sum over $n \geq L$ and $|m| \leq n$, use $\lambda_n \geq L(L+1)/R^2$, and apply Parseval in the angular variable. This proves (674); the uniform lower bound gives (675). $\square$

This leaves only a nonlinear bootstrap: a putative counterexample must first concentrate substantial mass using lower modes, then use a small rough scalar component where that mass is already concentrated, and finally undo the entire excursion on a fixed energy band. Neither a direct rough kick nor a target-scale composite can do the job.

The phrase “first concentrate” can itself be charged exactly. Write $P_{\leq N}$ for the sum of the spherical-harmonic sectors of degrees at most N , $Q_{>N} = I - P_{\leq N}$, and

$$a(v) := \|M_v e_0\|_2 = |\mathbb{S}^2|^{-1/2} \|v\|_2. \quad (679)$$

Proposition F.17 (The charged gate precedes nonlinear high-sector amplification). *For every bounded real v and every $N \geq 0$,*

$$\|M_v P_{\leq N}\|_{\text{op}} \leq \|M_v P_{\leq N}\|_{\text{HS}} = (N+1)a(v). \quad (680)$$

Moreover, every normalized $f \in H^1(\mathbb{S}^2)$ satisfies

$$\|M_v f\|_2^2 \leq C \|v\|_\infty a(v) (1 + \|A_3^{1/2} f\|_2). \quad (681)$$

In particular, if $\|v\|_\infty \leq B$ and $\|M_v f\|_2 \geq \eta$, then

$$1 + \|A_3^{1/2} f\|_2 \geq \frac{c\eta^2}{B a(v)}. \quad (682)$$

If ψ_s solves the frozen laboratory (737) with $\|\psi_s(0)\|_2 = 1$ and $\psi_s(0) \in \text{Ran } P_{\leq N}$, then, without any restriction on the evolution inside $\text{Ran } Q_{>N}$,

$$\arcsin \|Q_{>N} \psi_s(T)\|_2 \leq s(N+1) \int_0^T a(v(x)) dx. \quad (683)$$

Thus $\|Q_{>N}\psi_s(T)\|_2 \geq \eta$ implies

$$\int_0^T a(v(x)) \, dx \geq \frac{\arcsin \eta}{s_1(N+1)}, \quad T\mathcal{E}_\ell \geq \frac{(\arcsin \eta)^2}{s_1^2(N+1)^2}. \quad (684)$$

Proof. The addition theorem gives, pointwise,

$$\sum_{n=0}^N \sum_{m=-n}^n |Y_{n,m}(\omega)|^2 = \frac{(N+1)^2}{|\mathbb{S}^2|}. \quad (685)$$

Consequently

$$\|M_v P_{\leq N}\|_{\text{HS}}^2 = \frac{(N+1)^2}{|\mathbb{S}^2|} \|v\|_2^2, \quad (686)$$

which proves (680). Next, $\|v\|_4^2 \leq \|v\|_\infty \|v\|_2$ and the two-dimensional Gagliardo–Nirenberg inequality on the sphere gives

$$\|f\|_4^2 \leq C \|f\|_2 (\|A_3^{1/2} f\|_2 + \|f\|_2). \quad (687)$$

Hölder's inequality and $\|f\|_2 = 1$ prove (681).

For the dynamical claim, split the Hamiltonian into its block-diagonal part relative to $P_{\leq N} \oplus Q_{>N}$ and its off-diagonal part. Passing to the unitary interaction frame generated by the block-diagonal part leaves an off-diagonal block of norm $s\|Q_{>N} M_{v(x)} P_{\leq N}\|_{\text{op}}$. The standard two-subspace angle estimate therefore gives

$$\left| \frac{d}{dx} \arcsin \|Q_{>N}\psi_s(x)\|_2 \right| \leq s\|Q_{>N} M_{v(x)} P_{\leq N}\|_{\text{op}}. \quad (688)$$

Now use (680) and integrate. The last assertion follows from Cauchy–Schwarz and $\mathcal{E}_\ell = \int_0^T a(v(x))^2 \, dx$. $\square$

Estimate (683) is nonlinear: the entire high–high block, including arbitrarily rough scalar amplification and refocusing, has been retained in the block-diagonal propagator. It says that this amplification cannot occur before charged mass has crossed the chosen spectral cut. It does not sum over N , because one charged kick may cross many cuts at once. In fact, the transport-free scalar model disproves every attempt to recover the quadratic cost merely by adding these gates.

Proposition F.18 (Wrapped phases disprove charged-gate additivity). *Fix $s_* > 0$ and $0 < \delta < 1$, and put $S_\delta = [s_*(1 - \delta), s_*(1 + \delta)]$. In azimuthal coordinates define the bounded real mask*

$$\Theta_L(\theta, \phi) := \text{Arg}(e^{iL\phi}) \in (-\pi, \pi], \quad L \geq 1. \quad (689)$$

Delete the centrifugal term and solve

$$i\partial_x \psi_{\epsilon,L,s} = sM_{v_{\epsilon,L}}(x) \psi_{\epsilon,L,s}, \quad v_{\epsilon,L}(x, \omega) = \epsilon \Theta_L(\omega) \mathbf{1}_{[0, (s_*\epsilon)^{-1}]}(x), \quad \psi_{\epsilon,L,s}(0) = e_0. \quad (690)$$

Then

$$\mathcal{E}_{\epsilon,L} = \int \|M_{v_{\epsilon,L}}(x) e_0\|_2^2 \, dx = \frac{\pi^2}{3s_*} \epsilon \longrightarrow 0, \quad (691)$$

uniformly in L , but at the terminal time $T_\epsilon = (s_\epsilon)^{-1}$,*

$$\inf_{s \in S_\delta} \|P_{\geq L} \psi_{\epsilon,L,s}(T_\epsilon)\|_2 \geq \frac{\sin(\pi\delta)}{\pi\delta} > 0. \quad (692)$$

Thus no exact nonlinear high-band estimate, nor any summation of the first-crossing gates, can be controlled solely by the charged quadratic cost after centrifugal transport has been deleted.

Proof. The evolution is explicit:

$$\psi_{\epsilon,L,s}(T_\epsilon) = e^{-i(s/s_*)\Theta_L} e_0. \quad (693)$$

Also $|\mathbb{S}^2|^{-1}\|\Theta_L\|_2^2 = \pi^2/3$, which proves (691). Set $a = s/s_*$. The azimuthal Fourier coefficient of $e^{-ia\Theta_L}$ at frequency $-L$ is

$$c_{-L}(a) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-i(a-1)y} dy = \frac{\sin(\pi(a-1))}{\pi(a-1)}. \quad (694)$$

Every spherical harmonic with azimuthal frequency $-L$ has degree at least L . Orthogonality in ϕ therefore gives $\|P_{\geq L}\psi_{\epsilon,L,s}(T_\epsilon)\|_2 \geq |c_{-L}(a)|$. For $|a-1| \leq \delta < 1$ this proves (692). $\square$

This is an exact fixed-band, all-orders failure, not a Born calculation. It does not contradict Proposition F.17: (683) charges the nonvanishing pulse area $\int a(v_{\epsilon,L}) dx$, whereas (691) makes only its quadratic cost vanish. It also does not produce the converter, because the density remains uniform. The centrifugal/density coupling is therefore indispensable.

There is a second exact toll. Scalar multiplication changes phase but not density, so all density concentration is generated by the centrifugal flow.

Proposition F.19 (Exact density-concentration action). *Let ψ_s be a normalized H^1 solution of the frozen laboratory (737), starting from e_0 , with the action below finite, and assume that for some fixed $\varepsilon < 1$,*

$$\inf_{\gamma \in \mathbb{R}} \|\psi_s(T) - e^{i\gamma} h_\ell\|_2 \leq \varepsilon. \quad (695)$$

Then, uniformly for $s \in S$,

$$\int_0^T \int_{\mathbb{S}^2} |\psi_s(x, \omega)|^2 |\nabla_\omega \psi_s(x, \omega)|^2 d\omega dx \geq c_{S,\varepsilon} R^2 \ell^{1/2}. \quad (696)$$

For the variable-coefficient first-order model

$$i\partial_r \psi_s = s[r^{-2}A_3 + M_{v(r)}]\psi_s, \quad (697)$$

the corresponding necessary condition is

$$\int_{R_1}^{R_2} \frac{1}{r^2} \int_{\mathbb{S}^2} |\psi_s|^2 |\nabla_\omega \psi_s|^2 d\omega dr \geq c_{S,\varepsilon} \ell^{1/2}. \quad (698)$$

In particular, ideal bounded phase masks contribute exactly zero to the increase of $\|\psi_s\|_4^4$; every phase-mask construction must create the required concentration during its free-propagation pieces.

Proof. Put $\rho = |\psi_s|^2$ and $J = \Im(\overline{\psi_s} \nabla_\omega \psi_s)$. Since the scalar potential is real, it cancels from the continuity equation:

$$\partial_x \rho + \frac{2s}{R^2} \operatorname{div}_\omega J = 0. \quad (699)$$

For $Y(x) = \int_{\mathbb{S}^2} \rho(x)^2$, integration by parts and the pointwise bounds $|\nabla \rho| \leq 2|\psi_s||\nabla \psi_s|$ and $|J| \leq |\psi_s||\nabla \psi_s|$ give

$$|Y'(x)| \leq \frac{8s}{R^2} \int_{\mathbb{S}^2} |\psi_s|^2 |\nabla_\omega \psi_s|^2. \quad (700)$$

For the real highest-weight harmonic, direct integration of $|\sin^\ell \theta \cos(\ell\phi)|^p$ shows

$$\|h_\ell\|_p \asymp_p \ell^{1/4-1/(2p)}, \quad 1 \leq p < \infty. \quad (701)$$

Choosing the phase in (695) and applying Hölder,

$$1 - \frac{\varepsilon^2}{2} \leq |\langle h_\ell, \psi_s(T) \rangle| \leq \|h_\ell\|_{4/3} \|\psi_s(T)\|_4, \quad (702)$$

so $Y(T) = \|\psi_s(T)\|_4^4 \geq c_\varepsilon \ell^{1/2}$, whereas $Y(0) = |\mathbb{S}^2|^{-1}$. Integrating (700) proves (696). Repeating the computation with R^{-2} replaced by r^{-2} proves (698). Finally, multiplication by a unimodular scalar phase leaves ρ and hence Y unchanged. $\square$

For a smooth ideal product (741), let $\psi_{j,k}(\tau)$ denote the state during the j th free leg, $0 \leq \tau \leq \sigma_{j,\ell}$. The same calculation, now with angular coefficient $(2k)^{-1}$, shows that

$$\sum_{j=1}^m \int_0^{\sigma_{j,\ell}} \int_{\mathbb{S}^2} |\psi_{j,k}|^2 |\nabla_\omega \psi_{j,k}|^2 d\omega d\tau \geq c_{K,\varepsilon} \ell^{1/2} \quad (703)$$

whenever the product reaches the target with error at most ε , uniformly for k in a compact interval $K \Subset (0, \infty)$. Thus a wrapped phase may cheaply create enormous angular derivatives, but focusing still has to convert those derivatives into the density action in (703). This condition does not exclude the construction: the same cheap masks can make the integrand enormous during a short free leg. The statement extends to rough masks whenever the left-hand side is finite by regular approximation; an infinite action already satisfies the necessary condition.

The centrifugal term also gives a monotone radial energy. This rules out the stationary long weak slab even when its large centrifugal action is retained rather than treated as an error.

Proposition F.20 (Radial energy/BV obstruction).

Let v belong to $W^{1,1}([R_1, R_2]; L^\infty(\mathbb{S}^2; \mathbb{R}))$, and let ψ_s be a normalized H^1 solution of (697) with $\psi_s(R_1) = e_0$. Put

$$\Gamma_v := \|v(R_1)\|_\infty + \sup_{R_1 \leq r \leq R_2} \|v(r)\|_\infty + \int_{R_1}^{R_2} \|\partial_r v(r)\|_\infty dr. \quad (704)$$

Then, uniformly in $s > 0$,

$$\frac{1}{r^2} \|A_3^{1/2} \psi_s(r)\|_2^2 \leq \Gamma_v, \quad (705)$$

$$\int_{R_1}^{R_2} \frac{1}{r^2} \int_{\mathbb{S}^2} |\psi_s| |\nabla_\omega \psi_s| d\omega dr \leq \sqrt{\Gamma_v} \log \frac{R_2}{R_1}. \quad (706)$$

Consequently, fix $\Sigma = [s_0, s_1] \Subset (0, \infty)$. If $s \in \Sigma$ and the endpoint is within a sufficiently small fixed L^2 error of h_ℓ , then, for all sufficiently large ℓ ,

$$\Gamma_v \log^2 \frac{R_2}{R_1} \geq c_\Sigma > 0. \quad (707)$$

In particular, consider stationary weak slabs

$$v_n(r, \omega) = \epsilon_n w_n(\omega), \quad R_n \leq r \leq R_n + T_n, \quad (708)$$

where $R_n \geq R_* > 0$, $\|w_n\|_\infty \leq B$, and $a(w_n) \geq b > 0$. If

$$\mathcal{E}_n = \epsilon_n^2 T_n a(w_n)^2 \longrightarrow 0, \quad (709)$$

then the left-hand side of (706) tends to zero. Hence no stationary long weak slab, nor a version with switch-on and switch-off ramps of total L^∞ variation $O(\epsilon_n)$, can perform the scalar conversion in the variable-coefficient first-order model.

Proof. Define the instantaneous angular energy

$$\mathcal{H}_s(r) := \langle \psi_s(r), [r^{-2}A_3 + M_{v(r)}] \psi_s(r) \rangle. \quad (710)$$

Differentiating along the Schrödinger flow cancels the commutator with the Hamiltonian and gives the exact identity

$$\mathcal{H}'_s(r) = -\frac{2}{r^3} \|A_3^{1/2} \psi_s(r)\|_2^2 + \langle \psi_s(r), M_{\partial_r v(r)} \psi_s(r) \rangle. \quad (711)$$

The factor s cancels. Since $\mathcal{H}_s(R_1) = \langle e_0, M_{v(R_1)} e_0 \rangle$, integration of (711), followed by $|\langle \psi, M_v \psi \rangle| \leq \|v\|_\infty$, proves (705). Cauchy–Schwarz on the sphere then gives

$$\frac{1}{r^2} \int |\psi_s| |\nabla_\omega \psi_s| \leq \frac{1}{r^2} \|A_3^{1/2} \psi_s\|_2 \leq \frac{\sqrt{\Gamma_v}}{r}, \quad (712)$$

and integration proves (706).

For the conversion toll, apply the continuity equation associated with (697) to the one-Lipschitz test function $\varphi(\theta, \phi) = |\theta - \pi/2|$. The uniform density and the highest-weight density have moments differing by $\pi/2 - 1 - O(\ell^{-1/2})$. Endpoint L^2 error controls the density error in L^1 , while the continuity equation gives

$$\left| \int_{\mathbb{S}^2} \varphi(|\psi_s(R_2)|^2 - |e_0|^2) d\omega \right| \leq 2s \int_{R_1}^{R_2} \frac{1}{r^2} \int_{\mathbb{S}^2} |\psi_s| |\nabla_\omega \psi_s| d\omega dr. \quad (713)$$

Since $s \leq s_1$, a sufficiently accurate conversion forces the current in (706) to be bounded below uniformly for $s \in \Sigma$. This proves (707).

For (708), $\Gamma_{v_n} \leq 2B\epsilon_n$. Put $\eta_n := \mathcal{E}_n/b^2$. Since $a(w_n) \geq b$, $T_n \leq \eta_n/\epsilon_n^2$. Therefore, with $t_n := \eta_n/(R_*\epsilon_n^2)$,

$$\sqrt{\Gamma_{v_n}} \log \left(1 + \frac{T_n}{R_n} \right) \leq C_B \left(\frac{\eta_n}{R_*} \right)^{1/4} t_n^{-1/4} \log(1 + t_n) \leq C_{B,R_*} \eta_n^{1/4} \longrightarrow 0, \quad (714)$$

where $\sup_{t>0} t^{-1/4} \log(1 + t) < \infty$. Equation (706) completes the proof. Adding ramps of total variation $O(\epsilon_n)$ leaves $\Gamma_{v_n} = O(\epsilon_n)$. $\square$

Corollary F.21 (Radial superoscillation toll). *Assume the gap-cancelled converter hypotheses of Proposition F.12, now in the variable-coefficient first-order model on $[R_\ell, R_\ell + T_\ell]$, with $R_\ell \asymp \ell$, and assume the same duration bound*

$$T_\ell \leq C\mathcal{E}_\ell \sqrt{\ell}. \quad (715)$$

If the conversion succeeds with fixed sufficiently small error, then

$$\Gamma_{v_\ell} \geq c \frac{R_\ell^2}{T_\ell^2} \geq c \frac{\ell}{\mathcal{E}_\ell^2}. \quad (716)$$

If in addition $\|v_\ell\|_\infty \leq B$ and its radial dependence has m_ℓ bounded-amplitude monotone layers, then

$$m_\ell \geq c_B \frac{\ell}{\mathcal{E}_\ell^2}. \quad (717)$$

Thus every vanishing-cost gap-cancelled converter is forced to be super-rough both angularly and radially.

Proof. Equation (707) and $\log(1 + T_\ell/R_\ell) \leq T_\ell/R_\ell$ give the first inequality in (716); use (715) and $R_\ell \asymp \ell$ for the second. A bounded profile with m_ℓ monotone layers has $\Gamma_{v_\ell} \leq CB(1 + m_\ell)$, which proves (717). $\square$

The preceding lower bound concerns raw variation. Raw variation by itself is not a dynamical resource and cannot be placed on the cost side of the desired estimate.

Proposition F.22 (Raw radial variation is not a cost proxy). *Fix $I = [R_1, R_2]$, a compact $\Sigma = [s_0, s_1] \Subset (0, \infty)$, and a real $v \in W^{1,1}(I; L^\infty(\mathbb{S}^2))$. Let U_s^v denote the propagator of (697). There are real smooth perturbations z_n , supported away from the endpoints of I , such that*

$$\int_I a(z_n(r))^2 dr \rightarrow 0, \quad \int_I \|z_n(r)\|_\infty dr \rightarrow 0, \quad (718)$$

$$\int_I \|\partial_r z_n(r)\|_\infty dr \rightarrow \infty, \quad \int_I \|\nabla_\omega z_n(r)\|_2^2 dr \rightarrow 0, \quad (719)$$

while

$$\sup_{s \in \Sigma} \|U_s^{v+z_n}(R_2, R_1) - U_s^v(R_2, R_1)\|_{\text{op}} \rightarrow 0. \quad (720)$$

Moreover,

$$\left| \int_I a(v(r) + z_n(r))^2 dr - \int_I a(v(r))^2 dr \right| \rightarrow 0. \quad (721)$$

Consequently, Γ_{v+z_n} can be inflated without appreciable charged cost or dynamical effect. No coercive bridge from raw radial variation to the constant-column cost can close the converter problem.

Proof. Choose nonzero real $w \in C^\infty(\mathbb{S}^2)$ and $0 \leq \chi \in C_c^\infty((R_1, R_2))$, and put

$$z_n(r, \omega) = \delta_n \chi(r) \sin(\nu_n r) w(\omega), \quad \delta_n \downarrow 0, \quad \delta_n \nu_n \rightarrow \infty. \quad (722)$$

The first, second, and fourth assertions follow directly from the factor δ_n . Periodic averaging gives $\int_I \chi(r) |\cos(\nu_n r)| dr \rightarrow (2/\pi) \int_I \chi(r) dr$; hence the term $\delta_n \nu_n \chi(r) \cos(\nu_n r) w$ in $\partial_r z_n$ proves the third assertion. Since v has finite radial variation, the reverse triangle inequality also gives $\Gamma_{v+z_n} \rightarrow \infty$. Bounded-perturbation Duhamel gives

$$\sup_{s \in \Sigma} \|U_s^{v+z_n}(R_2, R_1) - U_s^v(R_2, R_1)\|_{\text{op}} \leq s_1 \int_I \|z_n(r)\|_\infty dr, \quad (723)$$

which proves (720). The cost assertion follows from Cauchy–Schwarz and $\int_I a(z_n)^2 \rightarrow 0$. $\square$

The dynamically relevant replacement is the radial work tested against the evolving density. It admits an exact scalar continuity representation and therefore yields a first, genuinely coupled cost–action bridge.

Proposition F.23 (Effective radial-work cost–action bridge). *Let*

$$v \in W^{1,1}([R_1, R_2]; L^\infty(\mathbb{S}^2)) \cap L^2([R_1, R_2]; H^1(\mathbb{S}^2)) \quad (724)$$

be real, with $v(R_1) = v(R_2) = 0$, and let ψ_s solve (697) from e_0 . Define

$$\mathfrak{W}_s := \int_{R_1}^{R_2} \langle \psi_s, M_{\partial_r v} \psi_s \rangle dr, \quad (725)$$

$$\mathfrak{R}_v := \int_{R_1}^{R_2} \|\nabla_\omega v(r)\|_2^2 dr, \quad \mathfrak{D}_s := \int_{R_1}^{R_2} \frac{1}{r^2} \int_{\mathbb{S}^2} |\psi_s|^2 |\nabla_\omega \psi_s|^2 d\omega dr. \quad (726)$$

Assume $\mathfrak{D}_s < \infty$; the lower bounds below are automatic if it is infinite. Then the exact identities

$$\mathfrak{W}_s = \frac{1}{R_2^2} \|A_3^{1/2} \psi_s(R_2)\|_2^2 + 2 \int_{R_1}^{R_2} \frac{1}{r^3} \|A_3^{1/2} \psi_s(r)\|_2^2 dr, \quad (727)$$

$$\mathfrak{W}_s = -2s \int_{R_1}^{R_2} \frac{1}{r^2} \int_{\mathbb{S}^2} \nabla_\omega v \cdot \Im(\overline{\psi_s} \nabla_\omega \psi_s) d\omega dr \quad (728)$$

hold. Fix $\Sigma = [s_0, s_1] \Subset (0, \infty)$ and constants $0 < c_R < C_R < \infty$. If

$$c_R \ell \leq R_1 \leq R_2 \leq C_R \ell, \quad s \in \Sigma, \quad \inf_{\gamma \in \mathbb{R}} \|\psi_s(R_2) - e^{i\gamma} h_\ell\|_2 \leq \varepsilon_0 \quad (729)$$

for a sufficiently small fixed ε_0 , then

$$\mathfrak{W}_s \geq c > 0, \quad \mathfrak{R}_v \mathfrak{D}_s \geq c R_1^2. \quad (730)$$

If, in addition, $v(r)$ has angular harmonic degree at most $L \geq 1$ for almost every r , then its charged cost

$$\mathcal{E}(v) := \int_{R_1}^{R_2} a(v(r))^2 dr \quad (731)$$

satisfies

$$\mathcal{E}(v) \mathfrak{D}_s \geq c \frac{R_1^2}{L(L+1)}. \quad (732)$$

Thus, at $R_1 \asymp \ell$ and target-scale bandwidth $L \lesssim \ell$, a vanishing-cost converter would have to create density action at least $\mathfrak{D}_s \gtrsim 1/\mathcal{E}(v)$.

Proof. Integrating (711), using the endpoint conditions on v , $A_3 e_0 = 0$, and $\psi_s(R_1) = e_0$, proves (727). Next put $\rho_s = |\psi_s|^2$. Its exact continuity equation is

$$\partial_r \rho_s + \frac{2s}{r^2} \operatorname{div}_\omega \Im(\overline{\psi_s} \nabla_\omega \psi_s) = 0. \quad (733)$$

Integrating $\int \rho_s \partial_r v$ by parts first in r and then on the sphere proves (728).

The endpoint hypothesis gives $|\langle h_\ell, \psi_s(R_2) \rangle| \geq 1 - \varepsilon_0^2/2$. Therefore

$$\|A_3^{1/2} \psi_s(R_2)\|_2^2 \geq \ell(\ell+1) |\langle h_\ell, \psi_s(R_2) \rangle|^2, \quad (734)$$

and (727) gives $\mathfrak{W}_s \geq c$. By (728) and Cauchy–Schwarz,

$$|\mathfrak{W}_s| \leq 2s_1 \sqrt{\mathfrak{R}_v} \left(\int_{R_1}^{R_2} \frac{1}{r^4} \int_{\mathbb{S}^2} |\psi_s|^2 |\nabla_\omega \psi_s|^2 d\omega dr \right)^{1/2} \leq \frac{2s_1}{R_1} \sqrt{\mathfrak{R}_v \mathfrak{D}_s}. \quad (735)$$

This proves (730). Finally, angular bandlimiting gives

$$\mathfrak{R}_v \leq L(L+1) \int_{R_1}^{R_2} \|v(r)\|_2^2 dr = 4\pi L(L+1) \mathcal{E}(v), \quad (736)$$

which proves (732). $\square$

These results expose the nonlinear bootstrap without closing it. A small charged component can act strongly only after the state has acquired large angular regularity by (681); the first escape through every fixed spectral cut is nevertheless charged by (683); and matching the highest-weight state forces the independent density action (696). Proposition F.18 eliminates multiscale additivity based only on those gates, while Proposition F.20 excludes stationary and radially low-variation weak slabs once centrifugal transport is restored. Corollary F.21 moreover forces radial variation and bounded-amplitude layer count $\gtrsim \ell/\mathcal{E}_\ell^2$ under gap cancellation, but Proposition F.22 proves that this raw BV quantity can be inflated at negligible cost and with negligible dynamical effect. Proposition F.23 replaces it by the exact state-correlated work and proves the product bridge (732). What is still missing is a nonlinear estimate controlling this effective work when both the scalar bandwidth and the evolving density action are super-rough. Without such an estimate, this phase is a sharper reduction, not a proof of SSNP_ℓ .

Remark F.24 (Controllability warning). The refusal to promote the preceding tolls to a no-go theorem is essential. In other bilinear Schrödinger systems, directions lost by the linearized or quadratic dynamics can be recovered by a cubic mechanism for rougher controls [21], and phase control can combine with Lie brackets to generate small-time approximate density transport by diffeomorphisms [22]. Those results do not provide the charged L^2 -cost decay, fixed-band uniformity, or reciprocal conversion required here. They do show that Born recoil plus a first-crossing estimate cannot be declared nonlinear-complete. Conversely, known bilinear Schrödinger obstructions use spacetime smoothing effects [23], which supports the present conclusion that the missing bridge should be a uniform smoothing or nonlinear-Plancherel estimate that controls the state-correlated work (725), not raw radial variation.

Proof obligation F.25 (Physically scaled scalar nonlinear-Plancherel and refocusing test). First consider the frozen angular laboratory

$$\begin{aligned} i\partial_x U_{\ell,s}(x) &= s[D_{\ell,R} + M_{v(x)}]U_{\ell,s}(x), \quad U_{\ell,s}(0) = \mathbf{I}, \\ D_{\ell,R} &= R^{-2}A_3. \end{aligned} \quad (737)$$

where $s \in S = [s_0, s_1] \subseteq (0, \infty)$, v is real and bounded, and its endpoint values vanish. Put

$$B_{\ell,v}(s) = \langle h_\ell, U_{\ell,s}(T)e_0 \rangle. \quad (738)$$

Decide whether the scalar-multiplication estimate

$$\int_S -\log(1 - |B_{\ell,v}(s)|^2) ds \leq C_S \int_0^T \|M_{v(x)}e_0\|_2^2 dx \quad (739)$$

denoted SSNP_ℓ , holds uniformly in ℓ , $R \asymp \ell$, and T . Proposition F.8 proves that scalar multiplication is essential, while Proposition F.9 excludes the direct ground-state adiabatic counterexample. A proof of (739), followed by a controlled freezing and directional-mode argument, excludes the converter. A failure must exhibit a nonadiabatic or rotating-frame scalar multimode mechanism.

As a constructive stress test, let

$$\mathcal{P}_{w,k}f = e^{-iw/(2k)}f, \quad \mathcal{F}_{\sigma,k}f = e^{-i\sigma A_3/(2k)}f. \quad (740)$$

Decide whether bounded real masks $w_{j,\ell}$ and nonnegative propagation times $\sigma_{j,\ell}$ can produce products

$$\mathcal{U}_{\ell,k} = \mathcal{P}_{w_{m,\ell,k}}\mathcal{F}_{\sigma_{m,\ell,k}} \cdots \mathcal{P}_{w_{1,\ell,k}}\mathcal{F}_{\sigma_{1,\ell,k}} \quad (741)$$

such that, on a fixed positive-measure energy interval,

$$\sup_{k \in K} \inf_{\gamma(k)} \|\mathcal{U}_{\ell,k}e_0 - e^{i\gamma(k)}h_\ell\|_2 = o(1), \quad (742)$$

with the reciprocal directional conversion needed by the barrier argument. If so, realize the masks by separated weak slabs as in Proposition F.5 and justify the full second-order radial error including A_3/r^2 and the incompatibility in Proposition F.6. Equivalently, replace the ideal product by a coupled weak-slab evolution and prove refocusing directly. If this cannot be done, prove an obstruction which uses uniformity in k or bidirectionality together with the radial action budget. Angular Sobolev growth alone cannot work because of Proposition F.5, but Proposition F.12 shows that angular growth combined with the scalar cost already excludes every shallow gap-cancelled composite. Corollary F.14 now excludes every target-scale Lipschitz realization and forces any cheap gap-cancelled control into angular degrees $\gtrsim \ell/\mathcal{E}_\ell^2$. Proposition F.16 excludes direct fixed-band injection into that scale at first order, while Proposition F.11 excludes the natural adiabatic adjacent-rung bootstrap. Propositions F.17 and F.19 now charge the first spectral escape, strong action by a small charged component, and the density

concentration needed to reach h_ℓ . Proposition F.18 shows that the charged gates cannot be made additive without centrifugal transport, while Proposition F.20 excludes stationary and radially low-variation weak slabs when that transport is restored. Corollary F.21 shows that any surviving gap-cancelled profile is doubly super-rough: its radial variation or bounded layer count, as well as its angular bandwidth, is at least $\gtrsim \ell/\mathcal{E}_\ell^2$. Proposition F.22 shows that this raw variation can be dynamically inert, whereas Proposition F.23 proves the exact bandlimited bridge (732). The remaining stronger full-space target is to control the state-correlated work (725) by the charged cost for a super-bandwidth radially oscillatory or moving-support profile, using fixed-band uniformity and reciprocal refocusing to prevent simultaneous blow-up of $\mathfrak{R}_v$ and $\mathfrak{D}_s$. The estimate must remain uniform through the scalar–centrifugal excursion and its reverse passage. By Propositions F.8 and F.9, its proof must use scalar multiplication, the cost of leaving the ground bundle, and the directional rotation isolated by Proposition F.13; no abstract ensemble inequality can suffice.

Remark F.26 (Finite-grid diagnostic only). The idealized product (741) was stress-tested on a periodic one-dimensional angular grid with generator $-\partial_x^2$. The target was a normalized localized oscillatory packet of nominal degree 5, and the masks and free-propagation times were optimized simultaneously at five values of the common dispersion parameter $s = 1/(2k)$. On a dense 41-point check over $s \in [0.95, 1.05]$, a six-mask run reached minimum fidelity 0.987 and mean fidelity 0.995; over the wider interval $[0.75, 1.25]$, a four-mask run reached minimum fidelity 0.779 and mean fidelity 0.910. The optimizer hit its iteration limit, so these are neither certified optima nor asymptotic evidence. They serve only as a falsification warning: modest-band dependence on $1/k$ does not by itself give an obvious algebraic no-go. This model has no radial coordinate, is not $\mathbb{S}^2$, and does not satisfy the action budget of Proposition F.6; it therefore provides no evidence for a full scalar converter. The reproducible diagnostic is recorded in `tmp/numerical_verification/phase_mask_circle_ensemble.py`.

Remark F.27 (Full-modal barrier/composite diagnostic only). The scale obstruction in Proposition F.12 was also checked against an explicit compressed composite pulse. Alternating X and Z rotations with seven recorded pulse areas give minimum fidelity 0.999996 on a dense check of $s \in [0.9, 1.1]$ in the ideal two-state model. The same pulse areas were then realized by a normalized equatorial scalar barrier and the real highest-weight mixer $h_\ell/\|h_\ell\|_\infty$ in the spherical-harmonic sector consisting of degrees $n \leq 4\ell$ and azimuthal orders in $\ell\mathbb{Z}$. At $\ell = 8$ this is a 137-dimensional Galerkin system. Its minimum, mean, and maximum fidelities were respectively 0.000203, 0.0375, and 0.184. This dramatic collapse is exactly the complementary-sector effect erased by compression.

The calculation is not a lower bound: the barrier is not asymptotic at $\ell = 8$, the recorded pulse was optimized only in the compressed model, and a different deep cascade could behave differently. It is therefore only a falsification check on the ordinary finite-depth composite idea, consistent with the rigorous frequency requirement (641). The reproducible calculation is recorded in `tmp/numerical_verification/barrier_composite_sphere.py`.

G A fixed-potential Riesz–Talbot counterexample to the buffered $N = 1$ entropy route

The preceding converter restrictions leave one genuinely nonlinear escape: concentrate the selected angular state on a set of exponentially small area, apply a translated reflector there, and use reciprocity to refocus the transmitted state. This section proves that the escape is real. It gives a single bounded zonal scalar potential in X_3 for which the selected $N = 1$ entropy diverges along every cofinal positively smoothed, radially truncated, double-buffered diagonal. The result disproves the proposed fixed-potential selected-channel theorem. It does not determine the spectral multiplicity of the limiting operator and therefore does not disprove Simon’s

conjecture.

G.1 Translated barriers and the exact Hardy isometry

Lemma G.1 (Translated-barrier Hardy isometry). *Let $\mathcal{H}$ be finite dimensional. Suppose that $A, B \in \mathcal{B}(\mathcal{H})$ satisfy*

$$A^*A - B^*B = I, \quad A \text{ is invertible}, \quad \|BA^{-1}\| < 1, \quad (743)$$

and let S be unitary. For $|z| < 1$ put

$$F(z) = (A + zSB)^{-1}. \quad (744)$$

Then

$$\frac{1}{2\pi} \int_0^{2\pi} F(e^{i\phi})^* F(e^{i\phi}) d\phi = I, \quad (745)$$

and consequently, for every $y \neq 0$,

$$\frac{1}{2\pi} \int_0^{2\pi} \log \frac{\|y\|}{\|F(e^{i\phi})y\|} d\phi \geq 0. \quad (746)$$

Let P be an orthogonal projection fixed by conjugation and let

$$A = I + (\cosh \rho - 1)P, \quad B = (\sinh \rho)P, \quad \rho > 0. \quad (747)$$

If $S\bar{y} = y$ and, for some $|\lambda| = 1$,

$$\|(I - P)y\| + \|\bar{y} - \lambda y\| \leq \epsilon \|y\|, \quad (748)$$

then

$$\frac{1}{2\pi} \int_0^{2\pi} \log \frac{\|y\|}{\|F(e^{i\phi})y\|} d\phi \geq \log \cosh \rho - C_\rho \epsilon. \quad (749)$$

Proof. Set $C = BA^{-1}$ and $T = SC$. Factoring on the right gives

$$F(z) = A^{-1}(I + zT)^{-1} = A^{-1} \sum_{n \geq 0} (-zT)^n. \quad (750)$$

The flux identity gives

$$A^{-*}A^{-1} = I - C^*C = I - T^*T. \quad (751)$$

Parseval and telescoping therefore yield

$$\frac{1}{2\pi} \int F^*F = \sum_{n \geq 0} T^{*n}(I - T^*T)T^n = I. \quad (752)$$

Jensen applied to $\log \|F(e^{i\phi})y\|^2$ proves (746).

If $Py = y$ and $\bar{y} = \lambda y$, then $Sy = \bar{\lambda}y$ and

$$F(e^{i\phi})y = \frac{y}{\cosh \rho + e^{i\phi} \bar{\lambda} \sinh \rho}. \quad (753)$$

The phase mean of the logarithm of the denominator is $\log \cosh \rho$. In the approximate case the inverse in (744) has norm at most e^ρ . The resolvent identity and (748) compare $F(e^{i\phi})y$ uniformly with the preceding scalar expression with error $C_\rho \epsilon \|y\|$. Both norms lie between fixed multiples of $\|y\|$, so the logarithm is uniformly Lipschitz and (749) follows. $\square$

G.2 Radial realization at summable charged cost

The angular operators constructed later in this section are realized by one scalar radial cell as follows. The order in which the scales are chosen is important. First fix the finite incoming angular space, the number of Riesz factors, the desired error, and the finite integer ratios between their frequencies. Next choose the weak slab lengths. Only then choose a common odd carrier so large that all of the frequencies are simultaneously far above the inverse radial scale while the entire cell remains far below the square of the smallest frequency. This leaves room for both the forward and reverse Talbot flights.

Proposition G.2 (A scalar reciprocal Riesz cell). *Let $K \Subset (0, \infty)$, let $\mathcal{K}$ be a finite-dimensional, conjugation-invariant zonal space containing e_0 , and fix $m \geq 1$, $\rho > 0$, and $\epsilon > 0$. Outside any prescribed radius R_- there is a bounded, compactly supported, real zonal potential $v_{m,\epsilon}$ with vanishing spherical mean at every radius such that the following hold.*

1. *At a chosen tuning point $k_\star \in \text{int } K$, its forward Riesz train, localized hyperbolic reflector, and reverse train satisfy the hypotheses of Proposition G.6 on $\mathcal{K}$, with cell error at most ϵ .*
2. *If $s = k_\star/k$, the error is at most*

$$C_{K,\mathcal{K}}m|s - 1| + \epsilon. \quad (754)$$

3. *Inserting a sufficiently distant free collar before the cell changes its transfer, up to error ϵ , only through the scalar circle phase $z = e^{i\vartheta(k)}$; the derivative of ϑ may be made arbitrarily large, uniformly on K .*
4. *Its charged cost obeys*

$$\int \int_{\mathbb{S}^2} |v_{m,\epsilon}(r, \omega)|^2 d\omega dr \leq C_\rho e^{-cm} + \epsilon. \quad (755)$$

The supports of any prescribed countable collection of such cells can be chosen disjoint.

Proof. We give the scale ledger, including the two approximations used repeatedly below. On a slab $[R, R + T]$ put $T^{-1}w(\omega)$, where w is a fixed finite-band real multiplier of degree at most D . Variation of constants, uniformly for $k \in K$, gives

$$\mathsf{T}_{T,w}(k) = \exp\left(-\frac{i}{2k}w\right) + O_K\left(T^{-1} + \frac{D^2T}{R^2} + \frac{T}{R}\right), \quad (756)$$

$$\mathsf{R}_{T,w}(k) = O_K\left(T^{-1} + \frac{D^2T}{R^2} + \frac{T}{R}\right), \quad (757)$$

$$\int_R^{R+T} \|T^{-1}w\|_2^2 dr = T^{-1}\|w\|_2^2. \quad (758)$$

The first error is the one-dimensional long-weak-slab error, the second comes from the angular generator on the slab, and the third from replacing r by R . The estimates also hold after one k derivative. On a free collar $[r_0, r_1]$, the outgoing evolution on every fixed angular block is

$$\exp\left[-\frac{iA_3}{2k}\left(\frac{1}{r_0} - \frac{1}{r_1}\right)\right] + O_K(r_0^{-1}), \quad (759)$$

with the error uniform after the block has been fixed. These are the standard Volterra and outgoing Hankel asymptotics; their uniformity is exactly the finite-block uniformity imposed in the statement.

Apply Proposition G.8 and first record its finite odd frequency ratios $1 = q_1 < \dots < q_m = Q$. Choose the mask-slab lengths so large that the sum of (758), except for the reflector, is below

$\epsilon/10$. Now choose a common odd carrier L_* so large that, with $L_t = L_* q_t$ and with $T_{\max}$ the largest slab length, there is a radial interval satisfying

$$R_- + \text{length}(\text{cell}) \ll R, \quad QL_* \sqrt{T_{\max}} \ll R \ll L_*^2. \quad (760)$$

This interval is nonempty once $L_* \gg Q\sqrt{T_{\max}}$. Thus the centrifugal error in every weak mask slab tends to zero, while (759) can realize both $\pi/(2L_t^2)$ and $3\pi/(2L_t^2)$ in the forward and reverse trains. A successive enlargement of L_* , R , and the slab lengths makes the sum of all transmission, reflection, spherical-Talbot, and outgoing errors below ϵ .

At the focus insert a fixed smooth radial bump Q_0 multiplying

$$\chi_m - \langle \chi_m \rangle_{\mathbb{S}^2}. \quad (761)$$

Choose the bump strength so that the induced two-port coefficient on the concentrated state is (747). Subtracting the spherical mean changes only the scalar channel; a compensating reflectionless scalar phase slab absorbs that change. Equations (800)–(801) and the stability of finite-dimensional transfer matrices show that the resulting cell has error $O(\epsilon + e^{-cm})$ and reflector cost $C_\rho e^{-cm}$. All masks have zero mean because the frequencies are odd, and (761) has zero mean by construction.

The reverse train gives (802); differentiating the finite product in $s = k_*/k$ gives (754). A free collar of length comparable to its starting radius conjugates the reflection coefficient by e^{2ikd} , while its angular action is $O((1/r_0 - 1/r_1)D^2) = o(1)$ on the already fixed block. Rechoose the common carrier beyond that collar and absorb this error into ϵ . This proves the phase assertion. Finally choose the next cell beyond the end of the preceding one. This proves disjointness and (755). $\square$

G.3 Uniform lacunary placement and the fixed potential

The radial translations must work for a continuum of energies and for all later positive smoothings and Galerkin cutoffs. The required uniformity is a compact-family form of the Riemann–Lebesgue lemma.

Lemma G.3 (Uniform torus placement). *Let I be a compact interval and let $\mathcal{F} \subseteq L^1(I; C(\mathbb{T}^q))$. Given $\delta > 0$ and $M > 0$, there are real frequencies $M < n_1 < \dots < n_q$ such that, uniformly for $F \in \mathcal{F}$,*

$$\left| \int_I F(k, e^{in_1 k}, \dots, e^{in_q k}) dk - \int_I \int_{\mathbb{T}^q} F(k, z) dz dk \right| < \delta. \quad (762)$$

The frequencies may be chosen successively: once $n_1, \dots, n_{q-1}$ have been fixed to meet earlier finite lists of inequalities, n_q can be taken arbitrarily large without disturbing those inequalities. The same conclusion holds simultaneously for finitely many compact families.

Proof. Approximate the compact family by finitely many functions which are trigonometric polynomials in z with $L^1(I)$ coefficients. Choose n_1 so that every nonconstant z_1 Fourier mode is small. Having chosen $n_1, \dots, n_{t-1}$, every Fourier mode involving a nonzero power of z_t has frequency $\ell_t n_t + O(1)$ and tends to zero as $n_t \rightarrow \infty$ by the Riemann–Lebesgue lemma. Its zero power is the z_t average. Induct on t and absorb the finite-net errors. Earlier inequalities do not contain the new frequency and are therefore unchanged. Taking the maximum of finitely many thresholds proves the final assertion. $\square$

Theorem G.4 (Failure of fixed-potential buffered selected entropy). *There are a bounded real zonal potential $V \in X_3$, with*

$$\int_{\mathbb{S}^2} V(r, \omega) d\omega = 0 \quad \text{for a.e. } r, \quad (763)$$

and a compact interval $K \Subset (0, \infty)$ with the following property. For every choice

$$B_n, R_n, L_n \rightarrow \infty, \text{quad} \frac{L_n}{B_n + R_n + 1} \rightarrow \infty, \quad (764)$$

let

$$V_n = \mathbf{1}_{[1, R_n]} S_{B_n} V, \quad (765)$$

and let $J_n(k)$ be the Jost matrix of $H_{R_n, L_n}(V_n)$. Then

$$\int_K \log^- \|J_n(k)^{-1} e_0\|^2 dk \rightarrow +\infty. \quad (766)$$

In particular, the $N = 1$ hypothesis (343) is false even when one is free to choose the cofinal positive smoothing, radial truncation, and double-buffered Galerkin diagonal.

Proof. Fix $k_\star > 0$ and a small compact interval K about it. Take $m_j = \lceil C_0 \log(j+1) \rceil$, with C_0 large enough that

$$\sum_j e^{-cm_j} < \infty. \quad (767)$$

Use Proposition G.2 with errors ϵ_j satisfying $\sum_j \epsilon_j^\beta < \infty$, and put the resulting cells on successive disjoint shells. Their amplitudes are uniformly bounded; by (755) and (767),

$$\|V\|_{X_3}^2 = \sum_j \|v_{m_j, \epsilon_j}\|_{L_{r, \omega}^2}^2 < \infty. \quad (768)$$

The mean-zero and reality assertions hold cell by cell.

Choose the free radial translations successively by Lemma G.3. At the j th choice the earlier cells, a finite range of smoothing degrees, all radial cuts through the current cell, and all Galerkin cutoffs which resolve its finite angular band form finitely many compact transfer-matrix families. A diagonal choice with error at most 2^{-j} therefore makes the physical energy average agree with the independent circle-phase average, uniformly for every later member of those families. Increasing the finite range at each step covers every positive smoothing degree, every radial cut, and every finite cutoff. The high-channel Schur estimate (328) supplies the remaining uniform passage from a resolved finite active block to the full double-buffered Galerkin model.

Group the cells into dyadic blocks

$$\mathcal{B}_q = \{j : 2^{q-1} < j \leq 2^q\}. \quad (769)$$

For $N = 2^q$, Proposition G.6 applies on

$$|s - 1| \leq w_N = \frac{c}{N \log(N+1) \log \log(N+2)}. \quad (770)$$

Its conditional version, obtained by starting the proof at the left endpoint of $\mathcal{B}_q$, gives circle-phase mean loss at least $c2^q - C$. Because $s = k_\star/k$ is bi-Lipschitz near $k_\star$, the corresponding energy window has length comparable to w_{2^q} . On the complement, the conditional phase mean is nonnegative by Lemma G.1. Hence every resolved dyadic block contributes at least

$$c2^q w_{2^q} - C2^{-q} \geq \frac{c'}{q \log(q+2)} - C2^{-q} \quad (771)$$

to the energy-averaged selected loss. The uniform placement error is included in the summable final term.

A smoothing or radial cut may leave one cell only partially resolved. Its circle-phase mean cannot decrease the accumulated loss, again by (746); uniform placement bounds its actual negative contribution by the assigned summable error. Cofinality of (764) implies that every

fixed dyadic block is eventually fully resolved and remains far below the Galerkin edge. Therefore (771) and

$$\sum_{q \geq 2} \frac{1}{q \log q} = +\infty \quad (772)$$

give divergence of the signed selected loss.

Finally, pointwise for $t > 0$ one has $-\log t \leq \log^- t$. Apply this with $t = \|J_n(k)^{-1}e_0\|^2$. The divergent lower bound for the signed selected loss therefore implies (766) directly. This proves the theorem. $\square$

Corollary G.5 (What the counterexample decides). *The universal endpoint estimates SCI_1 and HFI_1 are false in every form which implies the fixed selected-channel entropy bound (343). The exact signed centrifugal and angular heat interpolation identities remain valid, but their proposed uniform endpoint bounds do not. Theorem G.4 does not disprove Simon's conjecture: it controls one fixed probe only, while absolutely continuous spectral density may survive in moving angular directions. Any continuation of the conjecture must therefore be basis-free (for example, a moving-probe or full-operator entropy), or use a mechanism other than selected-channel entropy.*

G.4 Reciprocal phase synchronization

The earlier obstruction to stacking localized barriers was a possible exponential amplification of a small non-real tail. Reciprocal barriers do the opposite in phase average. Let $a = \cosh \rho$, $b = \sinh \rho$, and let H be the symmetric unitary reciprocal phase on an active block. An ideal refocused barrier acts by

$$\Phi_z(H) = (aI + zbH)^{-1}(aH + \bar{z}bI), \quad |z| = 1. \quad (773)$$

For $h, \lambda \in \mathbb{T}$, direct cross multiplication gives

$$|\Phi_z(h) - \Phi_z(\lambda)| = \frac{|h - \lambda|}{|a + zbh| |a + zb\lambda|}. \quad (774)$$

At a synchronized phase the projective derivative is

$$L_z(\lambda) = |a + zb\lambda|^{-2}, \quad \frac{1}{2\pi} \int_0^{2\pi} L_{e^{i\phi}}(\lambda) d\phi = 1. \quad (775)$$

For every $0 < \beta < 1$, strict Jensen gives

$$q_\beta := \frac{1}{2\pi} \int_0^{2\pi} L_{e^{i\phi}}(\lambda)^\beta d\phi < 1. \quad (776)$$

The value is independent of λ .

Proposition G.6 (Robust reciprocal-cascade entropy). *Consider N successive real reciprocal cells on nested finite active blocks. Suppose that, after the forward and reverse converter gauges, the j th cell has transfer coefficients*

$$A_j(s) = aI + E_{j,+}(s), \quad B_j(s) = bI + E_{j,-}(s) \quad (777)$$

on the active block, with the exact flux relations and

$$\|E_{j,+}(s)\| + \|E_{j,-}(s)\| \leq \eta_j(s), \quad \eta_j(s) \leq Cm_j|s - 1| + \epsilon_j, \quad (778)$$

where $m_j \leq C_0 \log(j+1)$ and $\sum_j \epsilon_j^\beta$ is sufficiently small. Starting from the free constant channel, let $J_N(s, z)$ be the resulting Jost block. There are $c_0, C_1 > 0$ such that, for

$$|s - 1| \leq w_N, \quad w_N = \frac{c_0}{N \log(N+1) \log \log(N+2)}, \quad (779)$$

one has

$$\int_{\mathbb{T}^N} -\log \|J_N(s, z)^{-1}e_0\| dz \geq c_0 N - C_1. \quad (780)$$

Every circle carries normalized Haar measure.

Proof. With zero error and $s = 1$, the refocused active Jost block has the scalar recurrence

$$d_j = ad_{j-1} + z_j b \overline{d_{j-1}}, \quad d_0 = 1. \quad (781)$$

Thus

$$\int_{\mathbb{T}} \log |ad + zb\bar{d}| dz = \log |d| + \log a, \quad (782)$$

and the exact mean loss is $N \log a$.

For the perturbed cascade put $H_j = J_j^{-1} \overline{J_j}$ in the refocused active frame. Its ideal update is (773); the operator-norm perturbation is $O(\eta_j)$. If the spectrum of H_j lies in an arc I , then the ideal image lies in $\Phi_z(I)$ and

$$|\Phi_z(I)| = \int_I |a + zbe^{i\vartheta}|^{-2} d\vartheta. \quad (783)$$

Its phase mean is $|I|$. Strict concavity and (776) give, uniformly for arcs of length at most a fixed $\ell_0 < 2\pi$,

$$\mathbb{E}_z |\Phi_z(I)|^\beta \leq q |I|^\beta, \quad q < 1. \quad (784)$$

There is only one way a short arc can make a macroscopic excursion: the random repelling point of Φ_z must fall within $O(|I| + \eta_j)$ of it. Equivalently, apply the maximal inequality to the nonnegative arc-length martingale generated by (783). An error arc injected with length $C\eta_j$ has probability at most $C\eta_j/\ell_0$ of ever reaching ℓ_0 . Hence the excursion probability before time N is at most

$$C \sum_{j \leq N} \eta_j(s) \leq \frac{C}{\log \log(N+2)} + C \sum_j \epsilon_j, \quad (785)$$

where (779) and $\sum_{j \leq N} m_j = O(N \log N)$ were used. On the complement, (784) and subadditivity of t^β keep the reciprocal phase in an arbitrarily short arc, except for a set of phase histories of arbitrarily small measure.

If H_j lies in an arc of length $o(1)$ around λ_j , then $H_j \bar{y}_j = y_j$ implies

$$\|\bar{y}_j - \bar{\lambda}_j y_j\| = o(1) \|y_j\|. \quad (786)$$

Lemma G.1 and (778) give a conditional phase-mean loss $\log a - o(1)$ at such a stage. At the exceptional stages the conditional mean remains nonnegative by (746). Choose the summable error budget small, discard finitely many initial stages, and sum the conditional means. This proves (780). $\square$

For a physical inner Jost matrix J , a reflectionless real converter with transmission U , and $x = J^{-1}e_0$, the state at the reflector is $y = Ux$ and the reciprocal phase is

$$S = UJ^{-1}\overline{J}U^T. \quad (787)$$

The half-line scattering identity makes S unitary and gives the exact outgoing-square relation

$$S\bar{y} = y. \quad (788)$$

G.5 The zonal quarter-Talbot Riesz factor

Put $D = -\partial_x^2$ on the circle and fix $0 < a < \pi/4$. Define

$$g_s(x) = e^{-i\pi s D/2} e^{-ias \cos x}. \quad (789)$$

Even and odd Fourier modes have multipliers 1 and $-i$ at $s = 1$, so

$$g_1(x) = \cos(a \cos x) - \sin(a \cos x) > 0, \quad (790)$$

$$p(x) = g_1(x)^2 = 1 - \sin(2a \cos x), \quad \frac{1}{2\pi} \int_0^{2\pi} p(x) dx = 1. \quad (791)$$

Strict Jensen gives the positive tilted entropy

$$D_p := \frac{1}{2\pi} \int_0^{2\pi} p \log p dx > 0. \quad (792)$$

Proposition G.7 (Zonal spherical Talbot factor and reciprocal inverse). *Let $\mathcal{K} \subset C^\infty(\mathbb{S}^2)$ be finite dimensional, zonal, and invariant under conjugation. Uniformly for s in a fixed compact neighbourhood of 1 and odd integers $L \rightarrow \infty$,*

$$e^{-i\pi s A_3/(2L^2)}(e^{-ias \cos(L\theta)} f) = g_s(L\theta) f + \mathcal{R}_{L,s} f, \quad \|\mathcal{R}_{L,s}\|_{\mathcal{K} \rightarrow L^2} \leq C_{\mathcal{K},a} \frac{\sqrt{\log L}}{L}. \quad (793)$$

A sign-reversed mask following a free flight of scaled duration $3\pi/(2L^2)$ gives an operator $W_{L,s}$ such that, for the forward factor $U_{L,s}$ above,

$$\|W_{L,1} U_{L,1} - \mathbf{I}\|_{\mathcal{K} \rightarrow L^2} \leq C_{\mathcal{K},a} \frac{\sqrt{\log L}}{L}, \quad (794)$$

$$\|W_{L,s} U_{L,s} - W_{L,1} U_{L,1}\|_{\mathcal{K} \rightarrow L^2} \leq C_{\mathcal{K},a} |s - 1|. \quad (795)$$

Proof. The identity $\cos(L\theta) = T_L(\cos \theta)$ makes the mask globally smooth. On zonal functions

$$A_3 = -\partial_\theta^2 - \cot \theta \partial_\theta. \quad (796)$$

Compare its scaled evolution with that of $-\partial_\theta^2$ by Duhamel. For every smooth periodic H ,

$$\|\cot \theta \partial_\theta H(L\theta)\|_{L^2(\mathbb{S}^2)} \leq C_H L \sqrt{\log L}. \quad (797)$$

The evolution time is $O(L^{-2})$, giving (793); derivatives falling on the fixed envelope cost only $O(L^{-1})$. In the flat fast variable $e^{-i3\pi D/2} = e^{i\pi D/2}$. Thus the sign-reversed mask after the $3\pi/2$ flight is the inverse factor at $s = 1$. The same Duhamel comparison proves (794); differentiation in s on the scaled finite fast-frequency block proves (795). $\square$

Proposition G.8 (Uniform Riesz concentration on a finite incoming block). *Fix $0 < \alpha < D_p$, a finite-dimensional conjugation-invariant zonal space $\mathcal{K}$, an integer m , and $\epsilon > 0$. There are odd, superlacunary frequencies $L_1 < \dots < L_m$, a real finite-band function $0 \leq \chi_m \leq 1$, and forward and reverse Talbot products $U_{m,s}$ and $W_{m,s}$ such that, with*

$$G_m(\theta) = \prod_{t=1}^m g_1(L_t \theta), \quad E_m = \left\{ \sum_{t=1}^m \log p(L_t \theta) \geq \alpha m \right\}, \quad (798)$$

one has

$$\|U_{m,1} f - G_m f\|_2 \leq \epsilon \|f\|_2, \quad (799)$$

$$\|(1 - \chi_m) U_{m,1} f\|_2 \leq (\epsilon + C e^{-cm}) \|f\|_2, \quad (800)$$

$$\|\chi_m\|_2^2 \leq C e^{-cm}, \quad (801)$$

$$\|W_{m,1} U_{m,1} - \mathbf{I}\|_{\mathcal{K} \rightarrow L^2} \leq \epsilon, \quad (802)$$

$$\|W_{m,s} U_{m,s} - W_{m,1} U_{m,1}\| + \|U_{m,s} - U_{m,1}\| \leq C_{\mathcal{K},a} m |s - 1|. \quad (803)$$

The last two norms are from $\mathcal{K}$ to L^2 .

Proof. Choose the frequencies successively. Riemann–Lebesgue, uniformly on the unit sphere of $\mathcal{K}$, makes every exponential moment used below as close as desired to the product moment of independent copies of $\log p$. On E_m , $\prod p(L_t\theta) \geq e^{\alpha m}$, hence $|E_m| \leq Ce^{-\alpha m}$. Under the tilted density $p(x) dx/(2\pi)$, the mean of $\log p$ is $D_p > \alpha$. Chernoff therefore gives, uniformly for $f \in \mathcal{K}$,

$$\int_{E_m^c} |G_m|^2 |f|^2 d\omega \leq Ce^{-cm} \|f\|_2^2. \quad (804)$$

Apply the positive finite-band spherical smoother (314) to a slightly enlarged typical set. For sufficiently large smoothing degree it gives $0 \leq \chi_m \leq 1$, (801), and the same weighted-tail estimate. Proposition G.7, with still more separated frequencies, proves (799) and (800). Apply the reciprocal factors in reverse order to obtain (802). Telescoping the two products and using (795) proves (803). $\square$

H Notation and normalization checks

- (i) $P_N \leq P_L$ means inclusion of ranges; it is not equivalent to the numerical statement $L \geq N$ unless the harmonic enumeration has been fixed accordingly.
- (ii) The physical probe contains the factor $r^{-(d-1)/2}$ in (126); the flattened probe does not.
- (iii) The Jost normalization is fixed by (54). Calling e^{ikr} “unit flux” without a factor $(2k)^{-1/2}$ would be inconsistent.
- (iv) The exterior matrix measure is not identified with the full-space measure until the multiplicity preserving bridge in Theorem 12.1(d) is supplied.
- (v) The proposed FCEN/PSR statement is a sufficient strengthening of Simon’s conjecture, not a known equivalent reformulation.

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